INCIDENT RESPONSE REPORT

Forensic analysis: test03

Case	test03
Main system	?
Verdict	COMPROMISED (probable, network-only evidence): CITADEL-DC01 (10.42.85.10) and DESKTOP-SDN1RPT (10.42.85.115) in domain C137.LOCAL were compromised. Access came through an RDP brute force of the domain Administrator account from 194.61.24.102. The attacker then ran a Meterpreter stager (coreupdater.exe) on both hosts with C2 to 203.78.103.109:443. Logon-triggered C2 re-staging was observed, and C2 was still active when the capture ended.
Trace period (UTC)	2020-09-18 21:58:07.470323 → 2020-09-19 05:25:21.448949 (99.8 % of the events; full range 2020-09-18 21:58:07.470323 → 2020-09-19 05:38:57.828520)
Incident window (UTC)	2020-09-18 20:59:58 → 2020-09-19 06:38:51
Findings	Critical: 6, High: 7, Medium: 4, Low: 2, Informational: 3
Report date	2026-10-20 17:56 UTC
Classification	CONFIDENTIAL: TLP:AMBER
Production	Report generated by Sleuthline (analyze.py) with the assistance of a language model. Model: anthropic / claude-opus-5-5.

Disclaimer: this report was produced automatically. Every finding is linked to verifiable records of the case database ("Reference" column) and to the original path of the artifact. It must be reviewed and validated by an analyst before any distribution or decision.

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1. Executive summary

Verdict: COMPROMISED (probable, network-only evidence): CITADEL-DC01 (10.42.85.10) and DESKTOP-SDN1RPT (10.42.85.115) in domain C137.LOCAL were compromised. Access came through an RDP brute force of the domain Administrator account from 194.61.24.102. The attacker then ran a Meterpreter stager (coreupdater.exe) on both hosts with C2 to 203.78.103.109:443. Logon-triggered C2 re-staging was observed, and C2 was still active when the capture ended.

The only evidence in this case is one network capture (case001.pcap, 2020-09-18 21:58:07 → 2020-09-19 05:38:57 UTC). It shows an external attacker taking control of the domain controller CITADEL-DC01 and the workstation DESKTOP-SDN1RPT in domain C137.LOCAL.

The attack started at 02:19 UTC on 2020-09-19:
- 194.61.24.102 scanned the DC's Internet-exposed RDP service with Nmap.
- It then guessed the domain Administrator password: 95 failed NLA attempts in 21 s, then a successful authentication at 02:21:47 UTC.
- At 02:22:36 the attacker opened an interactive RDP session on the DC. They used Internet Explorer to download a 7 KB program, "coreupdater.exe", from their own web server, and ran it. This opened a Metasploit/Meterpreter remote-control channel to a second attacker server, 203.78.103.109, on TCP 443.
- At 02:35:55 the attacker used the DC to RDP into the workstation as Administrator, downloaded the same file with Edge, and ran it at 02:40:49.
- Later, Meterpreter started again on its own about 30 s after new interactive logons, including a legitimate logon by user ricksanchez at 04:08:15. This strongly suggests the attacker installed persistence that runs at logon.
- When the capture ended, three attacker C2 sessions were still open (one on the DC, two on the workstation). The attacker therefore still had access.

Impact:
- The domain Administrator account and the domain controller are compromised, so the whole AD domain must be treated as compromised.
- No large data theft was seen on the wire. However, the encrypted RDP and C2 channels (≈0.5–0.7 MB uploaded, 12.8 MB of RDP from the DC) could have carried credentials or small files.

Confidence: high for the attack sequence on the wire (Suricata alerts plus matching flow, HTTP, fileinfo, RDP and Kerberos records). Execution and persistence are only probable, because no host artefacts (EVTX, registry, Prefetch, memory) were provided to confirm them.

Key figures

Findings	22 (critical: 6, high: 7, medium: 4, low: 2, informational: 3)
Key timeline events	24
Indicators of compromise	12
Questions answered	0 / 0 asked in the guidance
First / last key event (UTC)	2020-09-18 21:59:58 → 2020-09-19 05:38:51

Main findings

ID	Severity	Confidence	Finding	Period (UTC)
F-013	**Critical**	probable	Administrator password guessed over RDP/NLA: 95 failed CredSSP attempts in 21 s then first successful authentication at 02:21:47 UTC from 194.61.24.102	2020-09-19 02:19:26.619607 → 2020-09-19 02:22:36.664454
F-003	**Critical**	probable (model: confirmed, lowered by the evidence	RDP brute-force against CITADEL-DC01 from 194.61.24.102 followed by long interactive Administrator RDP sessions	2020-09-19 02:19:26.792622 → 2020-09-19 02:56:03.539280

ID	Severity	Confidence	Finding	Period (UTC)
		policy)		
F-022	Critical	probable	Consolidated incident chronology (network-only): RDP brute force → DC compromise → Meterpreter → RDP pivot to workstation → logon-triggered C2 persistence, ongoing at end of capture	2020-09-19 02:19:26.792622 → 2020-09-19 04:08:45.483780
F-004	Critical	probable (model: confirmed, lowered by the evidence policy)	Download of /coreupdater.exe (PE, 7168 B) from 194.61.24.102 by the DC (IE11) and the workstation (Edge)	2020-09-19 02:24:06.940063 → 2020-09-19 02:39:58.412735
F-007	Critical	probable	Execution of coreupdater.exe (reverse-TCP Meterpreter stager) inferred on CITADEL-DC01 and DESKTOP-SDN1RPT from download→outbound C2 sequence	2020-09-19 02:24:06.940063 → 2020-09-19 02:56:38.402867
F-005	Critical	probable (model: confirmed, lowered by the evidence policy)	Metasploit/Meterpreter stage delivery and C2 over raw TCP 443 from 203.78.103.109 to DC and workstation	2020-09-19 02:25:18.565676 → 2020-09-19 04:08:45.783080
F-012	High	probable	Logon-triggered re-launch of Meterpreter on both hosts without re-download – suspected autostart persistence (Run key / Startup folder / logon task), mechanism not observable	2020-09-18 21:59:39.797903 → 2020-09-19 05:03:14.869462
F-016	High	probable	Remote-access / lateral-movement channel inventory: RDP is the only remote-control channel (attacker→DC via non-Windows CredSSP client; DC→workstation via native mstsc with RDP-UDP); no PsExec/WinRM/WMI/admin$/RMM traffic	2020-09-19 02:21:26.114786 → 2020-09-19 04:09:52.725024
F-018	High	probable	Web origin of coreupdater.exe: attacker manually browsed to http://194.61.24.102/ with IE11 (DC) and Edge (workstation); SmartScreen app-reputation checks mark the user-launched execution on the workstation at 02:40:45 UTC	2020-09-19 02:23:09.692112 → 2020-09-19 02:40:49.938222
F-010	High	probable (model: confirmed, lowered by the evidence policy)	C2 channel characterisation: 5 raw-TCP Meterpreter sessions to 203.78.103.109:443 (no TLS); 3 alive at end of capture; no bulk exfiltration seen on the wire	2020-09-19 02:25:18.565676 → 2020-09-19 04:08:45.483780
F-006	High	probable	Lateral movement: RDP from CITADEL-DC01 (10.42.85.10) to DESKTOP-SDN1RPT (10.42.85.115) during attacker session	2020-09-19 02:35:55.285340 → 2020-09-19 02:36:23.515604
F-008	High	probable	First interactive logon of domain Administrator on DESKTOP-SDN1RPT (profile creation) at 02:36:24 UTC during RDP from the DC	2020-09-19 02:35:55.285340 → 2020-09-19 02:36:26.768288
F-009	High	probable	Meterpreter re-launched on DESKTOP-SDN1RPT 30 s after local logon of user ricksanchez (04:08:15 UTC) – logon-triggered persistence suspected	2020-09-19 02:40:49.936293 → 2020-09-19 04:08:45.483780
F-011	Medium	confirmed	RDP brute-force tooling fingerprint: OpenSSL-based non-Windows client (JA3 75fe5199…, TLS 1.3) plus Nmap SSL probes from 194.61.24.102	2020-09-18 21:58:07.470323 → 2020-09-19 02:19:26.792998
F-017	Medium	possible	FileShare activity on CITADEL-DC01 seen over SMB2: workstation browsed the root only, no file read or write; share root last modified 2020-09-19 03:34:18 UTC, during the attacker's C2 presence on the DC	2020-09-18 22:10:08.218493 → 2020-09-19 04:49:47.678833
F-021	Medium	probable	Collection/exfiltration assessment: no bulk exfiltration on the wire; only Meterpreter C2 uploads (~0.68 MB total) and the attacker's RDP channel could have carried data	2020-09-19 00:44:00.322000 → 2020-09-19 04:08:45.483780
F-019	Medium	probable	Defense evasion by masquerading: Meterpreter stager named 'coreupdater.exe' and C2 over raw (non-TLS) TCP on port 443	2020-09-19 02:24:06.940653 → 2020-09-19 02:29:49.290306
F-015	Low	possible	Domain Administrator AS-REP encrypted with RC4-HMAC (all other principals AES256) – Administrator account likely lacks AES keys / legacy password	2020-09-18 21:59:58.148531 → 2020-09-19 04:08:15.868598
F-020	Low	possible	Windows Defender cloud protection (MAPS, wdcp.microsoft.com) contacted once by the workstation during the coreupdater.exe download; the stager still ran – Defender did not block it (cause undeterminable)	2020-09-19 02:29:49.290306 → 2020-09-19 02:40:01.538456

Technical assessment

Environment (F-001/F-014):
- AD domain C137.LOCAL.
- DC CITADEL-DC01, 10.42.85.10, Windows Server 2012 R2, with TCP 3389 reachable from the Internet.
- Workstation DESKTOP-SDN1RPT, 10.42.85.115, Windows 10 build 19041.
- Accounts seen in Kerberos: DESKTOP-SDN1RPT$ (boot at 21:59:39), mortysmith (legitimate console logon at 21:59:58), Administrator, ricksanchez.

1. Reconnaissance and initial access (F-003, F-011, F-013; T1595, T1110.001, T1133, T1078.002):
- 194.61.24.102 sent an Nmap RDP probe at 02:19:26 (mstshash=nmap, JA3 3bdfb64d…).
- About 29.2k TLS-only flows followed at 02:19–02:20 from an OpenSSL/non-Windows client (JA3 75fe5199…, JA4 t13i781000_ab95583b6d39_d41ae481755e).
- Real password guessing: 95 CredSSP failures with mstshash=Administrator at 02:21:26–02:21:46, then the first successful NLA authentication at 02:21:47.
- Interactive RDP session from 02:22:36 to 02:52:47 (17.5 MB), and a second session from 02:56:03 to 02:57:41.
- Administrator's AS-REP used RC4-HMAC while all other principals used AES256 (F-015). This points to a legacy, never-rotated password.

2. Tool transfer and execution on the DC (F-004, F-007, F-018, F-019; T1105, T1204.002, T1059, T1036.005):
- IE11 started at 02:23:09 and requested GET http://194.61.24.102/ at 02:23:41 (228 B landing page, SHA256 6978f8fa… 2d77).
- GET /coreupdater.exe at 02:24:06 returned a 7168-byte PE: SHA256 10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6, SHA1 fd153c66386ca93ec9993d66a84d6f0d129a3a5c, MD5 eed41b4500e473f97c50c7385ef5e374.
- 02:25:18: outbound raw TCP to 203.78.103.109:443 receiving a ~0.5–0.9 MB Meterpreter stage (ET Metasploit Bind_API / PE over raw TCP alerts).
- That session was closed at 02:29:36 and re-staged at 02:29:49.

3. C2 (F-005, F-010; T1071, T1571, T1095, T1620):
- Five raw (non-TLS) TCP sessions to 203.78.103.109:443:
  - DC at 02:25:18, 02:29:49 and 02:56:38;
  - workstation at 02:40:49 and 04:08:45.
- Neither attacker IP was resolved through DNS.
- Three sessions were still ESTABLISHED at 05:38:51 (T-023).

4. Lateral movement (F-006, F-008, F-016; T1021.001):
- RDP from the DC to the workstation ran from 02:35:55 to 02:52:14. It negotiated hybrid_ex, carried no cookie and added an RDP-UDP channel, which is consistent with native mstsc.exe running inside the attacker's DC session.
- Administrator obtained a Kerberos TGT at 02:36:24 and logged on to the workstation for the first time (NETLOGON 'Default User.V6' lookup, protected_storage DPAPI pipe).
- Edge started at 02:39:04, loaded the landing page at 02:39:26 and downloaded coreupdater.exe at 02:39:58.
- Defender MAPS was contacted at 02:39:46 (F-020).
- SmartScreen checkappexec lookups at 02:40:45 and 02:40:49.938; the C2 connection opened 2 ms before the second one, at 02:40:49.936.
- No PsExec, SVCCTL, ATSVC, WMI, WinRM or admin$/C$ traffic and no executable transferred over SMB.

5. Persistence (F-009, F-012; T1547.001 / T1053.005 suspected):
- Meterpreter was re-staged with no new download about 30–35 s after interactive logons:
  - DC: RDP at 02:56:03, C2 at 02:56:38.
  - Workstation: ricksanchez console logon at 04:08:15, C2 at 04:08:45.

- The mechanism (Run key, Startup folder or logon task) cannot be seen in the capture.
- GPO files were unchanged (constant MD5s), and there was no remote service or task creation.

6. Other activity:
- The DC FileShare root was modified at 03:34:18 while the DC C2 was live (F-017). The cause cannot be determined.
- After the ricksanchez logon, the workstation only browsed the share root.
- Exfiltration (F-021): no bulk upload, cloud-storage use, DNS tunnelling or HTTP POST of data. C2 uploads total ≈0.5–0.7 MB, and the attacker RDP channel sent 12.76 MB from the DC. Clipboard or drive-redirection theft cannot be excluded.

Consolidated chronology: F-022, timeline events T-001 to T-024.

Noise excluded: Discord, device-metadata/USB-metadata user-agent alerts, the IE UA-CPU header, OneDrive client setup and the analyst/collection context.

2. Scope, sources and methodology

2.1 Analyzed system

Observed host names	?
Observed accounts	?
Machine time zone	ActiveTimeBias = None min (UTC = local time + bias)
Time convention of the report	All times are UTC.
Available traces	99,868 timeline events, from 2020-09-18 21:58:07.470323 to 2020-09-19 05:38:57.828520

2.2 Collection and processing

KAPE collection (!SANS_Triage target); parsing by recognized tools orchestrated by orchestrator.py; normalization into DuckDB by normalize.py; analysis by analyze.py following the playbook. The raw tool outputs are kept in C:\Sleuthline\cases\test03\parsed and every cited piece of evidence points to them.

Job	Artifact / tool	Status	Tool and version	Duration
suricata	Suricata - network captures (alerts and protocol metadata)	ok	suricata Suricata 8.0.7	12 s

Analysis database	C:\Sleuthline\cases\test03\db\case.duckdb
Pipeline version	commit 5d46b65c0f4f; content sha256 507779379B984D32; deployed 2026-10-20 18:46
Collection integrity	not checked (run the inventory again with a recent version of the pipeline)
Normalization	normalize.py v1.1 on 2026-10-20T16:50:08Z; mappings v2.2 (sha256 8a32b59741d3439e…)
Normalization warnings	none
Normalization errors	none
Outputs without a validated mapping	none

2.3 Methodology

The analysis follows the playbook below (PLAYBOOK.md), one step at a time. At each step, the agent queries the read-only database, correlates the sources and records a finding only if all its evidence references exist in the database (automatic check). Artifact dates are interpreted according to their evidential value (e.g. ShimCache = presence, Prefetch = execution).

Step	Purpose	Status	Tool calls	Duration
1	System context and scope	done	19	64 s

Step	Purpose	Status	Tool calls	Duration
2	Triage of automatic detections	done	41	743 s
3	Program execution	done	18	92 s
4	Memory (RAM)	done	2	9 s
5	Network traffic (PCAP)	done	28	772 s
6	Persistence	done	14	88 s
7	Accounts, authentication and privileges	done	22	141 s
8	Remote access and lateral movement	done	25	132 s
9	User activity on files and folders	done	15	87 s
10	Web browsing, downloads and e-mail	done	19	1327 s
11	Defense evasion and anti-forensics	done	17	75 s
12	Collection, exfiltration and communications	done	15	73 s
13	Incident timeline and correlation	done	15	66 s
14	Summary, answers and recommendations	done	3	65 s

2.4 Guidance provided by the analyst

Context	none
Period of interest	not specified
Questions	none
Provided IOCs	none
Suspects	none
Exclusions	none
Instructions	none

2.5 Limitations of the analysis

- The case holds only one network capture (case001.pcap, SHA256 09ABF49E...791CB) parsed by Suricata 8.0.7.
- There is no disk image or KAPE collection: no EVTX (Security 4624/4625/4688/4720/7045/1102, TerminalServices, Defender, Sysmon), no registry, Prefetch, Amcache, ShimCache, BAM/UserAssist, SRUM, $MFT/$J, LNK/JumpLists or browser history.
- There is no memory image (no raw.vol_* tables).
- The offline hashset was empty, so hash_hits gives no reputation information.

Consequences:
- Execution of coreupdater.exe, the persistence mechanism, the privileges obtained, and any post-exploitation (credential dumping, account creation, NTDS access, Defender tampering, log clearing) are inferred from network timing or not observable at all.
- Every critical/high finding is therefore rated "probable" (single source).
- Meterpreter and RDP payloads are encrypted, so commands, clipboard contents and files moved through them cannot be seen.

Coverage gaps:
- Nothing is visible before 21:58:07 or after 05:38:57 UTC.
- There is a 2 h 24 min pre-attack gap from 22:10:48 to 00:34:48 UTC.
- Only traffic that crossed the capture point is visible, so purely local host activity and other hosts or segments are out of scope.

Other caveats:
- The machine time zone is unknown (machine_active_time_bias_min = None). The raw EVE timestamps were in +0200 and are reported here in UTC.
- The cause of the 03:34:18 FileShare modification and the reason Defender did not block the stager are undetermined.

- Confidence lowered by the evidence policy (insufficient corroboration or unvalidated mapping): F-003, F-004, F-005, F-010.

- Step 1 (System context and scope): No registry/EVTX/MFT/memory: OS install date, time zone, local accounts/SIDs, profiles, USB/volume history, Panther logs unavailable. OS versions inferred from User-Agents/SMB only. Corroboration limited to a single source class (network).
- Step 2 (Triage of automatic detections): All evidence comes from a single network capture: critical findings cannot reach 'confirmed' (no host logs, Prefetch, Amcache, memory). RDP is TLS-encrypted: authentication result and accounts in sessions are inferred from volume/timing. Execution of coreupdater.exe is inferred from subsequent Meterpreter staging, not observed. C2 content is not decoded.
- Step 3 (Program execution): I could not establish the process name, path, PID, parent, run count, SHA1 from Amcache, or the persistence mechanism, because no disk, EVTX, registry or memory data was provided. Encrypted traffic (TLS, RDP, the Meterpreter channels) hides any commands the attacker ran. hash_hits is empty because no hashset was loaded. Each conclusion rests on the pcap alone, so it cannot be raised to 'confirmed'.
- Step 4 (Memory (RAM)): No memory image was provided, so this step could not show which process owned the Meterpreter C2 sockets to 203.78.103.109:443 on CITADEL-DC01 (10.42.85.10) or DESKTOP-SDN1RPT (10.42.85.115). It also could not reveal any injected or fileless code, a migrated Meterpreter process, a logon-triggered persistence mechanism (suspected in F-009), console command history or credential-access activity (e.g. lsass access). Live captures of both hosts would close these gaps; given the three C2 sessions still established at 05:38 UTC, memory acquisition with WinPmem/DumpIt is recommended before any remediation reboot.
- Step 5 (Network traffic (PCAP)): The case has no disk, EVTX, registry, SRUM, browser or memory artefacts. So no connection can be tied to an owning process (vol_netscan, Sysmon 3, SRUM), and the creation and execution of coreupdater.exe cannot be checked on disk (Amcache SHA1, Zone.Identifier, Prefetch). For that reason high findings stay 'probable' with a single network source. The Meterpreter TLV content and the RDP session content are encrypted, so commands run and data moved through RDP clipboard or drive redirection cannot be seen. What the attacker did after 05:38:57 UTC is outside the capture.
- Step 6 (Persistence): I could not identify the persistence mechanism itself (key, value, path, task or service name), when it was created, or whether it survives a reboot. That needs the SOFTWARE/NTUSER hives, C:\Windows\System32\Tasks, the Security, System and TaskScheduler EVTX, the Startup folders and the $MFT, and none of these were collected. Because the evidence comes from a single network source, F-012 stays probable.
- Step 7 (Accounts, authentication and privileges): No host logs were available: 4624/4625/4634/4648/4672/4720-4738/4740/4768-4776, RDP 21/22/24/25/1149/1024, the SAM, local accounts and the DC's own interactive logons. Kerberos for those logons stays on the DC, so it never appears on the wire. That means no conclusion is possible on account creation or group changes made locally on the DC during the attacker's session, or on privilege escalation (4672). The account behind the successful NLA at 02:21:47 is inferred from the RDP cookie, not from an authenticated identity. The RC4 interpretation could not be checked against pwdLastSet or msDS-SupportedEncryptionTypes.
- Step 8 (Remote access and lateral movement): Several sources asked for in this step do not exist in the case, which holds only the pcap:
  - No host artefacts: TerminalServices LSM/RCM EVTX, Security 4624 type 10 / 4648 / 5140 / 5145 / 7045, WinRM/PowerShell logs, mstsc MRU / Terminal Server Client registry, Default.rdp, bitmap cache, raw.dissect_remoteaccess_* and raw.dissect_mru_networkdrive, ShellBags, $MFT (for RMM folders).
  - No memory image (vol_netscan).

  So session IDs, RDP accounts and source names on the hosts cannot be confirmed, and Restricted Admin use cannot be checked.

  Some activity is invisible on the wire:
  - anything carried inside the encrypted/raw Meterpreter channel (pivoting, port forwarding, token or credential use);
  - the content of the RDP sessions;
  - any activity outside 21:58–05:38 UTC.

  The DC's own CredSSP Kerberos request for TERMSRV/DESKTOP-SDN1RPT is not on the wire because the KDC is local to the DC.

- Step 9 (User activity on files and folders): The case has no disk, registry or EVTX artefacts, so I could not check files opened, created, renamed or deleted on either host, Recycle Bin contents, removable media, or the coreupdater.exe on-disk path and Zone.Identifier. Suricata does not log SMB2 QUERY_DIRECTORY results, so the contents of FileShare and the entry changed at 03:34:18 cannot be identified. Anything done locally on the hosts or inside the Meterpreter channel is not visible.
- Step 10 (Web browsing, downloads and e-mail): Without Edge/IE history and download databases, MFT/$J, Zone.Identifier ADS or Prefetch, I can't establish where coreupdater.exe was saved, its MOTW HostUrl/ReferrerUrl, or a host-side execution time. HTTPS content (search terms, pages) can't be seen in the pcap. The content of the landing page wasn't extracted (Suricata gave only hashes and size). Every conclusion rests on one source, so it stays probable.
- Step 11 (Defense evasion and anti-forensics): Log clearing (1102/104), RecordID gaps, Defender events and exclusions, registry-based disabling, timestomping (SI<FN), USN deletions, Prefetch deletion, PowerShell logging changes, time changes (4616) and memory injection or hidden processes (malfind/psxview) could not be assessed. No EVTX, registry, MFT/$J or memory image is in the case. Anti-forensics done locally on the DC or workstation over the established Meterpreter sessions would not show up on the wire.
- Step 12 (Collection, exfiltration and communications): There are no host artefacts: no MFT/UsnJrnl (archives, staging), 7-Zip history, SRUM network usage, BITS events, OneDrive ODL/SyncEngineDatabase, Sysmon 3/22, 5156 or firewall logs. The content of the Meterpreter and RDP channels is opaque (raw/encrypted). Three C2 sessions were still alive at the end of the capture (05:38 UTC), so later exfiltration cannot be assessed.
- Step 13 (Incident timeline and correlation): There are no host artefacts (EVTX, MFT, registry, Prefetch, memory). Because of that I could not measure clock skew between hosts or check for timestomping, and the execution and persistence mechanisms are inferred from network behaviour, not observed. Anything after 05:38:57 UTC, including the outcome of the three live C2 sessions, is unknown. The RDP and Meterpreter content is encrypted or opaque, so the attacker's actions inside those sessions are not visible.
- Step 14 (Summary, answers and recommendations): The case is network-only (one pcap). There are no EVTX, registry, filesystem or memory artefacts, so execution, persistence mechanism, privileges and post-exploitation remain inferred, and all critical/high findings stay 'probable'. RDP and C2 content is encrypted. Nothing is visible outside 21:58:07–05:38:57 UTC or during the 22:10:48–00:34:48 gap. The machine time zone is unknown.

3. Incident timeline

Key events established by the analysis (UTC), with their evidence. The complete timeline of the incident window (all artifacts) is provided in incident_timeline.csv.

Timestamp (UTC)	Phase	Event	Description	MITRE	References
2020-09-18 21:59:58	context	mortysmith interactive logon on DESKTOP-SDN1RPT (pre-attack baseline)	Kerberos AS-REP (aes256) + TGS host/desktop-sdn1rpt.c137.local for mortysmith; legitimate user activity before the attack.		suricata/case001.pcap/eve_krb5.jsonl#38 suricata/case001.pcap/eve_krb5.jsonl#36
2020-09-19 02:19:26	initial access	Nmap RDP scan then RDP brute-force on DC from 194.61.24.102	Nmap RDP probe (cookie 'nmap') then ~29,000 short RDP connections with cookie 'Administrator' to 10.42.85.10:3389 (02:19-02:21 UTC).	T1046 T1110.001	suricata/case001.pcap/eve_alert.jsonl#25 suricata/case001.pcap/eve_alert.jsonl#21 suricata/case001.pcap/eve_alert.jsonl#27
2020-09-19 02:21:26	credential access	RDP/NLA password guessing against Administrator on DC (95 failures in 21 s)	95 byte-identical CredSSP failure flows (2613/2257 B) from 194.61.24.102 to 10.42.85.10:3389 with mstshash=Administrator, 02:21:26.11-02:21:46.95 UTC.	T1110.001	suricata/case001.pcap/eve_flow.jsonl#15690 suricata/case001.pcap/eve_rdp.jsonl#3
2020-09-19 02:21:47	initial access	First successful NLA authentication as Administrator from 194.61.24.102 (password found)	Flow 194.61.24.102:40234 → 10.42.85.10:3389 (6373/5222 B, shutdown after 0.8 s) differs from all failures: successful CredSSP validation; followed by short session 02:22:08 and interactive session 02:22:36.	T1110.001 T1078.002 T1133	suricata/case001.pcap/eve_flow.jsonl#32271 suricata/case001.pcap/eve_flow.jsonl#19321
2020-09-19 02:22:36	initial access	Attacker interactive RDP session on DC begins (until 02:52:47)	194.61.24.102:40238 -> 10.42.85.10:3389, 30 min, ~17.5 MB: successful Administrator RDP logon inferred.	T1021.001 T1078.002	suricata/case001.pcap/eve_flow.jsonl#30035 suricata/case001.pcap/eve_alert.jsonl#402
2020-09-19 02:23:09	delivery	Attacker launches IE11 on DC and types http://194.61.24.102/ (02:23:41)	First IE11 start on CITADEL-DC01 in attacker RDP session (bing favicon, ieonline.microsoft.com), then direct navigation to the attacker landing page (228 B, SHA256 6978f8fa…2d77), followed by click on /coreupdater.exe at 02:24:06.	T1105	suricata/case001.pcap/eve_http.jsonl#125 suricata/case001.pcap/eve_http.jsonl#117 suricata/case001.pcap/eve_fileinfo.jsonl#98
2020-09-19 02:24:06	execution	DC downloads coreupdater.exe from 194.61.24.102	IE11 GET http://194.61.24.102/coreupdater.exe, 7168 B, SHA256 10f3b920…1cfda6.	T1105	suricata/case001.pcap/eve_http.jsonl#116 suricata/case001.pcap/eve_fileinfo.jsonl#100
2020-09-19 02:25:18	command and control	DC receives Meterpreter stage from 203.78.103.109:443 (first C2 session)	Raw TCP 10.42.85.10:62414 -> 203.78.103.109:443, PE + Metasploit Bind_API construct; implies execution of coreupdater.exe.	T1071 T1571	suricata/case001.pcap/eve_alert.jsonl#438 suricata/case001.pcap/eve_alert.jsonl#439 suricata/case001.pcap/eve_flow.jsonl#28844
2020-09-19 02:29:36	execution	First DC Meterpreter session closed; replaced 13 s later by second staging	Flow 10.42.85.10:62414 -> 203.78.103.109:443 (02:25:18.566 -> 02:29:36.435, 51 KB up / 933 KB down, state closed) ends; new session 10.42.85.10:62476 starts at 02:29:49.290 (584 KB stage, closed 02:52:46 with RDP logoff).	T1059 T1620	suricata/case001.pcap/eve_flow.jsonl#28844 suricata/case001.pcap/eve_flow.jsonl#26788
2020-09-19 02:29:49	command and control	Second Meterpreter staging on DC	10.42.85.10:62476 -> 203.78.103.109:443, new stage (until 02:52:46).		suricata/case001.pcap/eve_alert.jsonl#403 suricata/case001.pcap/eve_flow.jsonl#26788
2020-09-19 02:35:55	lateral movement	RDP from DC to DESKTOP-SDN1RPT (lateral movement)	10.42.85.10:62514/63121 -> 10.42.85.115:3389 until 02:52:14.	T1021.001	suricata/case001.pcap/eve_rdp.jsonl#297 suricata/case001.pcap/eve_flow.jsonl#29416
2020-09-19 02:36:24	lateral movement	Administrator first interactive logon on	Kerberos AS for Administrator from workstation, NETLOGON 'Default User.V6' lookup and	T1021.001	suricata/case001.pcap/

Timestamp (UTC)	Phase	Event	Description	MITRE	References
		DESKTOP-SDN1RPT (new profile)	DPAPI protected_storage pipe: new profile creation during RDP from DC.		eve_krb5.jsonl#60 suricata/case001.pcap/eve_smb.jsonl#151
2020-09-19 02:39:46	defense-evasion	Workstation Windows Defender MAPS (wdcp.microsoft.com) cloud lookup during coreupdater.exe download	Only Defender cloud-protection session in the capture (10.42.85.115 → 104.40.17.153:443). Defender was active, yet the stager ran and C2 began at 02:40:49.		suricata/case001.pcap/eve_tls.jsonl#16337 suricata/case001.pcap/eve_flow.jsonl#28228
2020-09-19 02:39:58	execution	Workstation downloads coreupdater.exe from 194.61.24.102	Edge GET /coreupdater.exe by 10.42.85.115, same SHA256.	T1105	suricata/case001.pcap/eve_http.jsonl#156 suricata/case001.pcap/eve_fileinfo.jsonl#136
2020-09-19 02:40:45	execution	SmartScreen checkappexec on workstation – user launch of downloaded coreupdater.exe	After Edge download (02:39:58) and download reputation check (02:40:01), DESKTOP-SDN1RPT performs checkappexec.microsoft.com lookups at 02:40:45.79 and 02:40:49.938, the latter 2 ms after Meterpreter connection to 203.78.103.109:443 – execution of the MOTW-tagged file from the shell.	T1204.002	suricata/case001.pcap/eve_tls.jsonl#16363 suricata/case001.pcap/eve_tls.jsonl#16331 suricata/case001.pcap/eve_tls.jsonl#16370
2020-09-19 02:40:49	command and control	Workstation Meterpreter C2 session to 203.78.103.109:443 (alive at end of capture)	10.42.85.115:50875 -> 203.78.103.109:443, Metasploit stage, established until 05:38:44.		suricata/case001.pcap/eve_alert.jsonl#446 suricata/case001.pcap/eve_flow.jsonl#27604
2020-09-19 02:52:46	persistence	DC Meterpreter session #2 terminates together with attacker RDP logoff	C2 flow 10.42.85.10:62476->203.78.103.109:443 closes at 02:52:46.39, 1 s before RDP session 194.61.24.102:40238->DC:3389 ends (02:52:47.38). The payload ran in the user session. A new C2 session follows 35 s after the next RDP logon (02:56:03 -> 02:56:38), which fits logon-triggered persistence or a manual relaunch.	T1547.001	suricata/case001.pcap/eve_flow.jsonl#26788 suricata/case001.pcap/eve_flow.jsonl#30035 suricata/case001.pcap/eve_flow.jsonl#18312
2020-09-19 02:56:03	command and control	Second attacker RDP session on DC and new Meterpreter staging (02:56:38)	194.61.24.102:40240 -> DC:3389 (02:56:03-02:57:41) followed by DC C2 session 10.42.85.10:62613 -> 203.78.103.109:443, alive until end of capture.		suricata/case001.pcap/eve_flow.jsonl#22204 suricata/case001.pcap/eve_alert.jsonl#452 suricata/case001.pcap/eve_flow.jsonl#18312
2020-09-19 02:57:41	command and control	Last attacker RDP session on DC ends (last contact from 194.61.24.102)	Flow 194.61.24.102 -> 10.42.85.10:3389 (02:56:03.538 -> 02:57:41.585 UTC, 266 KB up / 1.0 MB down) is the last flow involving 194.61.24.102 in the capture. From here on the attacker's access relies only on the Meterpreter C2 to 203.78.103.109:443 (DC session started 02:56:38 inside this RDP session).	T1021.001 T1078.002	suricata/case001.pcap/eve_flow.jsonl#22204 suricata/case001.pcap/eve_flow.jsonl#18312
2020-09-19 03:34:18	collection	FileShare root directory on CITADEL-DC01 modified (entry created/deleted/renamed)	The SMB2 CREATE response for \\CITADEL-DC01\FileShare\<share_root> (seen later, at 04:09:53) reports modified/changed = 1600486458 (03:34:18 UTC). No SMB client activity at that time, so the change was made locally on the DC while the attacker's Meterpreter session (since 02:56:38) was active. The cause cannot be determined from the pcap.	T1039	suricata/case001.pcap/eve_smb.jsonl#193
2020-09-19 04:08:15	persistence	ricksanchez local logon on workstation, followed 30 s later by new Meterpreter session	Kerberos AS-REP ricksanchez, FileShare mapped at 04:08:16; outbound 10.42.85.115:50972→203.78.103.109:443 at 04:08:45 – logon-triggered persistence suspected.	T1547.001	suricata/case001.pcap/eve_krb5.jsonl#78 suricata/case001.pcap/eve_flow.jsonl#19978
2020-09-19 04:08:16	context / lateral movement check	Workstation connects to \\CITADEL-DC01\FileShare after ricksanchez logon (Explorer browsing until ~05:09)	SMB2 TREE_CONNECT from 10.42.85.115 to \\CITADEL-DC01\FileShare at 04:08:16 UTC. Explorer-style Desktop.ini/AutoRun.inf/<share_root> opens follow at 04:09:52, 04:30:24 and 04:31:14. No file transfer of executables. Most likely the user's legitimate drive mapping. A Meterpreter session was live on the host at the same time.	T1021.002	suricata/case001.pcap/eve_smb.jsonl#188 suricata/case001.pcap/eve_smb.jsonl#191 suricata/case001.pcap/eve_smb.jsonl#205
2020-09-19 04:08:45	persistence / C2	Workstation re-stages Meterpreter (new C2 session)	10.42.85.115:50972 -> 203.78.103.109:443, possibly persistence-triggered, alive until end of capture.		suricata/case001.pcap/eve_alert.jsonl#458 suricata/case001.pcap/eve_flow.jsonl#19978
2020-09-19 05:38:51	command and control	End of capture – three Meterpreter C2 sessions still ESTABLISHED (DC +	Last observed packets of the live C2 sessions: workstation 10.42.85.115:50972 (started 04:08:45, last 05:38:42, 39 KB/489 KB), workstation 10.42.85.115:50875 (started 02:40:49,	T1071 T1571	suricata/case001.pcap/eve_flow.jsonl#18312

Timestamp (UTC)	Phase	Event	Description	MITRE	References
		workstation x2)	last 05:38:44, 280 KB/1.76 MB), DC 10.42.85.10:62613 (started 02:56:38, last 05:38:51, 89 KB/961 KB). All in flow_state=established, flow_reason=shutdown (Suricata end of pcap). Last event of the incident visible on the wire; compromise was ongoing when the capture stopped (05:38:57).		suricata/case001.pcap/eve_flow.jsonl#27604 suricata/case001.pcap/eve_flow.jsonl#19978

Detailed evidence timeline

All the evidence cited by the findings and the timeline, sorted by time.

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
	raw.pcap_files	{"file": "C:\\Sleuthline\\cases\\test03\\pcap\\case001.pcap", "name": "case001.pcap", "bytes": "197583252", "sha256": "09ABF49EFEA1852E047987D92907704D47F36D75F6C8056E2CAFA6CC027791CB"}	suricata/pcap_files.jsonl#1		F-001, F-002
2020-09-18 21:58:07.470323	TLS (Suricata)	TLS handshake 10.42.85.10 dest_ip=10.42.85.10 | dest_port=3389 | tls_version=TLS 1.3 | tls_ja3_hash=75fe51990656df4f7a249d5b86aa29ae | tls_ja4=t13i781000_ab95583b6d39_d41ae481755e	suricata/case001.pcap/ eve_tls.jsonl#932	case001.pcap	F-011, I-009
2020-09-18 21:59:39.458927	DNS (Suricata)	DNS query / answer [{"rrname": "_ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.C137.local", "rrtype": "SRV"}] dns_type=response | dns_rcode=NOERROR | src_ip=10.42.85.10 | dest_ip=10.42.85.115 | dns_answers=[{"rrname": "_ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.C137.local", "rrtype": "SRV", "ttl": 600, "srv": {"priority": 0, "weight": 100, "port": 389, "name": "citadel-dc01.c137.local"}}]	suricata/case001.pcap/ eve_dns.jsonl#9	case001.pcap	F-001
2020-09-18 21:59:39.479837	DNS (Suricata)	DNS query / answer [{"rrname": "citadel-dc01.c137.local", "rrtype": "A"}] dns_type=response | dns_rcode=NOERROR | src_ip=10.42.85.10 | dest_ip=10.42.85.115 | dns_answers=[{"rrname": "citadel-dc01.c137.local", "rrtype": "A", "ttl": 3600, "rdata": "10.42.85.10"}] | dns_grouped_A=["10.42.85.10"]	suricata/case001.pcap/ eve_dns.jsonl#6	case001.pcap	F-001
2020-09-18 21:59:39.797903	Network events (Suricata)	network event krb5 TCP 10.42.85.115:49673 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#18	case001.pcap	F-001, F-012
2020-09-18 21:59:39.801104	Network events (Suricata)	network event krb5 TCP 10.42.85.115:49677 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#20	case001.pcap	F-014
2020-09-18 21:59:40.016569	Network events (Suricata)	network event smb TCP 10.42.85.115:49683 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#2	case001.pcap	F-001
2020-09-18 21:59:40.017751	Network events (Suricata)	network event smb TCP 10.42.85.115:49683 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#3	case001.pcap	F-014
2020-09-18 21:59:40.687057	Files transferred (Suricata)	file transferred over the network C137.local\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\Machine\Registry.pol app_proto=smb | src_ip=10.42.85.10 | dest_ip=10.42.85.115 | fileinfo_size=2784 | fileinfo_sha256=7a64e146fe72b9cdf61a216bc454425b1837b3c5875fbaf2d2f3e8923fdafc9a | fileinfo_md5=d9b402ccf91f0e9c6558d9497624decd | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#2	case001.pcap	F-012
2020-09-18 21:59:58.142475	Network events (Suricata)	network event krb5 TCP 10.42.85.10:88 -> 10.42.85.115:49709	suricata/case001.pcap/ eve_krb5.jsonl#27	case001.pcap	F-014
2020-09-18 21:59:58.148531	Network events (Suricata)	network event krb5 TCP 10.42.85.115:49710 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#38	case001.pcap	F-001, F-014, F-015, T-016
2020-09-18 21:59:58.149910	Network events (Suricata)	network event krb5 TCP 10.42.85.115:49711 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#36	case001.pcap	F-014, T-016
2020-09-18 22:01:00.882490	HTTP (Suricata)	HTTP request /e7a5/i/2016/204/2/e/jessica_from_rick_and_morty_by_fluffymystic-dab4xm5.png http_hostname=img04.deviantart.net | http_http_method=GET | http_status=301 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=text/html | http_length=0 | src_ip=10.42.85.115 | dest_ip=52.26.146.4	suricata/case001.pcap/ eve_http.jsonl#14	case001.pcap	F-001

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
2020-09-18 22:10:08.218493	Network events (Suricata)	network event smb TCP 10.42.85.115:50742 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#55	case001.pcap	F-017
2020-09-19 00:44:00.322000	Network flows (Suricata)	network flow: start 52.114.132.91 TCP 10.42.85.115:50639 -> 52.114.132.91:443 tls state=closed bytes=180437/12740	suricata/case001.pcap/ eve_flow.jsonl#18404	case001.pcap	F-021
2020-09-19 00:44:00.438933	TLS (Suricata)	TLS handshake v10.events.data.microsoft.com tls_sni=v10.events.data.microsoft.com | dest_ip=52.114.132.91 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=C=US, ST=WA, L=Redmond, O=Microsoft, OU=WSE, CN=*.events.data.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, CN=Microsoft Secure Server CA 2011 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=678aeaf909676262acfb913ccb78a126 | tls_	suricata/case001.pcap/ eve_tls.jsonl#901	case001.pcap	F-021
2020-09-19 02:19:26.619607	Network events (Suricata)	network event rdp TCP 10.42.85.10:3389 -> 194.61.24.102:38100	suricata/case001.pcap/ eve_rdp.jsonl#1	case001.pcap	F-013
2020-09-19 02:19:26.792588	TLS (Suricata)	TLS handshake 10.42.85.10 dest_ip=10.42.85.10 | dest_port=3389 | tls_version=TLS 1.2 | tls_subject=CN=CITADEL-DC01.C137.local | tls_issuerdn=CN=CITADEL-DC01.C137.local | tls_ja3_hash=6dc99de941a8f76cad308d9089e793d7 | tls_ja3s_hash=5397c414a9ebeaff1bf18b70ca22eaa0 | tls_ja4=t12i0502h1_7f27215065b1_e5744cedd59f	suricata/case001.pcap/ eve_tls.jsonl#931	case001.pcap	F-011
2020-09-19 02:19:26.792622	Detection Suricata	low ET SCAN RDP Connection Attempt from Nmap | TCP 194.61.24.102:38100 -> 10.42.85.10:3389 | Detection of a Network Scan | 1:BJ5fdE2KUXlcHa7o2AcT6LKStwM= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#25	case001.pcap	F-003, F-022, I-001, I-012, T-001
2020-09-19 02:19:26.792649	TLS (Suricata)	TLS handshake 10.42.85.10 dest_ip=10.42.85.10 | dest_port=3389 | tls_version=TLS 1.2 | tls_subject=CN=CITADEL-DC01.C137.local | tls_issuerdn=CN=CITADEL-DC01.C137.local | tls_ja3_hash=3bdfb64d53404bacd8a47056c6a756be | tls_ja3s_hash=5397c414a9ebeaff1bf18b70ca22eaa0 | tls_ja4=t12i050200_7f27215065b1_4c3f1086ad7a	suricata/case001.pcap/ eve_tls.jsonl#920	case001.pcap	F-011, I-012
2020-09-19 02:19:26.792998	TLS (Suricata)	TLS handshake 10.42.85.10 dest_ip=10.42.85.10 | dest_port=3389 | tls_version=TLS 1.2 | tls_subject=CN=CITADEL-DC01.C137.local | tls_issuerdn=CN=CITADEL-DC01.C137.local | tls_ja3_hash=e26ff759048e07b164d8faf6c2a19f53 | tls_ja3s_hash=5397c414a9ebeaff1bf18b70ca22eaa0 | tls_ja4=t12i050000_7f27215065b1_d2e2adf7177b	suricata/case001.pcap/ eve_tls.jsonl#926	case001.pcap	F-011
2020-09-19 02:19:26.798289	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:38106 -> 10.42.85.10:3389 tls state=closed bytes=723/134	suricata/case001.pcap/ eve_flow.jsonl#455	case001.pcap	F-013
2020-09-19 02:19:26.918948	Detection Suricata	low ET SCAN Behavioral Unusually fast Terminal Server Traffic Potential Scan or Infection (Inbound) | TCP 194.61.24.102:38288 -> 10.42.85.10:3389 | Detection of a Network Scan | 1:VTsSoaES/lxoZ4cCIOR4ghG5WYE= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#21	case001.pcap	F-003, T-001
2020-09-19 02:21:26.112470	Detection Suricata	low ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request | TCP 194.61.24.102:40044 -> 10.42.85.10:3389 | Generic Protocol Command Decode | 1:sLbyebDh6LCv2TWmLwr5aZU4kbI= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#27	case001.pcap	F-003, T-001
2020-09-19 02:21:26.114786	Network events (Suricata)	network event rdp TCP 10.42.85.10:3389 -> 194.61.24.102:40044	suricata/case001.pcap/ eve_rdp.jsonl#3	case001.pcap	F-001, F-013, F-016, T-014
2020-09-19 02:21:26.114998	Network events (Suricata)	network event rdp TCP 194.61.24.102:40044 -> 10.42.85.10:3389	suricata/case001.pcap/ eve_rdp.jsonl#4	case001.pcap	F-016
2020-09-19 02:21:33.618634	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40112 -> 10.42.85.10:3389 rdp state=closed bytes=2613/2257	suricata/case001.pcap/ eve_flow.jsonl#15690	case001.pcap	F-013, T-014

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
2020-09-19 02:21:47.166264	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40234 -> 10.42.85.10:3389 rdp state=closed bytes=6373/5222	suricata/case001.pcap/ eve_flow.jsonl#32271	case001.pcap	F-013, T-015
2020-09-19 02:22:08.311754	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40236 -> 10.42.85.10:3389 rdp state=closed bytes=28260/62729	suricata/case001.pcap/ eve_flow.jsonl#19321	case001.pcap	F-013, T-015
2020-09-19 02:22:36.664454	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40238 -> 10.42.85.10:3389 rdp state=closed bytes=4802218/12756305	suricata/case001.pcap/ eve_flow.jsonl#30035	case001.pcap	F-003, F-012, F-013, F-021, F-022, I-001, T-002, T-013
2020-09-19 02:22:36.664968	Detection Suricata	low ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request | TCP 194.61.24.102:40238 -> 10.42.85.10:3389 | Generic Protocol Command Decode | 1:iingav/QQcKdY/xllC/FjNyxsRQ= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#402	case001.pcap	F-003, T-002
2020-09-19 02:23:09.692112	HTTP (Suricata)	HTTP request /favicon.ico http_hostname=www.bing.com | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 6.3; Win64; x64; Trident/7.0; rv:11.0) like Gecko | http_http_content_type=image/x-icon | http_length=237 | src_ip=10.42.85.10 | dest_ip=204.79.197.200	suricata/case001.pcap/ eve_http.jsonl#125	case001.pcap	F-001, F-018, T-019
2020-09-19 02:23:10.334278	TLS (Suricata)	TLS handshake ieonline.microsoft.com tls_sni=ieonline.microsoft.com | dest_ip=204.79.197.200 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=www.bing.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 2 | tls_ja3_hash=8de139912912d62c9e27d022ab05e070 | tls_ja3s_hash=2b90c93096725457ad0cf004d948ed32 | tls_ja4=t12d2409s3_9295eb73ea51_6ed2096ec1d1	suricata/case001.pcap/ eve_tls.jsonl#16174	case001.pcap	F-018
2020-09-19 02:23:41.734776	HTTP (Suricata)	HTTP request / http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; rv:11.0) like Gecko | http_http_content_type=text/html | http_length=228 | src_ip=10.42.85.10 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#117	case001.pcap	F-018, T-019
2020-09-19 02:23:41.735412	Files transferred (Suricata)	file transferred over the network / app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.10 | fileinfo_size=228 | fileinfo_sha256=6978f8fa2bf5afbe1a074401b923ead519ce89471d83376f5cfda50c12bc2d77 | fileinfo_md5=614645af648513231c283090faa5c469 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#98	case001.pcap	F-018, I-010, T-019
2020-09-19 02:24:06.940063	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; rv:11.0) like Gecko | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.10 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#116	case001.pcap	F-001, F-004, F-007, F-018, I-001, I-003, I-011, T-003
2020-09-19 02:24:06.940653	Detection Suricata	high ET INFO PE EXE or DLL Windows file download HTTP | TCP 194.61.24.102:80 -> 10.42.85.10:62410 | Potential Corporate Privacy Violation | 1:u/j8zDTczy431sRgtCbNVlYztwk= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#370	case001.pcap	F-004
2020-09-19 02:24:06.940653	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.10 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#100	case001.pcap	F-004, F-007, F-019, F-022, I-004, I-005, I-006, I-007, T-003
2020-09-19	Network flows (Suricata)	network flow: start	suricata/case001.pcap/	case001.pcap	F-005, F-007,

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
02:25:18.565676		203.78.103.109 TCP 10.42.85.10:62414 -> 203.78.103.109:443 failed state=closed bytes=51044/932653	eve_flow.jsonl#28844		F-010, F-019, F-021, F-022, T-004, T-024
2020-09-19 02:25:18.753116	Detection Suricata	low ET HUNTING PE EXE Download over raw TCP | TCP 203.78.103.109:443 -> 10.42.85.10:62414 | Misc activity | 1:AEtaQJ4Pfkk0lfr1mrIy2qsaW8Q= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#438	case001.pcap	F-005, T-004
2020-09-19 02:25:19.071794	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.10:62414 | A Network Trojan was detected | 1:AEtaQJ4Pfkk0lfr1mrIy2qsaW8Q= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#439	case001.pcap	F-005, F-019, I-002, T-004
2020-09-19 02:29:49.290306	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62476 -> 203.78.103.109:443 failed state=closed bytes=41589/583890	suricata/case001.pcap/ eve_flow.jsonl#26788	case001.pcap	F-007, F-010, F-012, F-019, F-020, F-021, T-005, T-013, T-024
2020-09-19 02:29:49.580408	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.10:62476 | A Network Trojan was detected | 1:SF4txCtiLpUx9hIdO5hqR1RPreU= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#403	case001.pcap	F-005, T-005
2020-09-19 02:35:55.285340	Network flows (Suricata)	network flow: start 10.42.85.115 TCP 10.42.85.10:62514 -> 10.42.85.115:3389 rdp state=closed bytes=875256/452944	suricata/case001.pcap/ eve_flow.jsonl#29416	case001.pcap	F-006, F-008, F-016, T-006
2020-09-19 02:35:55.347240	Network events (Suricata)	network event rdp TCP 10.42.85.115:3389 -> 10.42.85.10:62514	suricata/case001.pcap/ eve_rdp.jsonl#297	case001.pcap	F-006, F-016, F-022, T-006
2020-09-19 02:35:55.417848	Network events (Suricata)	network event rdp TCP 10.42.85.10:62514 -> 10.42.85.115:3389	suricata/case001.pcap/ eve_rdp.jsonl#298	case001.pcap	F-016
2020-09-19 02:36:23.467377	Network events (Suricata)	network event rdp TCP 10.42.85.10:62514 -> 10.42.85.115:3389	suricata/case001.pcap/ eve_rdp.jsonl#299	case001.pcap	F-006, F-016
2020-09-19 02:36:23.515604	Network flows (Suricata)	network flow: start 10.42.85.115 UDP 10.42.85.10:63121 -> 10.42.85.115:3389 failed state=established bytes=283842/6695207	suricata/case001.pcap/ eve_flow.jsonl#31428	case001.pcap	F-006, F-016
2020-09-19 02:36:24.912466	Network events (Suricata)	network event krb5 TCP 10.42.85.10:88 -> 10.42.85.115:50694	suricata/case001.pcap/ eve_krb5.jsonl#57	case001.pcap	F-014
2020-09-19 02:36:24.912570	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50694 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#58	case001.pcap	F-015
2020-09-19 02:36:24.913620	Network events (Suricata)	network event krb5 TCP 10.42.85.10:88 -> 10.42.85.115:50695	suricata/case001.pcap/ eve_krb5.jsonl#59	case001.pcap	F-015
2020-09-19 02:36:24.913767	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50695 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#60	case001.pcap	F-001, F-008, F-014, F-015, F-022, I-008, T-011
2020-09-19 02:36:24.915473	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50696 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#52	case001.pcap	F-008, F-016
2020-09-19 02:36:24.989968	Network events (Suricata)	network event smb TCP 10.42.85.115:50701 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#144	case001.pcap	F-014
2020-09-19 02:36:24.991912	Network events (Suricata)	network event smb TCP 10.42.85.115:50701 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#149	case001.pcap	F-008
2020-09-19 02:36:25.038929	Network events (Suricata)	network event smb TCP 10.42.85.115:50701 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#151	case001.pcap	F-008, T-011
2020-09-19 02:36:26.766248	Network events (Suricata)	network event smb TCP 10.42.85.115:50710 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#131	case001.pcap	F-008
2020-09-19 02:36:26.768288	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50712 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#54	case001.pcap	F-001, F-008, F-014
2020-09-19 02:37:00.954915	DNS (Suricata)	DNS query / answer [{"rrname": "oneclient.sfx.ms", "rrtype": "A"}] dns_type=request | dns_rcode=NOERROR | src_ip=10.42.85.115 | dest_ip=10.42.85.10	suricata/case001.pcap/ eve_dns.jsonl#1868	case001.pcap	F-021
2020-09-19	Network flows (Suricata)	network flow: start	suricata/case001.pcap/	case001.pcap	F-021

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
02:37:01.033644		104.119.185.124 TCP 10.42.85.115:50731 -> 104.119.185.124:443 tls state=closed bytes=446236/39409046	eve_flow.jsonl#28319		
2020-09-19 02:37:01.185518	TLS (Suricata)	TLS handshake oneclient.sfx.ms tls_sni=oneclient.sfx.ms | dest_ip=104.119.185.124 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=C=US, ST=WA, L=Redmond, O=Microsoft Corporation, OU=Microsoft Corporation, CN=*.sfx.ms | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 5 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=19e4a55cecd087d9ebf88da03db13a0f	suricata/case001.pcap/ eve_tls.jsonl#16200	case001.pcap	F-021
2020-09-19 02:39:04.440228	DNS (Suricata)	DNS query / answer [{"rrname": "microsoftedgewelcome.microsoft.com", "rrtype": "A"}] dns_type=request | dns_rcode=NOERROR | src_ip=10.42.85.115 | dest_ip=10.42.85.10	suricata/case001.pcap/ eve_dns.jsonl#1885	case001.pcap	F-018
2020-09-19 02:39:26.940438	HTTP (Suricata)	HTTP request / http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=text/html | http_length=228 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#154	case001.pcap	F-018
2020-09-19 02:39:26.941704	Files transferred (Suricata)	file transferred over the network / app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.115 | fileinfo_size=228 | fileinfo_sha256=6978f8fa2bf5afbe1a074401b923ead519ce89471d83376f5cfda50c12bc2d77 | fileinfo_md5=614645af648513231c283090faa5c469 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#135	case001.pcap	I-010
2020-09-19 02:39:27.067445	TLS (Suricata)	TLS handshake nav.smartscreen.microsoft.com tls_sni=nav.smartscreen.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=smartscreen.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 1 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=986571066668055ae9481cb84fda634a | tls_ja4=t12d1909h2_d83cc789557e_7af1ed941c26	suricata/case001.pcap/ eve_tls.jsonl#16283	case001.pcap	F-018, F-020
2020-09-19 02:39:46.424507	Network flows (Suricata)	network flow: start 104.40.17.153 TCP 10.42.85.115:50863 -> 104.40.17.153:443 tls state=closed bytes=3023/5345	suricata/case001.pcap/ eve_flow.jsonl#28228	case001.pcap	F-020, T-021
2020-09-19 02:39:46.509658	TLS (Suricata)	TLS handshake wdcp.microsoft.com tls_sni=wdcp.microsoft.com | dest_ip=104.40.17.153 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft Corporation, CN=wdcp.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, CN=Microsoft Secure Server CA 2011 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=17e97216fa7f4ec8c4309	suricata/case001.pcap/ eve_tls.jsonl#16337	case001.pcap	F-018, F-020, T-021
2020-09-19 02:39:58.411571	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#156	case001.pcap	F-004, F-007, F-018, I-003, I-011, T-007
2020-09-19 02:39:58.412735	Detection Suricata	high ET INFO PE EXE or DLL Windows file download HTTP | TCP 194.61.24.102:80 -> 10.42.85.115:50864 | Potential Corporate Privacy Violation | 1:jol6Y9tRe9v45mCCcFMHMI7rx+A= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#443	case001.pcap	F-004
2020-09-19 02:39:58.412735	Files transferred (Suricata)	file transferred over the network /coreupdater.exe	suricata/case001.pcap/ eve_fileinfo.jsonl#136	case001.pcap	F-004, F-018, I-004, I-007,

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.115 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED			T-007
2020-09-19 02:40:01.538456	TLS (Suricata)	TLS handshake nav.smartscreen.microsoft.com tls_sni=nav.smartscreen.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=ae4edc6faf64d08308082ad26be60767 | tls_ja4=t12d1909h2_d83cc789557e_7af1ed941c26	suricata/case001.pcap/ eve_tls.jsonl#16363	case001.pcap	F-018, F-020, T-020
2020-09-19 02:40:45.794820	TLS (Suricata)	TLS handshake checkappexec.microsoft.com tls_sni=checkappexec.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=smartscreen.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 1 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=986571066668055ae9481cb84fda634a | tls_ja4=t12d1909h2_d83cc789557e_7af1ed941c26	suricata/case001.pcap/ eve_tls.jsonl#16331	case001.pcap	F-018, T-020
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/ eve_flow.jsonl#27604	case001.pcap	F-005, F-007, F-009, F-010, F-012, F-021, F-022, I-002, T-008, T-023
2020-09-19 02:40:49.938222	TLS (Suricata)	TLS handshake checkappexec.microsoft.com tls_sni=checkappexec.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=smartscreen.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 1 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=986571066668055ae9481cb84fda634a | tls_ja4=t12d1909h2_d83cc789557e_7af1ed941c26	suricata/case001.pcap/ eve_tls.jsonl#16370	case001.pcap	F-018, T-020
2020-09-19 02:40:50.282950	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.115:50875 | A Network Trojan was detected | 1:pcNgXsCjc/vxVdVvPl9FwJBoPrA= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#446	case001.pcap	F-005, T-008
2020-09-19 02:56:03.538780	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40240 -> 10.42.85.10:3389 rdp state=closed bytes=266261/1008697	suricata/case001.pcap/ eve_flow.jsonl#22204	case001.pcap	F-003, F-012, F-021, F-022, T-009, T-022
2020-09-19 02:56:03.539280	Detection Suricata	low ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request | TCP 194.61.24.102:40240 -> 10.42.85.10:3389 | Generic Protocol Command Decode | 1:8umR/HOHr8IBq4+rlDX8uPsTIS8= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#447	case001.pcap	F-003
2020-09-19 02:56:38.402867	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62613 -> 203.78.103.109:443 failed state=established bytes=89314/960911	suricata/case001.pcap/ eve_flow.jsonl#18312	case001.pcap	F-005, F-007, F-010, F-012, F-021, F-022, T-009, T-013, T-022, T-023
2020-09-19 02:56:38.758942	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.10:62613 | A Network Trojan was detected | 1:86YQAICAKKRpBPNDZzyNhJemfLo= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#452	case001.pcap	F-005, T-009
2020-09-19 04:08:15.862233	Network events (Suricata)	network event krb5 TCP 10.42.85.10:88 -> 10.42.85.115:50948	suricata/case001.pcap/ eve_krb5.jsonl#69	case001.pcap	F-014
2020-09-19 04:08:15.868598	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50949 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#78	case001.pcap	F-001, F-009, F-012, F-014, F-015, T-012
2020-09-19	Network events (Suricata)	network event	suricata/case001.pcap/	case001.pcap	F-009, F-014

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
04:08:15.872164		krb5 TCP 10.42.85.115:50950 -> 10.42.85.10:88	eve_krb5.jsonl#80		
2020-09-19 04:08:16.583483	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 10.42.85.115:50957 -> 10.42.85.10:445 smb state=established bytes=15650/12757	suricata/case001.pcap/ eve_flow.jsonl#17430	case001.pcap	F-017
2020-09-19 04:08:16.637781	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#188	case001.pcap	F-009, F-016, F-017, T-017
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed state=established bytes=39409/488552	suricata/case001.pcap/ eve_flow.jsonl#19978	case001.pcap	F-005, F-009, F-010, F-012, F-021, F-022, T-010, T-012, T-023
2020-09-19 04:08:45.783080	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.115:50972 | A Network Trojan was detected | 1:shvkq1T5rI2BCKSFPByeBaHUCDE= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#458	case001.pcap	F-005, T-010
2020-09-19 04:09:52.725024	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#191	case001.pcap	F-016, F-017, T-017
2020-09-19 04:09:53.219790	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#193	case001.pcap	F-017, T-018
2020-09-19 04:30:24.440014	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#205	case001.pcap	T-017
2020-09-19 04:31:14.843891	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#211	case001.pcap	F-017
2020-09-19 04:49:47.678833	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#257	case001.pcap	F-017
2020-09-19 05:03:14.869462	Files transferred (Suricata)	file transferred over the network C137.local\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\Machine\Registry.pol app_proto=smb | src_ip=10.42.85.10 | dest_ip=10.42.85.115 | fileinfo_size=2784 | fileinfo_sha256=7a64e146fe72b9cdf61a216bc454425b1837b3c5875fbaf2d2f3e8923fdafc9a | fileinfo_md5=d9b402ccf91f0e9c6558d9497624decd | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#165	case001.pcap	F-012

4. Detailed findings

4.1 System context and scope

Scope: one pcap (case001.pcap, SHA256 09ABF49E…791CB) parsed by Suricata 8.0.7, window 2020-09-18 21:58:07 → 2020-09-19 05:38:57 UTC (raw EVE in +0200). Environment: AD domain C137.LOCAL; DC CITADEL-DC01 10.42.85.10 (Windows Server 2012 R2/NT 6.3, RDP exposed to Internet); workstation DESKTOP-SDN1RPT 10.42.85.115 (Windows 10 build 19041). Domain accounts observed in Kerberos: DESKTOP-SDN1RPT$, mortysmith, Administrator, ricksanchez. External 194.61.24.102 dominates traffic (≈29k RDP/TLS flows to DC:3389, cookies 'Administrator'/'nmap') and served /coreupdater.exe to the DC – to be analysed later. Findings F-001 (system description) and F-002 (scope limitation: no host artefacts).

F-002: Scope limitation – no host-based artefacts (disk/EVTX/registry/memory) in the case; network capture only

Severity	**Informational**
Confidence	confirmed
Independent sources	network
Category	context
MITRE ATT&CK	n/a
Period (UTC)	? → ?

Observed facts

The 'sources' table lists only Suricata outputs derived from case001.pcap (eve_alert, anomaly, dcerpc, dns, fileinfo, flow, http, krb5, other, rdp, smb, tls + pcap_files); raw eve.jsonl and suricata.log were ignored by design (re-loaded split by event type / tool log). The 'jobs' table contains a single job (suricata, status ok, no warnings). No raw.registry_*, raw.evtx, raw.reg_*, raw.native_*, raw.amcache_*, raw.vol_* table exists. Capture covers only 2020-09-18 21:58:07 → 2020-09-19 05:38:57 UTC.

Analysis

Execution, persistence, account creation, Windows install date, users' local profiles, USB devices and time zone cannot be established from host artefacts; all conclusions will rely on a single source class (network capture), so high/critical findings can at most be 'probable' under the corroboration policy unless two independent sources exist. Activity outside the capture window or on non-captured segments is invisible. File content transferred (fileinfo hashes) is available but not the files themselves unless stored.

Evidence (1)

UTC	Artifact	Detail	Reference	Original source
	raw.pcap_files	{"file": "C:\\Sleuthline\\cases\\test03\\pcap\\case001.pcap", "name": "case001.pcap", "bytes": "197583252", "sha256": "09ABF49EFEA1852E047987D92907704D47F36D75F6C8056E2CAFA6CC027791CB"}	suricata/pcap_files.jsonl#1	

F-001: System description – network-only scope: C137.local AD environment (CITADEL-DC01 10.42.85.10, DESKTOP-SDN1RPT 10.42.85.115)

Severity	**Informational**
Confidence	confirmed
Independent sources	network
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2020-09-18 21:59:39.458927 → 2020-09-19 04:08:15.868598

Observed facts

Evidence base: a single packet capture C:\Sleuthline\cases\test03\pcap\case001.pcap (197,583,252 bytes, SHA256 09ABF49EFEA1852E047987D92907704D47F36D75F6C8056E2CAFA6CC027791CB) processed by Suricata 8.0.7 (job 'suricata', status ok, 2026-10-20 16:48:40–16:48:52 UTC). Capture window (flow_start/flow_end): 2020-09-18 21:58:07 UTC → 2020-09-19 05:38:57 UTC (raw EVE timestamps carry the sensor offset +0200; timeline is normalised to UTC). 32,573 flows, 30,453 TLS, 2,650 DNS, 229 HTTP, 241 fileinfo, 370 SMB, 309 DCERPC, 302 RDP, 84 Kerberos, 461 alerts.
Hosts identified:
- 10.42.85.10 = CITADEL-DC01.C137.local, domain controller of AD domain C137.LOCAL (DNS SRV _ldap._tcp.dc._msdcs.C137.local → citadel-dc01.c137.local, A record 10.42.85.10; it answers Kerberos AS/TGS, LDAP, CIFS). OS: Windows Server 2012 R2 / NT 6.3 (HTTP UA 'Mozilla/5.0 (Windows NT 6.3; Win64; x64; Trident/7.0; rv:11.0)' and 'Microsoft-CryptoAPI/6.3'; SMB negotiated dialect 3.02 max). RDP (TCP/3389) is reachable from the Internet.
- 10.42.85.115 = DESKTOP-SDN1RPT.C137.local (Kerberos machine account DESKTOP-SDN1RPT$, AS-REP 2020-09-18 21:59:39 UTC), Windows 10 build 19041 (Edge UA 'Edge/18.19041', 'Microsoft-CryptoAPI/10.0'; client offers SMB 2.02–3.11).
Domain accounts seen in Kerberos from 10.42.85.115: DESKTOP-SDN1RPT$ (21:59:39 UTC), mortysmith (AS-REP 21:59:58 UTC, CIFS TGS 22:10:08 UTC), Administrator (AS-REP 2020-09-19 02:36:24 UTC, TGS host/desktop-sdn1rpt, LDAP, cifs/CITADEL-DC01, ProtectedStorage/CITADEL-DC01 at 02:36:26 UTC), ricksanchez (AS-REP 2020-09-19 04:08:15 UTC).
External host: 194.61.24.102 is the top talker (29,313 flows, ~26.5 MB to the DC), almost all to 10.42.85.10:3389 (RDP/TLS, 99 RDP initial requests with cookie 'Administrator', 1 with cookie 'nmap'); the DC also downloaded http://194.61.24.102/coreupdater.exe (HTTP row 116). These are analysed in later steps.

Analysis

The case contains no host artefact (no KAPE triage, EVTX, registry, MFT, memory image): host name, OS, accounts and time zone are inferred from protocol metadata only (Kerberos cnames/snames, DNS, HTTP User-Agents, SMB negotiation), which is reliable for identity of hosts/accounts but approximate for OS build. Time zone of the hosts cannot be read from the registry; the Suricata output was written with +0200 offsets (sensor/processing local time), all report times are UTC. No collection tool activity (KAPE etc.) exists in the capture to be excluded. Early indicators of hostile activity (Internet RDP brute-force against the DC from 194.61.24.102, download of coreupdater.exe from the same IP) are noted here as context and will be qualified in dedicated steps.

Evidence (13)

UTC	Artifact	Detail	Reference	Original source
	raw.pcap_files	{"file": "C:\\Sleuthline\\cases\\test03\\pcap\\case001.pcap", "name": "case001.pcap", "bytes": "197583252", "sha256": "09ABF49EFEA1852E047987D92907704D47F36D75F6C8056E2CAFA6CC027791CB"}	suricata/pcap_files.jsonl#1	
2020-09-18 21:59:39.479837	DNS (Suricata)	DNS query / answer [{"rrname": "citadel-dc01.c137.local", "rrtype": "A"}] dns_type=response | dns_rcode=NOERROR | src_ip=10.42.85.10 | dest_ip=10.42.85.115 | dns_answers=[{"rrname": "citadel-dc01.c137.local", "rrtype": "A", "ttl": 3600, "rdata": "10.42.85.10"}] | dns_grouped_A=["10.42.85.10"]	suricata/case001.pcap/eve_dns.jsonl#6	case001.pcap
2020-09-18 21:59:39.458927	DNS (Suricata)	DNS query / answer [{"rrname": "_ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.C137.local", "rrtype": "SRV"}] dns_type=response | dns_rcode=NOERROR | src_ip=10.42.85.10 | dest_ip=10.42.85.115 | dns_answers=[{"rrname": "_ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.C137.local", "rrtype": "SRV", "ttl": 600, "srv": {"priority": 0, "weight": 100, "port": 389, "name": "citadel-dc01.c137.local"}}]	suricata/case001.pcap/eve_dns.jsonl#9	case001.pcap
2020-09-18 21:59:39.797903	Network events (Suricata)	network event krb5 TCP 10.42.85.115:49673 -> 10.42.85.10:88	suricata/case001.pcap/eve_krb5.jsonl#18	case001.pcap
2020-09-18 21:59:58.148531	Network events (Suricata)	network event krb5 TCP 10.42.85.115:49710 -> 10.42.85.10:88	suricata/case001.pcap/eve_krb5.jsonl#38	case001.pcap

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:36:24.913767	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50695 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#60	case001.pcap
2020-09-19 02:36:26.768288	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50712 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#54	case001.pcap
2020-09-19 04:08:15.868598	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50949 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#78	case001.pcap
2020-09-18 22:01:00.882490	HTTP (Suricata)	HTTP request /e7a5/i/2016/204/2/e/jessica_from_rick_and_morty_by_fluffymystic-dab4xm5.png http_hostname=img04.deviantart.net | http_http_method=GET | http_status=301 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=text/html | http_length=0 | src_ip=10.42.85.115 | dest_ip=52.26.146.4	suricata/case001.pcap/ eve_http.jsonl#14	case001.pcap
2020-09-19 02:23:09.692112	HTTP (Suricata)	HTTP request /favicon.ico http_hostname=www.bing.com | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 6.3; Win64; x64; Trident/7.0; rv:11.0) like Gecko | http_http_content_type=image/x-icon | http_length=237 | src_ip=10.42.85.10 | dest_ip=204.79.197.200	suricata/case001.pcap/ eve_http.jsonl#125	case001.pcap
2020-09-19 02:24:06.940063	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; rv:11.0) like Gecko | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.10 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#116	case001.pcap
2020-09-18 21:59:40.016569	Network events (Suricata)	network event smb TCP 10.42.85.115:49683 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#2	case001.pcap
2020-09-19 02:21:26.114786	Network events (Suricata)	network event rdp TCP 10.42.85.10:3389 -> 194.61.24.102:40044	suricata/case001.pcap/ eve_rdp.jsonl#3	case001.pcap

4.2 Triage of automatic detections

Detections = Suricata only (461: 7 high, 4 medium, 450 low). No Hayabusa/Chainsaw/Zircolite/Defender data (no EVTX); hashset empty so hash_hits meaningless. True positives: (F-003) Nmap RDP scan + ~29k RDP brute-force connections with 'Administrator' cookie from 194.61.24.102 to DC 10.42.85.10:3389 at 02:19-02:21 UTC, followed by 30-min 17.5 MB interactive RDP session 02:22:36-02:52:47 and a second one 02:56:03-02:57:41 (successful Administrator logon inferred). (F-004) /coreupdater.exe (7168 B, SHA256 10f3b920…1cfda6, MD5 eed41b45…) downloaded from 194.61.24.102 by DC (IE11, 02:24:06) and workstation 10.42.85.115 (Edge, 02:39:58). (F-005) 5 Meterpreter stage deliveries/C2 sessions over raw TCP 203.78.103.109:443 (DC 02:25:18, 02:29:49, 02:56:38; workstation 02:40:49, 04:08:45), three still established at end of capture 05:38 UTC. (F-006, from pivot) RDP DC -> workstation 02:35:55-02:52:14. Timeline T-001..T-010, IOCs I-001..I-007. False positives/context: device-metadata/USB-metadata UA alerts, UA-CPU header (IE11 on DC), Discord traffic at 22:05 (pre-attack, low volume).

F-003: RDP brute-force against CITADEL-DC01 from 194.61.24.102 followed by long interactive Administrator RDP sessions

Severity	**Critical**
Confidence	probable (model: confirmed, lowered by the evidence policy)
Independent sources	network
Evidence policy	critical finding backed by 1 independent source(s) (network): 2 are required for 'confirmed'
Category	initial access
MITRE ATT&CK	T1110.001, T1046, T1133, T1021.001, T1078.002
Period (UTC)	2020-09-19 02:19:26.792622 → 2020-09-19 02:56:03.539280

Observed facts

Suricata alerts (case001.pcap): 2020-09-19 02:19:26.792 UTC 'ET SCAN RDP Connection Attempt from Nmap' 194.61.24.102:38100 -> 10.42.85.10:3389 (eve_alert#25; EVE RDP cookie 'nmap'); 02:19:26.918 'ET SCAN Behavioral

Unusually fast Terminal Server Traffic' 194.61.24.102:38288 -> 10.42.85.10:3389 (eve_alert#21). 198 'ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request' + 200 'ET INFO RDP - Response To External Host' between 02:19:26 and 02:56:03 UTC; EVE RDP: 99 initial_request with cookie 'Administrator' (02:21:26 -> 02:56:03 UTC). Flow volume 194.61.24.102 -> 10.42.85.10:3389: 17,006 flows in minute 02:19 and 12,204 in 02:20 (each ≤1.2 kB to client = failed/short attempts), 96 in 02:21. Then two long sessions: src_port 40238, 02:22:36.664 -> 02:52:47.379 UTC, 4,802,218 B to server / 12,756,305 B to client, 36,909 pkts (eve_flow#30035; alerts eve_alert#402/#404/#405/#407), and src_port 40240, 02:56:03.538 -> 02:57:41.585 UTC, 266,261 B / 1,008,697 B (eve_flow#22204; eve_alert#447-#450).

Analysis

True positive. Pattern = Nmap RDP probing, then ~29k short RDP connections in ~2 minutes using the 'Administrator' mstshash cookie (password spraying / brute force), immediately followed by a 30-minute, 17 MB bidirectional RDP session from the same IP: this volume is consistent with a successful interactive desktop session, i.e. compromise of the domain Administrator account on the DC exposed to the Internet. The success is inferred from traffic volume/duration (RDP payload is TLS-encrypted, no host 4624/4625 logs available), so the authentication outcome itself is a strong inference rather than directly observed. All references come from the single network capture (one source), hence corroboration from host logs is impossible in this case.

Evidence (7)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:19:26.792622	Detection Suricata	low ET SCAN RDP Connection Attempt from Nmap | TCP 194.61.24.102:38100 -> 10.42.85.10:3389 | Detection of a Network Scan | 1:BJ5fdE2KUXlcHa7o2AcT6LKStwM= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#25	case001.pcap
2020-09-19 02:19:26.918948	Detection Suricata	low ET SCAN Behavioral Unusually fast Terminal Server Traffic Potential Scan or Infection (Inbound) | TCP 194.61.24.102:38288 -> 10.42.85.10:3389 | Detection of a Network Scan | 1:VTsSoaES/lxoZ4cCIOR4ghG5WYE= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#21	case001.pcap
2020-09-19 02:21:26.112470	Detection Suricata	low ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request | TCP 194.61.24.102:40044 -> 10.42.85.10:3389 | Generic Protocol Command Decode | 1:sLbyebDh6LCv2TWmLwr5aZU4kbI= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#27	case001.pcap
2020-09-19 02:22:36.664968	Detection Suricata	low ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request | TCP 194.61.24.102:40238 -> 10.42.85.10:3389 | Generic Protocol Command Decode | 1:iingav/QQcKdY/xllC/FjNyxsRQ= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#402	case001.pcap
2020-09-19 02:22:36.664454	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40238 -> 10.42.85.10:3389 rdp state=closed bytes=4802218/12756305	suricata/case001.pcap/ eve_flow.jsonl#30035	case001.pcap
2020-09-19 02:56:03.538780	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40240 -> 10.42.85.10:3389 rdp state=closed bytes=266261/1008697	suricata/case001.pcap/ eve_flow.jsonl#22204	case001.pcap
2020-09-19 02:56:03.539280	Detection Suricata	low ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request | TCP 194.61.24.102:40240 -> 10.42.85.10:3389 | Generic Protocol Command Decode | 1:8umR/HOHr8IBq4+rlDX8uPsTIS8= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#447	case001.pcap

F-004: Download of /coreupdater.exe (PE, 7168 B) from 194.61.24.102 by the DC (IE11) and the workstation (Edge)

Severity	**Critical**
Confidence	probable (model: confirmed, lowered by the evidence policy)
Independent sources	network
Evidence policy	critical finding backed by 1 independent source(s) (network): 2 are required for 'confirmed'
Category	execution / ingress tool transfer
MITRE ATT&CK	T1105, T1204.002

Period (UTC)	2020-09-19 02:24:06.940063 → 2020-09-19 02:39:58.412735

Observed facts

HTTP (EVE): 2020-09-19 02:23:41 UTC 10.42.85.10 GET http://194.61.24.102/ (200, 228 B) and /favicon.ico (404) (eve_http#117/#118), then 02:24:06.940 UTC GET /coreupdater.exe from 10.42.85.10:62410, UA 'Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; rv:11.0) like Gecko', 200, application/x-msdos-program, 7168 B (eve_http#116). Alerts: 'ET INFO Executable Download from dotted-quad Host' (eve_alert#365), 'ET INFO PE EXE or DLL Windows file download HTTP' [high] (eve_alert#370), 'ET HUNTING SUSPICIOUS Dotted Quad Host MZ Response' (eve_alert#371). Fileinfo eve_fileinfo#100: MD5 eed41b4500e473f97c50c7385ef5e374, SHA1 fd153c66386ca93ec9993d66a84d6f0d129a3a5c, SHA256 10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6, state CLOSED (complete). Same file downloaded by 10.42.85.115 (DESKTOP-SDN1RPT): GET / at 02:39:26 UTC (eve_http#154) and /coreupdater.exe at 02:39:58.411 UTC from port 50864, UA Edge/18.19041 (eve_http#156; alerts eve_alert#442/#443/#444; fileinfo eve_fileinfo#136 – identical hashes).

Analysis

True positive. The download from the attacker IP occurs 1.5 min into the attacker's interactive RDP session on the DC (browser IE11 on Server 2012 R2 = attacker driving the GUI), and 4 min after the lateral RDP DC -> workstation for the second host. 'coreupdater.exe' is a masquerading name; 7168 B is the typical size of an msfvenom-generated stager. Execution is not directly observable (no host artefacts) but is strongly implied by the Metasploit stage downloads that follow from each host ~1 min later (see Metasploit finding). Single source (network capture).

Evidence (6)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:24:06.940063	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; rv:11.0) like Gecko | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.10 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#116	case001.pcap
2020-09-19 02:24:06.940653	Detection Suricata	high ET INFO PE EXE or DLL Windows file download HTTP | TCP 194.61.24.102:80 -> 10.42.85.10:62410 | Potential Corporate Privacy Violation | 1:u/j8zDTczy431sRgtCbNVlYztwk= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#370	case001.pcap
2020-09-19 02:24:06.940653	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.10 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#100	case001.pcap
2020-09-19 02:39:58.411571	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#156	case001.pcap
2020-09-19 02:39:58.412735	Detection Suricata	high ET INFO PE EXE or DLL Windows file download HTTP | TCP 194.61.24.102:80 -> 10.42.85.115:50864 | Potential Corporate Privacy Violation | 1:jol6Y9tRe9v45mCCcFMHMI7rx+A= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#443	case001.pcap
2020-09-19 02:39:58.412735	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 |	suricata/case001.pcap/ eve_fileinfo.jsonl#136	case001.pcap

UTC	Artifact	Detail	Reference	Original source
		src_ip=194.61.24.102 | dest_ip=10.42.85.115 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED		

F-005: Metasploit/Meterpreter stage delivery and C2 over raw TCP 443 from 203.78.103.109 to DC and workstation

Severity	**Critical**
Confidence	probable (model: confirmed, lowered by the evidence policy)
Independent sources	network
Evidence policy	critical finding backed by 1 independent source(s) (network): 2 are required for 'confirmed'
Category	command and control
MITRE ATT&CK	T1071, T1571, T1095, T1105, T1059
Period (UTC)	2020-09-19 02:25:18.565676 → 2020-09-19 04:08:45.783080

Observed facts

Five outbound TCP sessions to 203.78.103.109:443 with app_proto 'failed' (no TLS), each starting with 'ET HUNTING PE EXE Download over raw TCP' followed by 'ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)' [high]: (1) 10.42.85.10:62414, 02:25:18.565 -> 02:29:36 UTC, 51,044 B up / 932,653 B down (eve_flow#28844; eve_alert#438/#439); (2) 10.42.85.10:62476, 02:29:49 -> 02:52:46 UTC, 41,589/583,890 B (eve_flow#26788; eve_alert#378/#403); (3) 10.42.85.115:50875, 02:40:49.936 -> 05:38:44 UTC (still established at end of capture), 280,374/1,764,209 B (eve_flow#27604; eve_alert#445/#446); (4) 10.42.85.10:62613, 02:56:38 -> 05:38:51 UTC (established), 89,314/960,911 B (eve_flow#18312; eve_alert#451/#452); (5) 10.42.85.115:50972, 04:08:45 -> 05:38:42 UTC (established), 39,409/488,552 B (eve_flow#19978; eve_alert#457/#458). No DNS resolution of 203.78.103.109 was observed (direct IP).

Analysis

True positive. Non-TLS traffic on port 443 that begins with a PE transfer containing the Metasploit Bind_API construct is the signature of a reverse_tcp stager receiving the Meterpreter stage (metsrv DLL, ~750-900 kB matching the downloaded volumes). Session 1 on the DC starts 72 s after coreupdater.exe was downloaded; session 3 on the workstation 52 s after its download, which indirectly evidences execution of coreupdater.exe on both hosts. Re-staging (sessions 2, 4, 5) indicates re-launch / persistence or migration. Three C2 channels were still alive at the end of capture (05:38 UTC): hosts must be considered actively compromised. Single source (network capture).

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:25:19.071794	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.10:62414 | A Network Trojan was detected | 1:AEtaQJ4Pfkk0lfr1mrly2qsaW8Q= | case001.pcap	suricata/case001.pcap/eve_alert.jsonl#439	case001.pcap
2020-09-19 02:25:18.565676	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62414 -> 203.78.103.109:443 failed state=closed bytes=51044/932653	suricata/case001.pcap/eve_flow.jsonl#28844	case001.pcap
2020-09-19 02:29:49.580408	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.10:62476 | A Network Trojan was detected | 1:SF4txCtiLpUx9hIdO5hqR1RPreU= | case001.pcap	suricata/case001.pcap/eve_alert.jsonl#403	case001.pcap
2020-09-19 02:40:50.282950	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.115:50875 | A Network Trojan was detected |	suricata/case001.pcap/eve_alert.jsonl#446	case001.pcap

UTC	Artifact	Detail	Reference	Original source
		1:pcNgXsCjc/vxVdVvPl9FwJBoPrA= | case001.pcap		
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/ eve_flow.jsonl#27604	case001.pcap
2020-09-19 02:56:38.758942	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.10:62613 | A Network Trojan was detected | 1:86YQAICAKKRpBPNDZzyNhJemfLo= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#452	case001.pcap
2020-09-19 02:56:38.402867	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62613 -> 203.78.103.109:443 failed state=established bytes=89314/960911	suricata/case001.pcap/ eve_flow.jsonl#18312	case001.pcap
2020-09-19 04:08:45.783080	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.115:50972 | A Network Trojan was detected | 1:shvkq1T5rI2BCKSFPByeBaHUCDE= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#458	case001.pcap
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed state=established bytes=39409/488552	suricata/case001.pcap/ eve_flow.jsonl#19978	case001.pcap
2020-09-19 02:25:18.753116	Detection Suricata	low ET HUNTING PE EXE Download over raw TCP | TCP 203.78.103.109:443 -> 10.42.85.10:62414 | Misc activity | 1:AEtaQJ4Pfkk0lfr1mrIy2qsaW8Q= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#438	case001.pcap

F-006: Lateral movement: RDP from CITADEL-DC01 (10.42.85.10) to DESKTOP-SDN1RPT (10.42.85.115) during attacker session

Severity	**High**
Confidence	probable
Independent sources	network
Category	lateral movement
MITRE ATT&CK	T1021.001
Period (UTC)	2020-09-19 02:35:55.285340 → 2020-09-19 02:36:23.515604

Observed facts

EVE RDP: 2020-09-19 02:35:55.347 UTC RDP initial_request (empty cookie) on flow 10.42.85.10:62514 -> 10.42.85.115:3389, initial_response, TLS handshake at 02:36:23 (eve_rdp#297/#298/#299). Flows: 10.42.85.10:62514 -> 10.42.85.115:3389, 02:35:55 -> 02:52:14 UTC, 875,256/452,944 B (eve_flow#29416); 10.42.85.10:63121 -> 10.42.85.115:3389, 02:36:23 -> 02:52:14 UTC, 283,842/6,695,207 B (eve_flow#31428). No Suricata alert fired (internal traffic). The workstation then fetched coreupdater.exe at 02:39:58 UTC.

Analysis

Not an automatic detection but surfaced while pivoting on the alert timeline. An RDP session from the DC to the workstation inside the attacker's DC session (02:22-02:52), ending at 02:52:14 just before the attacker's DC session ended (02:52:47), followed 4 min later by the stager download on the workstation with Edge: consistent with the attacker using the DC (and stolen Administrator credentials) to RDP onto DESKTOP-SDN1RPT. The account used is not visible (TLS/CredSSP). Probable; network-only evidence.

Evidence (4)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:35:55.347240	Network events (Suricata)	network event rdp TCP 10.42.85.115:3389 -> 10.42.85.10:62514	suricata/case001.pcap/ eve_rdp.jsonl#297	case001.pcap
2020-09-19 02:36:23.467377	Network events (Suricata)	network event rdp TCP 10.42.85.10:62514 -> 10.42.85.115:3389	suricata/case001.pcap/ eve_rdp.jsonl#299	case001.pcap
2020-09-19 02:35:55.285340	Network flows (Suricata)	network flow: start 10.42.85.115	suricata/case001.pcap/ eve_flow.jsonl#29416	case001.pcap

UTC	Artifact	Detail	Reference	Original source
		TCP 10.42.85.10:62514 -> 10.42.85.115:3389 rdp state=closed bytes=875256/452944		
2020-09-19 02:36:23.515604	Network flows (Suricata)	network flow: start 10.42.85.115 UDP 10.42.85.10:63121 -> 10.42.85.115:3389 failed state=established bytes=283842/6695207	suricata/case001.pcap/ eve_flow.jsonl#31428	case001.pcap

4.3 Program execution

Host execution artefacts (Prefetch, Amcache, ShimCache, BAM, UserAssist, PCA, SRUM, 4688/Sysmon, PowerShell) are not in the case. The only evidence is one pcap, so I inferred execution from network behaviour. F-007 (critical/probable): coreupdater.exe (7168 B, SHA256 10f3b920…cfda6) was most likely executed as a reverse-TCP Meterpreter stager. The DC downloaded it at 02:24:06 and opened its own outbound connection to 203.78.103.109:443 at 02:25:18. The workstation downloaded it at 02:39:58 and connected out at 02:40:49. F-008 (high/probable): domain Administrator logged on interactively to DESKTOP-SDN1RPT for the first time at 02:36:24, seen as Kerberos AS-REP, a NETLOGON 'Default User.V6' lookup and the DPAPI protected_storage pipe. So the workstation execution most likely ran as Administrator. F-009 (high/probable): user ricksanchez logged on at the console at 04:08:15 (no RDP at that time), and a new Meterpreter session started 30 s later at 04:08:45. This points to something that starts automatically at logon (Run key, Startup folder or scheduled task). I found no sign of remote execution between machines: no SVCCTL/ATSVC/WMI in DCERPC, and no other executable transferred over HTTP or SMB. Added timeline events T-011 and T-012 and IOC I-008 (C137\Administrator).

F-007: Execution of coreupdater.exe (reverse-TCP Meterpreter stager) inferred on CITADEL-DC01 and DESKTOP-SDN1RPT from download→outbound C2 sequence

Severity	**Critical**
Confidence	probable
Independent sources	network
Category	execution
MITRE ATT&CK	T1204.002, T1105, T1059, T1571
Period (UTC)	2020-09-19 02:24:06.940063 → 2020-09-19 02:56:38.402867

Observed facts

No host execution artefacts exist in the case (no Prefetch/Amcache/ShimCache/BAM/UserAssist/SRUM/4688/Sysmon/PowerShell – network capture only, see F-002). Execution is inferred from network facts:
(1) CITADEL-DC01 10.42.85.10: HTTP GET http://194.61.24.102/coreupdater.exe with IE11 UA 'Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; rv:11.0) like Gecko' at 2020-09-19 02:24:06.94 UTC; file 7168 bytes, SHA256 10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 (eve_fileinfo). 71 s later, 02:25:18.57 UTC, the DC itself OPENS an outbound TCP connection 10.42.85.10:62414 → 203.78.103.109:443 (flow 28844, 51,044 B up / 932,653 B down, closed 02:29:36), on which Suricata flags a Meterpreter stage delivery (F-005). Further DC-initiated sessions: 62476 → 203.78.103.109:443 02:29:49–02:52:46 (flow 26788) and 62613 → 203.78.103.109:443 02:56:38 until end of capture (flow 18312, still established).
(2) DESKTOP-SDN1RPT 10.42.85.115: GET http://194.61.24.102/coreupdater.exe with Edge 18.19041 UA at 02:39:58.41 UTC (same SHA256). 51 s later, 02:40:49.94 UTC, the workstation opens 10.42.85.115:50875 → 203.78.103.109:443 (flow 27604, 280,374 B up / 1,764,209 B down, established until 05:38:44 = end of capture).
No other executable was transferred over HTTP/SMB in clear text (all eve_fileinfo objects are OCSP/CRL, MS metadata, images, SYSVOL GPO files, and the two coreupdater.exe copies).

Analysis

The connections to 203.78.103.109:443 are initiated by the internal hosts (reverse_tcp), so a process on each host must have run the stager; the short delay (51–71 s) after the download of a 7 KB PE – the typical size of a msfvenom windows/x64/meterpreter/reverse_tcp stager – makes coreupdater.exe the most likely executed program, launched interactively in the attacker's RDP sessions (DC: Administrator RDP session from 194.61.24.102 starting 02:22:36;

workstation: RDP from the DC as Administrator starting 02:35:55). Confidence 'probable': single source (pcap); the exact path (likely the browser Downloads folder of Administrator), process name, PID, parent and run count cannot be verified without Prefetch/Amcache/4688/Sysmon. Alternative (less likely): the stager was executed by another means (e.g. PowerShell) and the download was only a decoy – nothing supports this.

Evidence (7)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:24:06.940653	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.10 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/eve_fileinfo.jsonl#100	case001.pcap
2020-09-19 02:25:18.565676	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62414 -> 203.78.103.109:443 failed state=closed bytes=51044/932653	suricata/case001.pcap/eve_flow.jsonl#28844	case001.pcap
2020-09-19 02:29:49.290306	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62476 -> 203.78.103.109:443 failed state=closed bytes=41589/583890	suricata/case001.pcap/eve_flow.jsonl#26788	case001.pcap
2020-09-19 02:56:38.402867	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62613 -> 203.78.103.109:443 failed state=established bytes=89314/960911	suricata/case001.pcap/eve_flow.jsonl#18312	case001.pcap
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/eve_flow.jsonl#27604	case001.pcap
2020-09-19 02:24:06.940063	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; rv:11.0) like Gecko | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.10 | dest_ip=194.61.24.102	suricata/case001.pcap/eve_http.jsonl#116	case001.pcap
2020-09-19 02:39:58.411571	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/eve_http.jsonl#156	case001.pcap

F-008: First interactive logon of domain Administrator on DESKTOP-SDN1RPT (profile creation) at 02:36:24 UTC during RDP from the DC

Severity	**High**
Confidence	probable
Independent sources	network
Category	lateral movement
MITRE ATT&CK	T1021.001, T1078.002
Period (UTC)	2020-09-19 02:35:55.285340 → 2020-09-19 02:36:26.768288

Observed facts

At 2020-09-19 02:36:24.91 UTC, 29 s after the RDP connection 10.42.85.10:62514 → 10.42.85.115:3389 (flow 29416, 02:35:55–02:52:14), the workstation 10.42.85.115 performs Kerberos AS-REQ/AS-REP for cname 'Administrator' (krbtgt/C137.LOCAL, eve_krb5 #57-#60), TGS for host/desktop-sdn1rpt.c137.local (#51-#52), LDAP/CITADEL-DC01 (#61-#62), cifs/CITADEL-DC01 (#63-#64). It then tree-connects \\CITADEL-DC01\NETLOGON (eve_smb #149) and opens 'Default

User.V6' (STATUS_OBJECT_NAME_NOT_FOUND, #151) at 02:36:25.04, then opens the 'protected_storage' named pipe (#131, 02:36:26.77) after a TGS for ProtectedStorage/CITADEL-DC01.C137.local (#53-#54).

Analysis

The lookup of the network Default User profile in NETLOGON is performed by the User Profile Service only when a NEW local profile is created, and the protected_storage pipe (MS-BKRP, DPAPI domain backup key) is used when the new profile generates its DPAPI master key. Hence the Administrator account logged on interactively (RDP, LogonType 10) on DESKTOP-SDN1RPT for the first time at 02:36:24 UTC: the coreupdater.exe download (02:39:58) and execution (02:40:49) on the workstation therefore most likely ran in the Administrator context, in C:\Users\Administrator(.C137)\. Single source (pcap); to be verified with 4624/4778 and C:\Users\ creation time on the host.

Evidence (7)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:36:24.913767	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50695 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#60	case001.pcap
2020-09-19 02:36:24.915473	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50696 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#52	case001.pcap
2020-09-19 02:36:24.991912	Network events (Suricata)	network event smb TCP 10.42.85.115:50701 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#149	case001.pcap
2020-09-19 02:36:25.038929	Network events (Suricata)	network event smb TCP 10.42.85.115:50701 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#151	case001.pcap
2020-09-19 02:36:26.766248	Network events (Suricata)	network event smb TCP 10.42.85.115:50710 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#131	case001.pcap
2020-09-19 02:36:26.768288	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50712 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#54	case001.pcap
2020-09-19 02:35:55.285340	Network flows (Suricata)	network flow: start 10.42.85.115 TCP 10.42.85.10:62514 -> 10.42.85.115:3389 rdp state=closed bytes=875256/452944	suricata/case001.pcap/ eve_flow.jsonl#29416	case001.pcap

F-009: Meterpreter re-launched on DESKTOP-SDN1RPT 30 s after local logon of user ricksanchez (04:08:15 UTC) – logon-triggered persistence suspected

Severity	**High**
Confidence	probable
Independent sources	network
Category	execution
MITRE ATT&CK	T1547.001, T1053.005, T1037
Period (UTC)	2020-09-19 02:40:49.936293 → 2020-09-19 04:08:45.483780

Observed facts

At 2020-09-19 04:08:15.86 UTC the workstation 10.42.85.115 performs Kerberos AS-REQ/AS-REP for cname 'ricksanchez' (eve_krb5 #69-#82: krbtgt/C137, host/desktop-sdn1rpt.c137.local, LDAP, cifs/CITADEL-DC01), then at 04:08:16.64 tree-connects \\CITADEL-DC01\FileShare (eve_smb #188, flow 17430 kept open until end of capture). No RDP flow towards 10.42.85.115:3389 exists at that time (only flows 29416/31428 at 02:35–02:52), i.e. this is a console/local interactive logon. At 04:08:45.48 UTC – 30 s after the logon – the workstation opens a new outbound connection 10.42.85.115:50972 → 203.78.103.109:443 (flow 19978, 39,409 B up / 488,552 B down, established until 05:38:42), flagged as Meterpreter staging (F-005), while the earlier session (flow 27604, from 02:40:49) is still alive.

Analysis

A new stager connection immediately after a user's interactive logon, with no attacker remote session to the workstation at that time, strongly suggests an autostart mechanism executing the payload at logon (HKCU/HKLM Run key, Startup folder, logon scheduled task or service) planted during the Administrator session 02:36–02:52 – e.g. Meterpreter 'persistence'/'exploit/windows/local/persistence' modules. Alternative: the user ricksanchez manually ran

coreupdater.exe (less likely given timing). The 04:08 SMB mapping of FileShare is consistent with a mapped drive at logon. Single source; the persistence mechanism itself must be confirmed on the host (Run keys, Startup, Tasks, 7045/4698).

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 04:08:15.868598	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50949 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#78	case001.pcap
2020-09-19 04:08:15.872164	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50950 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#80	case001.pcap
2020-09-19 04:08:16.637781	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#188	case001.pcap
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed state=established bytes=39409/488552	suricata/case001.pcap/ eve_flow.jsonl#19978	case001.pcap
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/ eve_flow.jsonl#27604	case001.pcap

4.5 Network traffic (PCAP)

Network analysis of case001.pcap. Capture window: 2020-09-18 21:58:07 → 2020-09-19 05:38:57 UTC. Anything before or after that window is invisible on the wire.

***Alerts:** All 461 Suricata alerts were re-qualified, and the step-2 qualification stands.*
- True positives: Nmap RDP scan, RDP brute force with the 'Administrator' cookie, PE download from a dotted-quad host, Metasploit Bind_API, and PE over raw TCP.
- Noise: Discord, device-metadata/USB-metadata user agents, the UA-CPU header, and RDP responses to an external host.

Conversations:
- External peers are only the two attacker IPs plus Microsoft, CDN, OCSP and browsing sites.
- 194.61.24.102:
- RDP brute force at 02:19–02:20 (29,208 TLS flows).
- 100 RDP flows totalling 5.35 MB up / 14 MB down, until 02:57:41.
- HTTP / and /coreupdater.exe at 02:23:41/02:24:06 (DC) and 02:39:26/02:39:58 (workstation).
- 203.78.103.109:443: five raw-TCP Meterpreter sessions (new finding F-010).
- Each starts with a ~0.5–0.9 MB stage download.
- Total upload is ≈0.5 MB, so no bulk exfiltration is visible.
- Three sessions were still ESTABLISHED at the end of the capture.
- Neither attacker IP was resolved through DNS.

***Lateral movement:** Internal traffic shows only DC→workstation RDP (already F-006). SMB carried only sysvol GPO files, NETLOGON profile lookups and FileShare browsing after ricksanchez logged on. There was no admin$/C$, no executable transferred over SMB, and no SVCCTL/ATSVC/DCSync.*

***DNS:** Nothing algorithmic or newly seen that is malicious. 10.90.90.90 is a DNSBL sinkhole.*

***TLS:** New finding F-011 fingerprints the brute-force tool: JA3 75fe5199…/JA4 t13i781000_… (OpenSSL-based client offering TLS 1.3), plus three Nmap ssl-probe JA3s. Added as IOC I-009. RDP uses the DC's self-signed CN=CITADEL-DC01.C137.local certificate.*

***Files carved:** The only suspicious one is coreupdater.exe (already IOCs I-004..I-006). hash_hits is empty because no hash lists were loaded.*

***Notes:** Context was recorded for the unexplained Authenticode/Intel CRL check by the DC at 04:02 UTC during its live C2 session, the RC4 AS-REP for Administrator, and the OneDrive and telemetry traffic.*

F-010: C2 channel characterisation: 5 raw-TCP Meterpreter sessions to 203.78.103.109:443 (no TLS); 3 alive at end of capture; no bulk exfiltration seen on the wire

Severity	**High**
Confidence	probable (model: confirmed, lowered by the evidence policy)
Independent sources	network
Evidence policy	high finding backed by 1 independent source(s) (network): 2 are required for 'confirmed'
Category	
MITRE ATT&CK	T1071, T1095, T1571, T1105, T1041
Period (UTC)	2020-09-19 02:25:18.565676 → 2020-09-19 04:08:45.483780

Observed facts

raw.suricata_flow, all flows to 203.78.103.109:443 (app_proto 'failed', meaning Suricata could not identify a protocol; anomaly APPLAYER_PROTO_DETECTION_SKIPPED on each one; raw.suricata_tls has no TLS handshake for them):
- 10.42.85.10:62414, 02:25:18 → 02:29:36 UTC (258 s), 51,044 B up / 932,653 B down, closed (eve_flow#28844)
- 10.42.85.10:62476, 02:29:49 → 02:52:46 (1377 s), 41,589 up / 583,890 down, closed. It ends with the attacker's RDP session (02:52:47) (eve_flow#26788)
- 10.42.85.115:50875, 02:40:49 → 05:38:44 (10,675 s), 280,374 up / 1,764,209 down, still ESTABLISHED at end of capture (eve_flow#27604)
- 10.42.85.10:62613, 02:56:38 → 05:38:51 (9,733 s), 89,314 up / 960,911 down, ESTABLISHED (eve_flow#18312)
- 10.42.85.115:50972, 04:08:45 → 05:38:42 (5,397 s), 39,409 up / 488,552 down, ESTABLISHED (eve_flow#19978)
The total sent to the C2 by both hosts over ~3 h is ≈0.5 MB. Neither attacker IP (194.61.24.102, 203.78.103.109) shows up in any DNS answer (0 rows in raw.suricata_dns): both were reached by literal IP. The DC's only other external flows during the window are to Microsoft/CDN/OCSP endpoints, each under 6 KB upstream. Internal SMB between the workstation and the DC carried only sysvol GPO files (gpt.ini, GptTmpl.inf, Registry.pol), the NETLOGON 'Default User.V6' profile lookup and \\CITADEL-DC01\FileShare browsing (Desktop.ini/AutoRun.inf at 04:08–05:09 UTC). No SMB flow is larger than ~60 KB.

Analysis

Every session starts with the server sending ~0.5–0.9 MB (the Meterpreter stage: metsrv DLL plus extensions, which matches the 'Metasploit Bind_API' and 'PE EXE over raw TCP' alerts). After that the traffic looks like interactive C2. Uploads per session are small (39–280 KB). That fits interactive commands, screenshots, or harvesting small amounts of data such as credentials or hashdump output. It does not fit bulk exfiltration. However, Meterpreter's TLV traffic is XOR/AES-encrypted, so the content cannot be checked from the pcap. Data could also have left through the attacker's interactive RDP session (02:22–02:57, 5.35 MB client→server and 14 MB server→client on DC:3389), for example via clipboard or drive redirection, and that is invisible at this level. The workstation's 280 KB upload is the largest and may reflect post-exploitation on DESKTOP-SDN1RPT. Three sessions were still active when the capture ended, so the compromise was ongoing at 05:38 UTC and anything after that point is outside network visibility. Disk/memory confrontation (owning process via vol_netscan, Sysmon 3, SRUM) is impossible: no host or memory artefacts are in the case.

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:25:18.565676	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62414 -> 203.78.103.109:443 failed state=closed bytes=51044/932653	suricata/case001.pcap/ eve_flow.jsonl#28844	case001.pcap
2020-09-19 02:29:49.290306	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62476 -> 203.78.103.109:443 failed state=closed bytes=41589/583890	suricata/case001.pcap/ eve_flow.jsonl#26788	case001.pcap
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/ eve_flow.jsonl#27604	case001.pcap

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:56:38.402867	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62613 -> 203.78.103.109:443 failed state=established bytes=89314/960911	suricata/case001.pcap/ eve_flow.jsonl#18312	case001.pcap
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed state=established bytes=39409/488552	suricata/case001.pcap/ eve_flow.jsonl#19978	case001.pcap

F-011: RDP brute-force tooling fingerprint: OpenSSL-based non-Windows client (JA3 75fe5199..., TLS 1.3) plus Nmap SSL probes from 194.61.24.102

Severity	**Medium**
Confidence	confirmed
Independent sources	network
Category	
MITRE ATT&CK	T1110.001, T1046, T1133
Period (UTC)	2020-09-18 21:58:07.470323 → 2020-09-19 02:19:26.792998

Observed facts

raw.suricata_tls, dest 10.42.85.10:3389, src 194.61.24.102:
- 29,204 TLS ClientHellos with JA3 75fe51990656df4f7a249d5b86aa29ae, JA4 t13i781000_ab95583b6d39_d41ae481755e, TLS 1.3 offered, ~78 cipher suites including TLS_EMPTY_RENEGOTIATION_INFO_SCSV (255) (first record eve_tls#932). These records carry Suricata's default timestamp (2020-09-18T21:58:07 UTC); the matching flows are 02:19:26–02:20:26 UTC (29,208 TLS flows, 21.1 MB up / 3.9 MB down).
- Three single ClientHellos at 02:19:26.79 UTC with very short cipher lists (47-10-19-57-4): JA3 3bdfb64d53404bacd8a47056c6a756be (eve_tls#920), e26ff759048e07b164d8faf6c2a19f53 (eve_tls#926), 6dc99de941a8f76cad308d9089e793d7 (eve_tls#931). They coincide with the 'ET SCAN RDP Connection Attempt from Nmap' alert.
The server certificate is the DC's self-signed RDP certificate CN=CITADEL-DC01.C137.local.

Analysis

The native Windows mstsc client of 2020 did not offer TLS 1.3 or such a large OpenSSL-style cipher list. The JA3/JA4 matches an OpenSSL/FreeRDP-based brute-force tool (e.g. hydra/crowbar/ncrack/xfreerdp). The short-cipher hellos are Nmap ssl/rdp NSE probes. This tells us the brute force was automated from a Linux-type attacking host, and gives detection fingerprints. JA3 alone is not unique to malware, so it should be used as a hunting indicator together with the source IP.

Evidence (4)

UTC	Artifact	Detail	Reference	Original source
2020-09-18 21:58:07.470323	TLS (Suricata)	TLS handshake 10.42.85.10 dest_ip=10.42.85.10 | dest_port=3389 | tls_version=TLS 1.3 | tls_ja3_hash=75fe51990656df4f7a249d5b86aa29ae | tls_ja4=t13i781000_ab95583b6d39_d41ae481755e	suricata/case001.pcap/ eve_tls.jsonl#932	case001.pcap
2020-09-19 02:19:26.792649	TLS (Suricata)	TLS handshake 10.42.85.10 dest_ip=10.42.85.10 | dest_port=3389 | tls_version=TLS 1.2 | tls_subject=CN=CITADEL-DC01.C137.local | tls_issuerdn=CN=CITADEL-DC01.C137.local | tls_ja3_hash=3bdfb64d53404bacd8a47056c6a756be | tls_ja3s_hash=5397c414a9ebeaff1bf18b70ca22eaa0 | tls_ja4=t12i050200_7f27215065b1_4c3f1086ad7a	suricata/case001.pcap/ eve_tls.jsonl#920	case001.pcap
2020-09-19 02:19:26.792998	TLS (Suricata)	TLS handshake 10.42.85.10 dest_ip=10.42.85.10 | dest_port=3389 | tls_version=TLS 1.2 | tls_subject=CN=CITADEL-DC01.C137.local |	suricata/case001.pcap/ eve_tls.jsonl#926	case001.pcap

UTC	Artifact	Detail	Reference	Original source
		tls_issuerdn=CN=CITADEL-DC01.C137.local | tls_ja3_hash=e26ff759048e07b164d8faf6c2a19f53 | tls_ja3s_hash=5397c414a9ebeaff1bf18b70ca22eaa0 | tls_ja4=t12i050000_7f27215065b1_d2e2adf7177b		
2020-09-19 02:19:26.792588	TLS (Suricata)	TLS handshake 10.42.85.10 dest_ip=10.42.85.10 | dest_port=3389 | tls_version=TLS 1.2 | tls_subject=CN=CITADEL-DC01.C137.local | tls_issuerdn=CN=CITADEL-DC01.C137.local | tls_ja3_hash=6dc99de941a8f76cad308d9089e793d7 | tls_ja3s_hash=5397c414a9ebeaff1bf18b70ca22eaa0 | tls_ja4=t12i0502h1_7f27215065b1_e5744cedd59f	suricata/case001.pcap/ eve_tls.jsonl#931	case001.pcap

4.6 Persistence

Persistence analysis, network-only case. I confirmed that no host persistence artefact table exists (registry, EVTX, tasks, services, WMI, Startup folders, MFT). On the wire I checked:
- DCERPC interfaces: only EPM, DRSUAPI, LSARPC and NETLOGON, all normal for a domain workstation. No SVCCTL, ATSVC, WINREG or WMI, so no remote creation of a service, task or registry value.
- Default Domain Policy files over SMB: gpt.ini, Registry.pol and GptTmpl.inf have the same MD5 across the refreshes at 21:59, 02:16, 03:30 and 05:03, so the GPO was not tampered with.
- Reboots: machine-account Kerberos shows a single boot at 21:59:39, so no reboot happened during the capture.
- HTTP: no new download before the later C2 sessions.

New finding F-012 (high/probable): Meterpreter is re-staged 30-35 s after interactive logons, with no re-download:
- DC: RDP #2 at 02:56:03, C2 at 02:56:38. The previous C2 session died with the RDP logoff at 02:52:46/47 (timeline event T-013).
- Workstation: ricksanchez console logon at 04:08:15, C2 at 04:08:45.

This fits logon-triggered autostart persistence: a Run key (HKLM most likely), the All-Users Startup folder, or a logon scheduled task. The alternative is a manual relaunch, which is plausible on the DC but much less likely on the workstation.

F-012: Logon-triggered re-launch of Meterpreter on both hosts without re-download – suspected autostart persistence (Run key / Startup folder / logon task), mechanism not observable

Severity	**High**
Confidence	probable
Independent sources	network
Category	persistence
MITRE ATT&CK	T1547.001, T1053.005, T1037
Period (UTC)	2020-09-18 21:59:39.797903 → 2020-09-19 05:03:14.869462

Observed facts

Facts (case001.pcap, Suricata EVE, all UTC):
CITADEL-DC01 (10.42.85.10):
- Attacker RDP session #1 194.61.24.102:40238 -> 10.42.85.10:3389, 02:22:36.66 -> 02:52:47.38, closed/shutdown, 4.8 MB up / 12.8 MB down (eve_flow.jsonl#30035).
- Meterpreter session #2 10.42.85.10:62476 -> 203.78.103.109:443, 02:29:49.29 -> 02:52:46.39 closed (eve_flow.jsonl#26788): it ends 1 s before RDP session #1 ends.
- Attacker RDP session #2 194.61.24.102:40240 -> DC:3389, 02:56:03.54 -> 02:57:41.59 (eve_flow.jsonl#22204).
- New Meterpreter session 10.42.85.10:62613 -> 203.78.103.109:443 at 02:56:38.40, 35 s after RDP logon, still established at 05:38:51 (eve_flow.jsonl#18312), 961 kB downstream (stage).
DESKTOP-SDN1RPT (10.42.85.115):
- ricksanchez Kerberos AS-REP 04:08:15.868 and TGS host/desktop-sdn1rpt 04:08:15.872 (eve_krb5.jsonl#78, #80), console logon (no RDP at that time).
- New Meterpreter session 10.42.85.115:50972 -> 203.78.103.109:443 at 04:08:45.48, 30 s after logon, 488 kB stage, established at end of capture (eve_flow.jsonl#19978).

- The previous workstation session (50875, 02:40:49) is still established in parallel (eve_flow.jsonl#27604).
No re-download: HTTP log has no request to 194.61.24.102 or any other non-Microsoft host around 02:56 (DC) or 04:08 (workstation). Only OCSP/CRL, ctldl.windowsupdate, weather tiles and device metadata appear (e.g. eve_http.jsonl#162, #220). So the stager was started from a copy already on disk.
No reboot in the capture: the machine account DESKTOP-SDN1RPT$ requests AS/TGS only at 21:59:39-21:59:47 UTC, at boot (eve_krb5.jsonl#18, #42). There are no later machine-account AS-REQs. Persistence that survives a reboot therefore cannot be shown on the wire. What can be shown is a re-start at user logon.
Ruled out on the wire:
- Remote persistence creation: DCERPC holds only EPM (e1af8308), DRSUAPI DsBind/DsCrackNames (e3514235 opnums 0/1/12/13), LSARPC (12345778) and NETLOGON (12345678-...cffb), all workstation->DC and all expected for domain logon. There is no SVCCTL (367abb81), ATSVC/ITaskSchedulerService (1ff70682/86d35949), WINREG (338cd001) or WMI/DCOM.
- GPO-based persistence through Default Domain Policy: gpt.ini (22 B, MD5 b20996efc49d181769db184aeea375a2), Machine\Registry.pol (2784 B, MD5 d9b402ccf91f0e9c6558d9497624decd) and GptTmpl.inf (1098 B, MD5 f443c7b00e42c58336e9113c4b92a1ea) are identical in hash when read at 21:59:40, 02:16:21, 03:30:14 and 05:03:14 UTC (eve_fileinfo.jsonl#2, #90, #185, #165). No other GPO GUID was accessed.

Analysis

On both hosts the pattern repeats: an interactive logon (RDP Administrator on the DC at 02:56:03, console ricksanchez on the workstation at 04:08:15) is followed 30-35 s later by a fresh reverse-TCP Meterpreter stage from 203.78.103.109:443. No new download comes first. On the DC the earlier session died together with the RDP logoff (02:52:46 vs 02:52:47). That fits a payload running inside the interactive user session (user-context process) that is restarted at the next logon. The most likely mechanisms:
- HKLM/HKCU ...\CurrentVersion\Run(Once) value pointing to the copy of coreupdater.exe (the HKLM variant would explain the ricksanchez trigger, since ricksanchez never ran the file);
- All-Users Startup folder (C:\ProgramData\Microsoft\Windows\Start Menu\Programs\StartUp);
- a scheduled task with an ONLOGON trigger;
- less likely, Winlogon Userinit/Shell or a logon script.
A service-based persistence would start at boot rather than at logon, and no reboot occurred. Alternative explanation: the attacker relaunched the payload manually through the live sessions (RDP #2 on the DC, or the existing workstation session 50875 that was still up at 04:08). The DC case is ambiguous for that reason. The workstation case is the stronger one, because no attacker RDP was active at 04:08 and the timing matches ricksanchez's logon to the second. Confidence stays 'probable': only one source (network) exists. The exact mechanism (key name, file path, task name), creation time and whether it survives a reboot cannot be determined without host artefacts (SOFTWARE/NTUSER hives, Tasks folder, Security 4698/4657, System 7045, Startup folders, $MFT). This finding refines F-009 and extends the logon-triggered hypothesis to the DC.

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:22:36.664454	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40238 -> 10.42.85.10:3389 rdp state=closed bytes=4802218/12756305	suricata/case001.pcap/ eve_flow.jsonl#30035	case001.pcap
2020-09-19 02:29:49.290306	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62476 -> 203.78.103.109:443 failed state=closed bytes=41589/583890	suricata/case001.pcap/ eve_flow.jsonl#26788	case001.pcap
2020-09-19 02:56:03.538780	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40240 -> 10.42.85.10:3389 rdp state=closed bytes=266261/1008697	suricata/case001.pcap/ eve_flow.jsonl#22204	case001.pcap
2020-09-19 02:56:38.402867	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62613 -> 203.78.103.109:443 failed state=established bytes=89314/960911	suricata/case001.pcap/ eve_flow.jsonl#18312	case001.pcap
2020-09-19	Network events	network event	suricata/case001.pcap/	case001.pcap

UTC	Artifact	Detail	Reference	Original source
04:08:15.868598	(Suricata)	krb5 TCP 10.42.85.115:50949 -> 10.42.85.10:88	eve_krb5.jsonl#78	
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed state=established bytes=39409/488552	suricata/case001.pcap/ eve_flow.jsonl#19978	case001.pcap
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/ eve_flow.jsonl#27604	case001.pcap
2020-09-18 21:59:39.797903	Network events (Suricata)	network event krb5 TCP 10.42.85.115:49673 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#18	case001.pcap
2020-09-18 21:59:40.687057	Files transferred (Suricata)	file transferred over the network C137.local\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\Machine\Registry.pol app_proto=smb | src_ip=10.42.85.10 | dest_ip=10.42.85.115 | fileinfo_size=2784 | fileinfo_sha256=7a64e146fe72b9cdf61a216bc454425b1837b3c5875fbaf2d2f3e8923fdafc9a | fileinfo_md5=d9b402ccf91f0e9c6558d9497624decd | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#2	case001.pcap
2020-09-19 05:03:14.869462	Files transferred (Suricata)	file transferred over the network C137.local\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\Machine\Registry.pol app_proto=smb | src_ip=10.42.85.10 | dest_ip=10.42.85.115 | fileinfo_size=2784 | fileinfo_sha256=7a64e146fe72b9cdf61a216bc454425b1837b3c5875fbaf2d2f3e8923fdafc9a | fileinfo_md5=d9b402ccf91f0e9c6558d9497624decd | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#165	case001.pcap

4.7 Accounts, authentication and privileges

Accounts and authentication were rebuilt from network data only. The case has no Security/TerminalServices EVTX and no SAM (raw.reg_useraccounts absent), so I worked from the RDP, Kerberos, SMB2 and LDAP records in case001.pcap.

***RDP brute force:** The 29,202 TLS-only flows at 02:19-02:20 never reached authentication (no X.224/CredSSP exchange). The real password guessing was 95 identical CredSSP failure flows with mstshash=Administrator at 02:21:26-02:21:46. A larger, different flow at 02:21:47 most likely marks the first successful login, i.e. the moment the password was found. Short sessions followed at 02:22:08, then the interactive session at 02:22:36 (F-013, critical/probable; T-014, T-015).*

Per-account summary (F-014):
- DESKTOP-SDN1RPT$: boot at 21:59:39.
- mortysmith: legitimate console logon at 21:59:58 (T-016).
- Administrator: Kerberos logon on the workstation at 02:36:24 via RDP from the DC, which created a new profile and requested the DPAPI ProtectedStorage ticket (F-008).
- ricksanchez: console logon at 04:08:15.
- No Kerberos failures other than the normal PREAUTH_REQUIRED, and no lockout/revoked errors.
- All 24 SMB sessions and all 95 LDAP binds used Kerberos and succeeded. No NTLM and no SAMR were seen.

***Encryption (F-015, low/possible):** Administrator's AS-REP used RC4-HMAC while every other account used AES256. The likely cause is a legacy, never-rotated password with no AES keys.*

***Noise:** I recorded a note on the TLS flood so it is not miscounted as failed logons.*

F-013: Administrator password guessed over RDP/NLA: 95 failed CredSSP attempts in 21 s then first successful authentication at 02:21:47 UTC from 194.61.24.102

Severity	**Critical**
Confidence	probable
Independent sources	network
Category	credential access
MITRE ATT&CK	T1110.001, T1078.002, T1021.001
Period (UTC)	2020-09-19 02:19:26.619607 → 2020-09-19 02:22:36.664454

Observed facts

Flow-level breakdown of all TCP/3389 connections between 194.61.24.102 and CITADEL-DC01 10.42.85.10 (raw.suricata_flow, case001.pcap):
(1) 02:19:26-02:20:26 UTC: 29,202 byte-identical TLS-only flows (723 B to server / 134 B to client), no RDP X.224 negotiation and no CredSSP exchange: these are handshakes rejected by the server, not credential tests. They follow the single 'nmap' cookie RDP probe at 02:19:26 (eve_rdp #1/#2, server answered protocol hybrid_ex).
(2) 02:21:26.11-02:21:46.95 UTC: 95 RDP flows, all byte-identical (2,613 B to server / 2,257 B to client, 14/8 packets, ended by RST/timeout), each carrying an X.224 Connection Request with mstshash cookie 'Administrator' and negotiated protocol 'hybrid' (NLA/CredSSP) (eve_rdp #3 onwards, 99 'Administrator' initial_request records 02:21:26-02:56:03). The constant size is the signature of a CredSSP exchange ending with an authentication failure (equivalent on host side to Security 4625, LogonType 3, Status 0xC000006D) – e.g. eve_flow #15690 (194.61.24.102:40112, 02:21:33.62-02:21:33.83).
(3) 02:21:47.166 UTC: flow eve_flow #32271 (src port 40234) differs: 6,373 B / 5,222 B, 32/23 packets, closed by shutdown after 0.8 s – a CredSSP exchange that went further than the failures, i.e. the first successful credential validation (tool confirming the guessed password, then disconnecting).
(4) 02:22:08.31-02:22:17.34 UTC: eve_flow #19321 (port 40236), 28,260/62,729 B, 9 s – first short graphical session (logon screen/licensing), then 02:22:36.66 eve_flow #30035, 4.8 MB/12.76 MB, the 30-min interactive session already described in F-003, and 02:56:03 eve_flow #22204 (second session).
No Kerberos AS-REQ for Administrator from the DC is visible at this time (the KDC is local to the DC, so the RDP logon is not visible on the wire); no KDC_ERR_PREAUTH_FAILED error exists in the whole capture (eve_krb5: only KDC_ERR_PREAUTH_REQUIRED errors).

Analysis

Refines F-003: the actual password-guessing phase lasted ~21 s with ~95 failed attempts against the account named in the cookie (Administrator), and the password was found at ~02:21:47 UTC (≈ 04:21:47 local, the raw EVE carries +0200). This implies a weak/dictionary password on the built-in domain Administrator (RID 500 of C137.LOCAL, an account exempt from lockout by default, which explains why no lockout interrupted the attack). The RDP cookie only proves the username the client sent; the account actually authenticated is inferred from the cookie plus the subsequent Kerberos activity of 'Administrator' from the DC towards the workstation at 02:36:24 (F-008). Host confirmation (Security 4625/4624 type 10/3, 4776 or TerminalServices 1149/21/25 on CITADEL-DC01) is not available in the case. Alternative explanation for flow (3): a successful NLA by another account cannot be excluded on network data alone, but all 99 negotiation records carry 'Administrator'.

Evidence (7)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:21:33.618634	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40112 -> 10.42.85.10:3389 rdp state=closed bytes=2613/2257	suricata/case001.pcap/ eve_flow.jsonl#15690	case001.pcap
2020-09-19 02:21:47.166264	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40234 -> 10.42.85.10:3389 rdp state=closed bytes=6373/5222	suricata/case001.pcap/ eve_flow.jsonl#32271	case001.pcap
2020-09-19 02:22:08.311754	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40236 -> 10.42.85.10:3389 rdp state=closed bytes=28260/62729	suricata/case001.pcap/ eve_flow.jsonl#19321	case001.pcap
2020-09-19 02:22:36.664454	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40238 -> 10.42.85.10:3389 rdp state=closed bytes=4802218/12756305	suricata/case001.pcap/ eve_flow.jsonl#30035	case001.pcap
2020-09-19 02:21:26.114786	Network events (Suricata)	network event rdp TCP 10.42.85.10:3389 -> 194.61.24.102:40044	suricata/case001.pcap/ eve_rdp.jsonl#3	case001.pcap
2020-09-19 02:19:26.619607	Network events (Suricata)	network event rdp TCP 10.42.85.10:3389 -> 194.61.24.102:38100	suricata/case001.pcap/ eve_rdp.jsonl#1	case001.pcap
2020-09-19 02:19:26.798289	Network flows (Suricata)	network flow: start 10.42.85.10	suricata/case001.pcap/ eve_flow.jsonl#455	case001.pcap

UTC	Artifact	Detail	Reference	Original source
		TCP 194.61.24.102:38106 -> 10.42.85.10:3389 tls state=closed bytes=723/134		

F-015: Domain Administrator AS-REP encrypted with RC4-HMAC (all other principals AES256) – Administrator account likely lacks AES keys / legacy password

Severity	**Low**
Confidence	possible
Independent sources	network
Category	credential access
MITRE ATT&CK	T1558, T1078.002
Period (UTC)	2020-09-18 21:59:58.148531 → 2020-09-19 04:08:15.868598

Observed facts

eve_krb5 #57-#60, 02:36:24.912-.913 UTC: AS-REQ (cname Administrator, sname krbtgt/C137.LOCAL) from DESKTOP-SDN1RPT → KDC_ERR_PREAUTH_REQUIRED (#58) → second AS-REQ with pre-authentication (#59) → AS-REP (#60) with krb5_encryption = rc4-hmac (Suricata flag weak_encryption=true) while the TGT itself is aes256-cts-hmac-sha1-96. Every other AS-REP in the capture (DESKTOP-SDN1RPT$ #4, mortysmith #38, ricksanchez #78) and every TGS-REP, including those of Administrator (#52, #54, #62, #64, #66, #68), uses aes256-cts-hmac-sha1-96. The AS exchange follows the normal two-step preauth pattern issued by the Windows 10 client during the RDP logon from the DC (eve_rdp #297-#299).

Analysis

The AS-REP enc-part is encrypted with the client's long-term key; a Windows 10 client offers AES first, so the KDC selecting RC4 for Administrator only (from the same workstation that obtained AES for mortysmith and ricksanchez) most likely means the built-in Administrator account has no AES keys stored – i.e. its password has not been changed since the domain/forest was raised to 2008+ functional level, consistent with a weak, never-rotated password that was brute-forced (F-013). Alternative (less likely here): the client deliberately offered only RC4 (overpass-the-hash with an NT hash) – unlikely because this logon is a standard interactive RDP/CredSSP logon with the cleartext password just guessed by the attacker. Not an attack indicator by itself, but a hardening issue (RC4 usable for Kerberoasting/AS-REP cracking of this account). Host confirmation (msDS-SupportedEncryptionTypes, pwdLastSet of Administrator, 4768 TicketEncryptionType 0x17) not available.

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:36:24.913767	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50695 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#60	case001.pcap
2020-09-19 02:36:24.912570	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50694 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#58	case001.pcap
2020-09-19 02:36:24.913620	Network events (Suricata)	network event krb5 TCP 10.42.85.10:88 -> 10.42.85.115:50695	suricata/case001.pcap/ eve_krb5.jsonl#59	case001.pcap
2020-09-18 21:59:58.148531	Network events (Suricata)	network event krb5 TCP 10.42.85.115:49710 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#38	case001.pcap
2020-09-19 04:08:15.868598	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50949 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#78	case001.pcap

F-014: Per-account authentication summary reconstructed from Kerberos/SMB/LDAP on the wire (no Security EVTX/SAM available)

Severity	**Informational**
Confidence	confirmed
Independent sources	network
Category	context
MITRE ATT&CK	T1078.002

Period (UTC)	2020-09-18 21:59:39.801104 → 2020-09-19 04:08:15.872164

Observed facts

Kerberos exchanges between DESKTOP-SDN1RPT (10.42.85.115) and KDC CITADEL-DC01 (10.42.85.10), realm C137.LOCAL (eve_krb5, 84 records; note: Suricata labels the src/dst of KRB records inversely):
- DESKTOP-SDN1RPT$ (machine account): AS-REQ/AS-REP 21:59:39-21:59:47 UTC (boot), TGS for ldap/cifs/host SPNs; aes256-cts-hmac-sha1-96.
- mortysmith: AS-REQ 21:59:58.14 (after KDC_ERR_PREAUTH_REQUIRED), AS-REP 21:59:58.148 (aes256), TGS host/desktop-sdn1rpt.c137.local (=interactive logon on the workstation), LDAP 22:00:19, cifs+krbtgt 22:10:08. Pre-attack, legitimate user of the workstation.
- Administrator: AS-REQ x2 02:36:24.912/.913, AS-REP 02:36:24.913 (session key etype rc4-hmac, ticket aes256), TGS host/desktop-sdn1rpt (02:36:24.915), LDAP, cifs/CITADEL-DC01, krbtgt, ProtectedStorage/CITADEL-DC01 (02:36:26.768, DPAPI backup key), ldap 02:39:10. Coincides with the RDP DC→workstation session (eve_rdp #297-299, 02:35:55-02:36:23).
- ricksanchez: AS-REQ 04:08:15.862, AS-REP 04:08:15.868/.887 (aes256), TGS host/desktop-sdn1rpt 04:08:15.872, LDAP/cifs/krbtgt 04:08:16. No RDP at that time (console logon).
Failures: the only KRB_ERROR code in the capture is KDC_ERR_PREAUTH_REQUIRED (normal first-step error), for the machine account, mortysmith and ricksanchez. No KDC_ERR_PREAUTH_FAILED, KDC_ERR_CLIENT_REVOKED (lockout) or KDC_ERR_C_PRINCIPAL_UNKNOWN: no password spraying/guessing against Kerberos, no lockout.
SMB2 SESSION_SETUP: 24 sessions, all STATUS_SUCCESS, all Kerberos (realm C137.LOCAL, SPN cifs/CITADEL-DC01[.C137.local]); no NTLMSSP authentication observed. LDAP: 95 binds, all SASL GSS-SPNEGO with result success, no simple bind, no failure. DCERPC interfaces seen (step 6): EPM, DRSUAPI, LSARPC, NETLOGON only – no SAMR traffic on the wire, therefore no remote account creation/group modification (4720/4728/4732 equivalents) observed between the hosts.

Analysis

Account usage chronology: mortysmith (legitimate user, 21:59:58) → domain Administrator compromised by RDP brute force on the DC (F-013, 02:21:47) and reused from the DC to RDP into the workstation (02:36:24, first profile creation – F-008) → ricksanchez console logon at 04:08:15 (legitimate user; triggered Meterpreter re-launch F-009/F-012). Kerberos with aes256 for all users except the Administrator AS-REP session key (see separate finding). Blind spots: any account created or added to Administrators locally on the DC (where the attacker was interactive and SAMR calls would be local), local SAM accounts of either host, 4672/4648 privileged/explicit-credential logons and the DC's own logons (local KDC) are invisible on the wire. Absence of SAMR on the wire is not evidence that no account was created.

Evidence (12)

UTC	Artifact	Detail	Reference	Original source
2020-09-18 21:59:58.142475	Network events (Suricata)	network event krb5 TCP 10.42.85.10:88 -> 10.42.85.115:49709	suricata/case001.pcap/ eve_krb5.jsonl#27	case001.pcap
2020-09-18 21:59:58.148531	Network events (Suricata)	network event krb5 TCP 10.42.85.115:49710 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#38	case001.pcap
2020-09-18 21:59:58.149910	Network events (Suricata)	network event krb5 TCP 10.42.85.115:49711 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#36	case001.pcap
2020-09-19 02:36:24.912466	Network events (Suricata)	network event krb5 TCP 10.42.85.10:88 -> 10.42.85.115:50694	suricata/case001.pcap/ eve_krb5.jsonl#57	case001.pcap
2020-09-19 02:36:24.913767	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50695 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#60	case001.pcap
2020-09-19 02:36:26.768288	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50712 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#54	case001.pcap
2020-09-19 04:08:15.862233	Network events (Suricata)	network event krb5 TCP 10.42.85.10:88 -> 10.42.85.115:50948	suricata/case001.pcap/ eve_krb5.jsonl#69	case001.pcap
2020-09-19 04:08:15.868598	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50949 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#78	case001.pcap
2020-09-19 04:08:15.872164	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50950 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#80	case001.pcap

UTC	Artifact	Detail	Reference	Original source
2020-09-18 21:59:39.801104	Network events (Suricata)	network event krb5 TCP 10.42.85.115:49677 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#20	case001.pcap
2020-09-18 21:59:40.017751	Network events (Suricata)	network event smb TCP 10.42.85.115:49683 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#3	case001.pcap
2020-09-19 02:36:24.989968	Network events (Suricata)	network event smb TCP 10.42.85.115:50701 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#144	case001.pcap

4.8 Remote access and lateral movement

The DC was controlled remotely from the Internet, and the attacker then used it as a pivot to reach the workstation over RDP. RDP was the only remote-control channel seen on the wire, apart from the Meterpreter C2 already covered (F-005/F-010). New finding F-016 (high/probable), timeline event T-017, one note.

How this was checked (network data only, case001.pcap):
- Listed every internal flow pair and every flow on admin ports (445, 3389, 5985/5986, 135, 139, 22, 5900, 23).
- Read all eve_rdp negotiation records, the SMB tree connects, named pipes and file opens, the DCERPC interfaces, and the Kerberos service tickets.
- Searched DNS and TLS server names for RMM / remote-access tools.

Results:
*1. **Internet → DC (inbound RDP):** the RDP sessions from 194.61.24.102 to the DC used the cookies 'nmap' (1) and 'Administrator' (99), and always negotiated 'hybrid' (CredSSP). That fits the non-Windows client already fingerprinted in F-011.*
*2. **DC → workstation (lateral movement):** the RDP hop from 10.42.85.10 to 10.42.85.115:3389 ran from 02:35:55 to 02:52:14 UTC.*
- It negotiated 'hybrid_ex', sent no cookie, and added an RDP-UDP channel (UDP flow #31428, 6.7 MB to the client). That is typical of the native Windows mstsc.exe running inside the attacker's session on the DC.
- The workstation server also advertised restricted_admin, which does not show that mode was used.
- The Kerberos ticket for Administrator on the workstation was issued at 02:36:24.
- This refines F-006.
*3. **No other remote-execution or admin traffic:***
- No SVCCTL, task-scheduler (ATSVC), WMI/DCOM or registry (WINREG) RPC.
- No ADMIN$/C$ share access and no PSEXESVC pipe.
- No WinRM (5985/5986), SSH, VNC or Telnet.
- The DC opened no SMB/RPC connection to the workstation.
- No AnyDesk/TeamViewer/ScreenConnect/RustDesk/Splashtop/Atera/ngrok or similar tool in DNS or TLS.
*4. **Shares used:** only IPC$, sysvol and NETLOGON (Administrator profile creation at 02:36:24), plus FileShare after ricksanchez logged on at 04:08:16. On FileShare there was only Explorer browsing (Desktop.ini/AutoRun.inf), so most likely the user's normal drive mapping.*
*5. **Ruled out (note):** 10.90.90.90 is a DNS-blocklist sinkhole (its certificate reads 'dnsbl@example.com') and 192.168.45.1 is the DC's DNS forwarder; neither is a lateral-movement peer. The capture shows no other internal hosts.*

F-016: Remote-access / lateral-movement channel inventory: RDP is the only remote-control channel (attacker→DC via non-Windows CredSSP client; DC→workstation via native mstsc with RDP-UDP); no PsExec/WinRM/WMI/admin$/RMM traffic

Severity	**High**
Confidence	probable
Independent sources	network
Category	lateral movement
MITRE ATT&CK	T1021.001, T1133, T1078.002, T1570
Period (UTC)	2020-09-19 02:21:26.114786 → 2020-09-19 04:09:52.725024

Observed facts

Facts (case001.pcap, Suricata 8.0.7; raw EVE timestamps are +0200, converted to UTC):
1) Inbound RDP to CITADEL-DC01 10.42.85.10:3389 from 194.61.24.102. There is 1 X.224 request with cookie 'nmap' at 02:19:26 (eve_rdp #1/#2). Then 99 X.224 requests with cookie 'mstshash=Administrator' between 02:21:26 and 02:56:03 (eve_rdp #3/#4/#5), and the server always selected protocol 'hybrid' (CredSSP), never 'hybrid_ex'. No RDP flow to the

workstation from the Internet, and no outbound RDP from either internal host to the Internet.
2) Internal RDP from CITADEL-DC01 (10.42.85.10:62514) to DESKTOP-SDN1RPT (10.42.85.115:3389). X.224 request with an empty cookie at 02:35:55.347 (eve_rdp #297). The server selected 'hybrid_ex' and advertised extended_client_data, dynvc_gfx, restricted_admin and redirected_authentication (eve_rdp #298). The TLS handshake came 28 s later at 02:36:23.467, and the workstation certificate serial was 58b3b0ac0fd388b54c1f2528cd8052b3 (eve_rdp #299). Two flows carried the session:
- TCP flow #29416: 02:35:55–02:52:14, 875,256 B to server / 452,944 B to client.
- UDP 3389 flow #31428 (RDP-UDP transport, app_proto 'failed'): from 02:36:23.5, 283,842 B to server / 6,695,207 B to client, ending at the same instant, 02:52:14.487.

This is the only DC→workstation session on any admin port. On the workstation the Kerberos host/desktop-sdn1rpt TGS for Administrator was issued at 02:36:24 (krb5 #52).
3) All internal flow pairs (10.42.85.0/24) were enumerated. Apart from the RDP above, the only traffic is workstation→DC: DNS 53, LDAP 389, Kerberos 88, EPM 135 (40), dynamic RPC 49155/49158, SMB 445 (23), NTP 123, plus NetBIOS broadcasts. The DC opened no SMB/135/5985/5986 connection to the workstation, and there were no WinRM 5985/5986, SSH, VNC or Telnet flows.
4) DCERPC interfaces seen: EPM e1af8308, DRSUAPI e3514235, NETLOGON 12345678-...-cffb and LSARPC 12345778. No SVCCTL, ATSVC/ITaskSchedulerService, WMI/DCOM IRemoteActivation or WINREG.
5) SMB tree connects (all workstation→DC): \\CITADEL-DC01.C137.local\IPC$ (18), \sysvol (4), \\CITADEL-DC01\IPC$ and \NETLOGON at 02:36:24 (eve_smb #145, #149; Administrator profile creation), and \\CITADEL-DC01\FileShare at 04:08:16 (eve_smb #188) after ricksanchez logged on. On FileShare the only accesses were Explorer-style probes (Desktop.ini, AutoRun.inf, <share_root>) at 04:09:52, 04:30:24 and 04:31:14, with desktop.ini at 04:49, 05:06 and 05:09 (eve_smb #191–193, #205–211, #257, #269, #270). No ADMIN$/C$ connection, no PSEXESVC or other service pipe, and no file write/transfer of an executable.
6) There were no DNS queries or TLS SNI for remote-access/RMM products (AnyDesk, TeamViewer, ScreenConnect/ConnectWise, RustDesk, Splashtop, Atera, LogMeIn, ngrok, Kaseya, GoTo, BeyondTrust, DWService, Supremo, Ammyy, Radmin, VNC): 0 rows in suricata_dns and suricata_tls.

Analysis

Remote control of the environment ran through two channels. The first was interactive RDP. The second was the Meterpreter C2 already covered in F-005/F-010. The RDP protocol negotiation tells the two RDP hops apart:
- The Internet→DC sessions always negotiated 'hybrid' with an explicit mstshash cookie. This matches the OpenSSL/non-Windows client fingerprint in F-011 (e.g. xfreerdp-like tooling), used for both the brute force and the interactive session.
- The DC→workstation hop negotiated 'hybrid_ex', sent no cookie and opened an RDP-UDP side channel. This is typical of the native Windows mstsc.exe running inside the attacker's interactive session on the DC.

So the DC served as the pivot: the attacker reached DESKTOP-SDN1RPT using credentials of the compromised domain Administrator (Kerberos TGS at 02:36:24, profile creation via NETLOGON). The 28 s between the X.224 negotiation and the TLS/CredSSP handshake is consistent with a human entering credentials or accepting a prompt, but this is interpretation only.
The server advertising restricted_admin does not show that Restricted Admin mode was used.
No PsExec-style, WMI, WinRM, scheduled-task or service-based remote execution and no third-party RMM tool are visible on the wire during the capture window. That is absence of network evidence only: traffic encrypted inside the Meterpreter channel (e.g. pivoting, port forwarding, credential use) and activity outside 21:58–05:38 UTC are invisible.
The FileShare access after ricksanchez logged on looks like normal user/Explorer browsing (probably a drive mapping at logon), not attacker staging. A Meterpreter session was also live on the workstation at that time, so it cannot be fully excluded.
Host artefacts (TerminalServices-LocalSessionManager 21/24/25, RemoteConnectionManager 1149, Security 4624 type 10, Default.rdp/MRU on the DC, bitmap cache) are not in the case and would be needed to confirm accounts and session IDs.

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:21:26.114786	Network events (Suricata)	network event rdp TCP 10.42.85.10:3389 -> 194.61.24.102:40044	suricata/case001.pcap/ eve_rdp.jsonl#3	case001.pcap
2020-09-19 02:21:26.114998	Network events (Suricata)	network event rdp TCP 194.61.24.102:40044 -> 10.42.85.10:3389	suricata/case001.pcap/ eve_rdp.jsonl#4	case001.pcap
2020-09-19 02:35:55.347240	Network events (Suricata)	network event rdp TCP 10.42.85.115:3389 -> 10.42.85.10:62514	suricata/case001.pcap/ eve_rdp.jsonl#297	case001.pcap
2020-09-19 02:35:55.417848	Network events (Suricata)	network event rdp TCP 10.42.85.10:62514 -> 10.42.85.115:3389	suricata/case001.pcap/ eve_rdp.jsonl#298	case001.pcap
2020-09-19 02:36:23.467377	Network events (Suricata)	network event rdp TCP 10.42.85.10:62514 -> 10.42.85.115:3389	suricata/case001.pcap/ eve_rdp.jsonl#299	case001.pcap
2020-09-19 02:35:55.285340	Network flows (Suricata)	network flow: start 10.42.85.115 TCP 10.42.85.10:62514 -> 10.42.85.115:3389 rdp state=closed bytes=875256/452944	suricata/case001.pcap/ eve_flow.jsonl#29416	case001.pcap
2020-09-19 02:36:23.515604	Network flows (Suricata)	network flow: start 10.42.85.115 UDP 10.42.85.10:63121 -> 10.42.85.115:3389 failed state=established bytes=283842/6695207	suricata/case001.pcap/ eve_flow.jsonl#31428	case001.pcap
2020-09-19 02:36:24.915473	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50696 -> 10.42.85.10:88	suricata/case001.pcap/ eve_krb5.jsonl#52	case001.pcap
2020-09-19 04:08:16.637781	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#188	case001.pcap
2020-09-19 04:09:52.725024	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#191	case001.pcap

4.9 User activity on files and folders

Host artefacts for user file and folder activity are not in the case. information_schema has only raw.pcap_files and the raw.suricata_* tables, so there is no LNK, Jump List, ShellBag, MRU, WordWheelQuery, Windows Timeline, SIDR, thumbcache, Notepad, MFT, UsnJrnl, Recycle Bin, or USB/EMDMgmt/setupapi data. Instead I rebuilt file activity from SMB2 and HTTP on the wire:
- All SMB CREATE/READ/WRITE records and all SMB fileinfo records were reviewed.
- Sysvol GPO reads are routine, with constant MD5s.
- The NETLOGON 'Default User.V6' lookup and the protected_storage pipe were already covered in F-008.
- The samr pipe writes at 22:10 are pre-attack and normal. This corrects step 7's statement that no SAMR was seen.

New finding F-017 (medium/possible), from the ricksanchez session on the workstation:
- The workstation connected to \\CITADEL-DC01\FileShare at 04:08:16 UTC and only enumerated the root (Desktop.ini/AutoRun.inf probes, 12.7 KB to the client in 90 min). No file was read or written on the share.
- The share root's modified/changed time was 03:34:18 UTC. No SMB client was active then, and the DC Meterpreter session was live, so the directory was changed locally on the DC. Whether the attacker did it cannot be determined from the pcap (T-018).

Pre-attack browsing by mortysmith around 22:00 UTC is benign context (note recorded).

F-017: FileShare activity on CITADEL-DC01 seen over SMB2: workstation browsed the root only, no file read or write; share root last modified 2020-09-19 03:34:18 UTC, during the attacker's C2 presence on the DC

Severity	**Medium**
Confidence	possible
Independent sources	network
Category	collection / user file activity
MITRE ATT&CK	T1039, T1135, T1005
Period (UTC)	2020-09-18 22:10:08.218493 → 2020-09-19 04:49:47.678833

Observed facts

No host file-activity artefacts exist in the case. information_schema lists only raw.pcap_files plus the raw.suricata_* tables: no LNK, Jump Lists, ShellBags, MRU, MFT, UsnJrnl, RecycleBin, USB or thumbcache data. So file activity can only be rebuilt from SMB2 on the wire (case001.pcap).

Observed facts (UTC):
- SMB2 flow_id 254243920305440, 10.42.85.115:50957 -> 10.42.85.10:445 (dialect 3.02, session 65970764775533). It started at 04:08:16 and was still established at the end of the capture (05:38:57). It carried only 15,650 B to the server and 12,757 B to the client (eve_flow #17430).
- 04:08:16.637 TREE_CONNECT to \\CITADEL-DC01\FileShare (share type FILE, tree 1) succeeded (eve_smb #188). This is 1 s after ricksanchez's Kerberos logon.
- Explorer-style probes:
  - 04:09:52, 04:30:24 and 04:31:14: CREATE Desktop.ini and AutoRun.inf returned STATUS_OBJECT_NAME_NOT_FOUND (#191, #192, #205, #206, #209, #210).
  - 04:49:47, 05:06:38 and 05:09:16: CREATE desktop.ini returned not found (#257, #269, #270).
- 04:09:53, 04:30:24 and 04:31:14: CREATE <share_root> succeeded (#193, #207, #211). The root directory's MACB values never changed: created 1600404491 = 2020-09-18 04:48:11 UTC; modified/changed/accessed 1600486458 = 2020-09-19 03:34:18 UTC.
- None of the 370 eve_smb records shows an SMB2 READ or WRITE on a named file in FileShare. No eve_fileinfo record exists for that share (the 18 SMB fileinfo records are all sysvol GPO files with constant MD5s). The only SMB2 WRITEs in the capture are 16 writes on the 'samr' named pipe at 22:10:08 UTC (eve_smb #55-#71), before the attack.
- No client-to-DC:445 SMB flow is captured between 03:25 and 03:45 UTC. The DC's only traffic there is NetBIOS/LLMNR broadcasts at 03:36:04-03:37:25.

Analysis

Interpretation:
1. The workstation session (ricksanchez, starting 04:08) only listed the FileShare root. Desktop.ini/AutoRun.inf probes and about 12 KB of server data over 90 min fit directory enumeration in Explorer. No file content was pulled from the share over SMB, so no SMB-based staging or exfiltration from the workstation is visible.
2. The FileShare root directory was last modified/changed at 03:34:18 UTC. On NTFS this means an entry directly under the root was created, deleted or renamed at that moment. No SMB client caused it from the captured segment. At that time no interactive RDP session to the DC was active (the attacker's last RDP ended at 02:57:41), but the DC Meterpreter session started at 02:56:38 was still established. So the change was made locally on the DC: possibly by the attacker through Meterpreter (dropping, renaming or deleting a file in the share), or by a legitimate local process or user. This is a hypothesis: the pcap cannot show which entry changed.
3. The 22:10 samr pipe writes come from the workstation before the attack, in the normal domain-member context. They correct the step-7 statement that no SAMR was seen, but they are not malicious.
Verification needs host data from CITADEL-DC01: $MFT/$UsnJrnl entries under the FileShare path around 03:34:18 UTC, and Security 5145/4663 events if auditing was on.

Evidence (7)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 04:08:16.637781	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#188	case001.pcap
2020-09-19 04:09:53.219790	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#193	case001.pcap
2020-09-19 04:31:14.843891	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#211	case001.pcap
2020-09-19 04:09:52.725024	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#191	case001.pcap
2020-09-19 04:49:47.678833	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#257	case001.pcap
2020-09-19 04:08:16.583483	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 10.42.85.115:50957 -> 10.42.85.10:445 smb state=established bytes=15650/12757	suricata/case001.pcap/ eve_flow.jsonl#17430	case001.pcap
2020-09-18 22:10:08.218493	Network events (Suricata)	network event smb TCP 10.42.85.115:50742 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#55	case001.pcap

4.10 Web browsing, downloads and e-mail

The case has no browser, MFT or Zone.Identifier tables, so I rebuilt web activity from Suricata HTTP, DNS, TLS SNI and fileinfo records in case001.pcap.

New finding F-018 (high/probable): coreupdater.exe was delivered by hand-typed web browsing, with no search, phishing or e-mail step.

DC:
- 02:23:09: IE11 started for the first time in the attacker's RDP session.
- 02:23:41: GET http://194.61.24.102/, a 228 B landing page (SHA256 6978f8fa...2d77, IOC I-010).
- 02:24:06: /coreupdater.exe downloaded with that page as Referer.
- The DC made no SmartScreen lookups.

Workstation:
- 02:39:04: Edge started for the first time in the new Administrator profile.
- 02:39:26: same landing page loaded, with a SmartScreen nav check at 02:39:27.
- 02:39:58: download with the same Referer, followed by a download-reputation check at 02:40:01.
- 02:40:45 and 02:40:49.938: SmartScreen checkappexec lookups. The second is 2 ms after the Meterpreter connection to 203.78.103.109:443. This is strong network evidence that the file was launched from the shell, which supports F-007.

Also checked:
- No SMTP/IMAP/POP3 traffic and no webmail or cloud-storage use during the attack window.
- Browsing by mortysmith before the attack (Bing images, CNN, Office, Discord) is benign, as are checkappexec lookups at logon. Both are recorded in a note.

Added timeline events T-019 and T-020.

F-018: Web origin of coreupdater.exe: attacker manually browsed to http://194.61.24.102/ with IE11 (DC) and Edge (workstation); SmartScreen app-reputation checks mark the user-launched execution on the workstation at 02:40:45 UTC

Severity	**High**
Confidence	probable
Independent sources	network
Category	initial access / delivery
MITRE ATT&CK	T1105, T1204.002, T1608.001, T1071.001
Period (UTC)	2020-09-19 02:23:09.692112 → 2020-09-19 02:40:49.938222

Observed facts

No browser artefact exists in the case (no raw.chromium_*, firefox_*, hindsight, dissect_browser_*, dissect_iexplore_history, mft or Zone.Identifier tables). The browsing was rebuilt from Suricata HTTP/DNS/TLS/fileinfo in case001.pcap.
CITADEL-DC01 (10.42.85.10), inside the attacker's RDP session #1:
- 02:23:09.69: IE11 64-bit starts for the first time. GET www.bing.com/favicon.ico, UA 'Windows NT 6.3; Win64; x64; Trident/7.0; rv:11.0' (eve_http#125).
- 02:23:10: TLS go.microsoft.com and ieonline.microsoft.com, JA3 8de139912912d62c9e27d022ab05e070 (eve_tls#16172, #16174). go.microsoft.com/browserconfig.xml returns 404 at 02:23:34 (eve_http#129).
- 02:23:41.73: GET http://194.61.24.102/, 200 text/html, 228 B, MD5 614645af648513231c283090faa5c469, SHA256 6978f8fa2bf5afbe1a074401b923ead519ce89471d83376f5cfda50c12bc2d77 (eve_http#117, eve_fileinfo#98). Then /favicon.ico returns 404 (eve_http#118). UA is now 'WOW64; Trident/7.0' (32-bit IE tab).
- 02:24:06.94: GET /coreupdater.exe with Referer http://194.61.24.102/, 200 application/x-msdos-program, 7168 B, SHA256 10f3b920...1cfda6 (eve_http#116, eve_fileinfo#100).
- No DNS lookup for any search engine query or for the IP: the URL was typed directly. No SmartScreen/checkappexec traffic from the DC.

DESKTOP-SDN1RPT (10.42.85.115), Administrator session opened over RDP from the DC at 02:36:24:
- 02:39:04: first launch of Edge (Chromium) in the new profile. DNS microsoftedgewelcome.microsoft.com, www.msn.com and others (eve_dns#1885); TLS nav.smartscreen.microsoft.com (eve_tls#16210).
- 02:39:26.94: GET http://194.61.24.102/ returns the same 228 B page, SHA256 6978f8fa… (eve_http#154, eve_fileinfo#135). At 02:39:27.07 there is a nav.smartscreen URL-reputation lookup (eve_tls#16283).
- 02:39:46.5: TLS wdcp.microsoft.com, Defender cloud protection (eve_tls#16337).
- 02:39:58.41: GET /coreupdater.exe with Referer http://194.61.24.102/, Edge UA 'Windows NT 10.0; Win64; x64 … AppleWebKit/537.36', 7168 B, same SHA256 (eve_http#156, eve_fileinfo#136).
- 02:40:01.5: nav.smartscreen, the download-reputation check (eve_tls#16363).
- 02:40:45.79: checkappexec.microsoft.com, the SmartScreen app-reputation check that runs when a file carrying Mark-of-the-Web is launched from the shell (eve_tls#16331).
- 02:40:49.938: a second checkappexec (eve_tls#16370). This is 2 ms after the outbound Meterpreter connection to 203.78.103.109:443 at 02:40:49.936 (see F-005/F-007).
No SMTP/IMAP/POP3 flows (ports 25/465/587/110/143/993/995) and no webmail or cloud-storage TLS SNI were seen during the attack. The only file-sharing/chat traffic was Discord at 22:05, before the attack.

Analysis

Delivery vector, interpreted: the attacker hosted a minimal landing page (228 B, most likely a directory listing or a single link) and the payload on the same server they brute-forced from (194.61.24.102). From each interactive session they downloaded it by hand with the native browser: IE11 on the DC, then Edge on the workstation after the RDP hop. The Referer header confirms the download came from a click on that page. There was no phishing, no e-mail and no search-engine step. On the workstation, the SmartScreen sequence fits the normal Windows 10 chain: navigation check → download check → checkappexec at execution. It independently suggests that the user (attacker as Administrator) launched coreupdater.exe from Explorer or the Edge download bar at about 02:40:45. The file most likely ran under its MOTW and was allowed, since C2 started 4 s later. This strengthens F-007 for the workstation, but it is still a network-only inference. Without Edge History/Downloads, Prefetch or Zone.Identifier, the on-disk path cannot be given (probably C:\Users\Administrator\Downloads\coreupdater.exe on the workstation, and the IE default Downloads folder of the Administrator profile on the DC). Logon-time checkappexec lookups (22:00:15, 02:36:31, 04:08:34, each about 7-20 s after a logon) are baseline shell behaviour and are not attributed to the payload. Only one source (pcap), so the confidence is probable.

Evidence (14)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:23:09.692112	HTTP (Suricata)	HTTP request /favicon.ico http_hostname=www.bing.com | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 6.3; Win64; x64; Trident/7.0; rv:11.0) like Gecko | http_http_content_type=image/x-icon | http_length=237 | src_ip=10.42.85.10 | dest_ip=204.79.197.200	suricata/case001.pcap/ eve_http.jsonl#125	case001.pcap
2020-09-19 02:23:10.334278	TLS (Suricata)	TLS handshake ieonline.microsoft.com tls_sni=ieonline.microsoft.com | dest_ip=204.79.197.200 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=www.bing.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 2 | tls_ja3_hash=8de139912912d62c9e27d022ab05e070 | tls_ja3s_hash=2b90c93096725457ad0cf004d948ed32 | tls_ja4=t12d2	suricata/case001.pcap/ eve_tls.jsonl#16174	case001.pcap
2020-09-19 02:23:41.734776	HTTP (Suricata)	HTTP request / http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; rv:11.0) like Gecko | http_http_content_type=text/html | http_length=228 | src_ip=10.42.85.10 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#117	case001.pcap

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:24:06.940063	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; rv:11.0) like Gecko | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.10 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#116	case001.pcap
2020-09-19 02:23:41.735412	Files transferred (Suricata)	file transferred over the network / app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.10 | fileinfo_size=228 | fileinfo_sha256=6978f8fa2bf5afbe1a074401b923ead519ce89471d83376f5cfda50c12bc2d77 | fileinfo_md5=614645af648513231c283090faa5c469 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#98	case001.pcap
2020-09-19 02:39:04.440228	DNS (Suricata)	DNS query / answer [{"rrname": "microsoftedgewelcome.microsoft.com", "rrtype": "A"}] dns_type=request | dns_rcode=NOERROR | src_ip=10.42.85.115 | dest_ip=10.42.85.10	suricata/case001.pcap/ eve_dns.jsonl#1885	case001.pcap
2020-09-19 02:39:26.940438	HTTP (Suricata)	HTTP request / http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=text/html | http_length=228 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#154	case001.pcap
2020-09-19 02:39:27.067445	TLS (Suricata)	TLS handshake nav.smartscreen.microsoft.com tls_sni=nav.smartscreen.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=smartscreen.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 1 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=986571066668055ae9481cb84fda6	suricata/case001.pcap/ eve_tls.jsonl#16283	case001.pcap
2020-09-19 02:39:58.411571	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#156	case001.pcap
2020-09-19 02:40:01.538456	TLS (Suricata)	TLS handshake nav.smartscreen.microsoft.com tls_sni=nav.smartscreen.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=ae4edc6faf64d08308082ad26be60767 | tls_ja4=t12d1909h2_d83cc789557e_7af1ed941c26	suricata/case001.pcap/ eve_tls.jsonl#16363	case001.pcap
2020-09-19 02:40:45.794820	TLS (Suricata)	TLS handshake checkappexec.microsoft.com tls_sni=checkappexec.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=smartscreen.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 1 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=986571066668055ae9481cb84fda634a	suricata/case001.pcap/ eve_tls.jsonl#16331	case001.pcap
2020-09-19 02:40:49.938222	TLS (Suricata)	TLS handshake checkappexec.microsoft.com tls_sni=checkappexec.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=smartscreen.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 1 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=986571066668055ae9481cb84fda634a	suricata/case001.pcap/ eve_tls.jsonl#16370	case001.pcap
2020-09-19 02:39:46.509658	TLS (Suricata)	TLS handshake wdcp.microsoft.com	suricata/case001.pcap/ eve_tls.jsonl#16337	case001.pcap

UTC	Artifact	Detail	Reference	Original source
		tls_sni=wdcp.microsoft.com | dest_ip=104.40.17.153 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft Corporation, CN=wdcp.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, CN=Microsoft Secure Server CA 2011 | tls_ja3_hash=28a2c9bd18a11de089ef		
2020-09-19 02:39:58.412735	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.115 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#136	case001.pcap

4.11 Defense evasion and anti-forensics

Defense evasion was reviewed in a network-only case. The host checks this step asks for cannot be done: there is no EVTX (1102/104/4616/Defender 5001-5012), no MFT/$J/$LogFile (timestomping, mass deletion, wiping tools), no Defender exclusions table, no registry and no memory image. On the wire:
(1) New finding F-019 (medium/probable): masquerading. The stager is named coreupdater.exe, and Meterpreter C2 runs as raw non-TLS TCP on port 443 with no DNS resolution of the attacker IPs. The stage is loaded reflectively in memory (T1036.005/T1571/T1620).
(2) New finding F-020 (low/possible): the only Defender MAPS (wdcp.microsoft.com) lookup came from the workstation at 02:39:46 UTC, during the download. Defender was therefore active but did not stop the C2 at 02:40:49. Whether the stager went undetected or the attacker tampered with Defender (T1562.001) cannot be determined. The DC shows no AV cloud traffic. Telemetry on the workstation kept running until the end of the capture. Timeline event added for 02:39:46.
(3) No SMB access to log files, Defender files or admin shares. The GPO files were unchanged (step 6).
(4) Capture continuity: a 2 h 24 min pre-attack gap (22:10:48 to 00:34:48 UTC) is recorded as a note and is not attacker activity. The attack window is covered continuously. NTP shows nothing anomalous.

F-019: Defense evasion by masquerading: Meterpreter stager named 'coreupdater.exe' and C2 over raw (non-TLS) TCP on port 443

Severity	**Medium**
Confidence	probable
Independent sources	network
Category	defense-evasion
MITRE ATT&CK	T1036.005, T1571, T1620
Period (UTC)	2020-09-19 02:24:06.940653 → 2020-09-19 02:29:49.290306

Observed facts

Facts (case001.pcap, UTC): the PE delivered from http://194.61.24.102/coreupdater.exe (7168 B, SHA256 10f3b920… 1cfda6) uses a generic 'updater' name (eve_fileinfo.jsonl#100). The five C2 sessions it triggers go to 203.78.103.109 on dest_port 443 but carry no TLS handshake (the eve_tls table has no record for these flows; Suricata alert 'PE over raw TCP' / Meterpreter stage, eve_alert.jsonl#439; flow eve_flow.jsonl#28844, DC, start 02:25:18; eve_flow.jsonl#26788, workstation, start 02:40:49). The attacker IPs were never resolved through DNS (step 5).

Analysis

Interpretation: the attacker uses a benign-sounding file name (T1036.005, match legitimate name) and the HTTPS port so that egress filtering and casual review treat the traffic as normal web traffic (T1571 non-standard protocol on a standard port; the protocol is raw Meterpreter TLV, not TLS). This is basic, low-sophistication evasion. A Meterpreter stage arrives as an in-memory reflective DLL (T1620), so it most likely left no file on disk beyond the 7 KB stager. That cannot be verified without host or memory data. Confidence is probable because it rests on network behaviour only.

Evidence (4)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:24:06.940653	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.10 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#100	case001.pcap
2020-09-19 02:25:19.071794	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.10:62414 | A Network Trojan was detected | 1:AEtaQJ4Pfkk0lfr1mrIy2qsaW8Q= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#439	case001.pcap
2020-09-19 02:25:18.565676	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62414 -> 203.78.103.109:443 failed state=closed bytes=51044/932653	suricata/case001.pcap/ eve_flow.jsonl#28844	case001.pcap
2020-09-19 02:29:49.290306	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62476 -> 203.78.103.109:443 failed state=closed bytes=41589/583890	suricata/case001.pcap/ eve_flow.jsonl#26788	case001.pcap

F-020: Windows Defender cloud protection (MAPS, wdcp.microsoft.com) contacted once by the workstation during the coreupdater.exe download; the stager still ran – Defender did not block it (cause undeterminable)

Severity	**Low**
Confidence	possible
Independent sources	network
Category	defense-evasion
MITRE ATT&CK	T1562.001
Period (UTC)	2020-09-19 02:29:49.290306 → 2020-09-19 02:40:01.538456

Observed facts

Facts: the only wdcp.microsoft.com TLS session in the whole capture comes from 10.42.85.115 to 104.40.17.153:443 at 2020-09-19 02:39:46.51 UTC (eve_tls.jsonl#16337; flow eve_flow.jsonl#28228, 02:39:46–02:41:36, 3023 B up / 5345 B down). It falls between the landing-page visit (02:39:26), the SmartScreen nav check (eve_tls.jsonl#16283, 02:39:27), the download (02:39:58) and the reputation check (eve_tls.jsonl#16363, 02:40:01). Meterpreter C2 began at 02:40:49 (eve_flow.jsonl#26788). The DC (Windows Server 2012 R2) made no SmartScreen or wdcp connection at all. Workstation telemetry (settings-win/v10.events.data.microsoft.com) continued until 05:38 UTC, so there is no network sign that telemetry or security services were disabled.

Analysis

Interpretation: the Defender MAPS lookup shows that Defender real-time and cloud protection were active on DESKTOP-SDN1RPT at 02:39 UTC. The 7 KB stager was not blocked, because the C2 session started 51 s later and lasted until the end of the capture. There are three possible explanations: (a) the stager was not detected (custom or encoded payload); (b) the attacker, running as Administrator, added an exclusion or turned off real-time protection (Set-MpPreference, registry), which would show as Defender 5001/5007 events; (c) the detection was only reported. Without the Defender operational log, the MpPreference registry or MPLog, none of these can be decided. No Defender lookup was seen for the logon-triggered relaunch at 04:08:45, which fits a cached verdict or an exclusion. The DC has no Defender-equivalent cloud traffic, which is normal for 2012 R2 without an AV product.

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
2020-09-19	TLS (Suricata)	TLS handshake	suricata/case001.pcap/	case001.pcap

UTC	Artifact	Detail	Reference	Original source
02:39:46.509658		wdcp.microsoft.com tls_sni=wdcp.microsoft.com | dest_ip=104.40.17.153 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft Corporation, CN=wdcp.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, CN=Microsoft Secure Server CA 2011 | tls_ja3_hash=28a2c9bd18a11de089ef	eve_tls.jsonl#16337	
2020-09-19 02:39:46.424507	Network flows (Suricata)	network flow: start 104.40.17.153 TCP 10.42.85.115:50863 -> 104.40.17.153:443 tls state=closed bytes=3023/5345	suricata/case001.pcap/ eve_flow.jsonl#28228	case001.pcap
2020-09-19 02:39:27.067445	TLS (Suricata)	TLS handshake nav.smartscreen.microsoft.com tls_sni=nav.smartscreen.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=smartscreen.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 1 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=986571066668055ae9481cb84fda6	suricata/case001.pcap/ eve_tls.jsonl#16283	case001.pcap
2020-09-19 02:40:01.538456	TLS (Suricata)	TLS handshake nav.smartscreen.microsoft.com tls_sni=nav.smartscreen.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=ae4edc6faf64d08308082ad26be60767 | tls_ja4=t12d1909h2_d83cc789557e_7af1ed941c26	suricata/case001.pcap/ eve_tls.jsonl#16363	case001.pcap
2020-09-19 02:29:49.290306	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62476 -> 203.78.103.109:443 failed state=closed bytes=41589/583890	suricata/case001.pcap/ eve_flow.jsonl#26788	case001.pcap

4.12 Collection, exfiltration and communications

Collection/exfiltration review (network only). I ranked all internal→external flows by bytes_toserver, reviewed POST/PUT in HTTP, all fileinfo transfers, DNS (length, distinct names, cloud-storage domains) and the RDP volumes. I found no bulk exfiltration, no archive transfer, no cloud-storage upload and no DNS tunnelling. Meterpreter C2 upload totals ≈0.50 MB across 5 sessions; the largest is the workstation Administrator session, 280 KB. The attacker RDP session sent 12.76 MB DC→attacker, which fits screen graphics; it is encrypted, so clipboard or drive redirection cannot be excluded. The OneDrive traffic (oneclient.sfx.ms, 39 MB down) is client setup for the new Administrator profile. New finding F-021, plus a noise note. Network IOCs (194.61.24.102, 203.78.103.109, URL) were already recorded.

F-021: Collection/exfiltration assessment: no bulk exfiltration on the wire; only Meterpreter C2 uploads (~0.68 MB total) and the attacker's RDP channel could have carried data

Severity	**Medium**
Confidence	probable
Independent sources	network
Category	exfiltration
MITRE ATT&CK	T1041, T1071, T1021.001
Period (UTC)	2020-09-19 00:44:00.322000 → 2020-09-19 04:08:45.483780

Observed facts

How this was checked: every outbound flow in case001.pcap (raw.suricata_flow, internal→external) was ranked by flow_bytes_toserver, both per destination and per flow. I also reviewed all non-GET HTTP requests, all fileinfo transfers, and DNS for long or tunnelling queries and for cloud-storage domains. Raw EVE timestamps are +0200; the times below are UTC.

(1) Meterpreter C2 to 203.78.103.109:443, bytes up / down:
- flow #28844 DC 02:25:18-02:29:36: 51,044 / 932,653
- #26788 DC 02:29:49-02:52:46: 41,589 / 583,890

- #27604 WS 02:40:49 → end of capture, still established: 280,374 / 1,764,209
- #18312 DC 02:56:38 → end, still established: 89,314 / 960,911
- #19978 WS 04:08:45 → end, still established: 39,409 / 488,552
Total upload ≈ 0.50 MB. The largest is the workstation's Administrator session (#27604, 280 KB in ~3 h). After each session's ~0.5–0.9 MB stage, the extra downstream points to extension or module loading.

(2) Attacker RDP from 194.61.24.102 to the DC on 3389:
- #30035, 02:22:36-02:52:47: 4.80 MB to the server, 12.76 MB from the DC to the attacker.
- #22204, 02:56:03-02:57:41: 266 KB / 1.0 MB.
The DC→attacker volume is consistent with 30 minutes of screen graphics. The RDP payload is encrypted (CredSSP/TLS), so clipboard or drive-redirection transfers cannot be ruled out.

(3) Largest non-attacker outbound flows are all Microsoft or browsing services:
- 104.119.185.124 (oneclient.sfx.ms, the OneDrive client binary CDN), #28319 at 02:37:01: 446 KB up / 39.4 MB down. DNS for oneclient.sfx.ms at 02:37:00 from both hosts. This was 37 s after Administrator's first logon on the workstation, so it is OneDrive per-user setup for the new profile, a download.
- 52.114.132.91 (v10.events.data.microsoft.com, Windows telemetry), #18404 at 02:44:00: 180 KB up.
- Bing/MSN 204.79.197.200 and 13.107.21.200: browsing.

(4) HTTP: the only POSTs are to dmd.metaservices.microsoft.com/metadata.svc (device metadata, 1.7 KB) and go.microsoft.com/fwlink LinkID=252669, plus one OCSP-like POST to o.ss2.us at 22:01 (pre-attack). There is no PUT and no POST to an attacker IP.

(5) fileinfo: the only executable is /coreupdater.exe (inbound). SMB files are sysvol GPO only. No archive (.zip/.7z/.rar) crossed the wire in either direction.

(6) DNS: 2,650 records, 378 distinct queries, and the longest query field is 100 characters including the JSON. There is no high-entropy or tunnelling pattern. The only cloud-storage domain is oneclient.sfx.ms: no onedrive.live.com sync, Dropbox, MEGA, Google Drive, WeTransfer, pastebin or ngrok.

Analysis

Fact: the capture window shows no bulk outbound transfer from either host to any destination. Interpretation: the only channels that could have carried stolen data are (a) the Meterpreter sockets, with 0.5 MB up in total, enough for credential dumps (hashdump/kiwi output, LSA secrets), small documents or screenshots but not mass data, and (b) the encrypted RDP session, where data viewed on screen, or copied via clipboard or a redirected drive, would sit inside the 12.76 MB DC→attacker stream and cannot be told apart. Because three C2 sessions were still established at 05:38 UTC, exfiltration after the capture ended cannot be excluded. The OneDrive traffic is client installation for the new Administrator profile, not file synchronisation. No host artefacts (MFT/UsnJrnl archives, 7-Zip history, SRUM network usage, BITS, ODL) exist to confirm or refute staging on disk. Confidence is probable for 'no bulk exfiltration during the window' and undetermined for small or credential exfiltration.

Evidence (12)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/ eve_flow.jsonl#27604	case001.pcap
2020-09-19 02:56:38.402867	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62613 -> 203.78.103.109:443 failed state=established bytes=89314/960911	suricata/case001.pcap/ eve_flow.jsonl#18312	case001.pcap
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed	suricata/case001.pcap/ eve_flow.jsonl#19978	case001.pcap

UTC	Artifact	Detail	Reference	Original source
		state=established bytes=39409/488552		
2020-09-19 02:25:18.565676	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62414 -> 203.78.103.109:443 failed state=closed bytes=51044/932653	suricata/case001.pcap/ eve_flow.jsonl#28844	case001.pcap
2020-09-19 02:29:49.290306	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62476 -> 203.78.103.109:443 failed state=closed bytes=41589/583890	suricata/case001.pcap/ eve_flow.jsonl#26788	case001.pcap
2020-09-19 02:22:36.664454	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40238 -> 10.42.85.10:3389 rdp state=closed bytes=4802218/12756305	suricata/case001.pcap/ eve_flow.jsonl#30035	case001.pcap
2020-09-19 02:56:03.538780	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40240 -> 10.42.85.10:3389 rdp state=closed bytes=266261/1008697	suricata/case001.pcap/ eve_flow.jsonl#22204	case001.pcap
2020-09-19 02:37:01.033644	Network flows (Suricata)	network flow: start 104.119.185.124 TCP 10.42.85.115:50731 -> 104.119.185.124:443 tls state=closed bytes=446236/39409046	suricata/case001.pcap/ eve_flow.jsonl#28319	case001.pcap
2020-09-19 02:37:01.185518	TLS (Suricata)	TLS handshake oneclient.sfx.ms tls_sni=oneclient.sfx.ms | dest_ip=104.119.185.124 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=C=US, ST=WA, L=Redmond, O=Microsoft Corporation, OU=Microsoft Corporation, CN=*.sfx.ms | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 5 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e	suricata/case001.pcap/ eve_tls.jsonl#16200	case001.pcap
2020-09-19 02:37:00.954915	DNS (Suricata)	DNS query / answer [{"rrname": "oneclient.sfx.ms", "rrtype": "A"}] dns_type=request | dns_rcode=NOERROR | src_ip=10.42.85.115 | dest_ip=10.42.85.10	suricata/case001.pcap/ eve_dns.jsonl#1868	case001.pcap
2020-09-19 00:44:00.322000	Network flows (Suricata)	network flow: start 52.114.132.91 TCP 10.42.85.115:50639 -> 52.114.132.91:443 tls state=closed bytes=180437/12740	suricata/case001.pcap/ eve_flow.jsonl#18404	case001.pcap
2020-09-19 00:44:00.438933	TLS (Suricata)	TLS handshake v10.events.data.microsoft.com tls_sni=v10.events.data.microsoft.com | dest_ip=52.114.132.91 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=C=US, ST=WA, L=Redmond, O=Microsoft, OU=WSE, CN=*.events.data.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, CN=Microsoft Secure Server CA 2011 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls	suricata/case001.pcap/ eve_tls.jsonl#901	case001.pcap

4.13 Incident timeline and correlation

I put the incident timeline together from the network capture alone (case001.pcap, Suricata).

Chronology (UTC):
*- **Start of the incident:** at 02:19:26, 194.61.24.102 sent an Nmap RDP probe to the domain controller (DC). RDP password guessing followed: 95 failed CredSSP logons between 02:21:26 and 02:21:46, then the first successful Administrator logon at 02:21:47.*
*- **DC compromise:** the attacker opened an interactive RDP session at 02:22:36. They then downloaded coreupdater.exe from http://194.61.24.102/ with IE11 at 02:24:06. Meterpreter sessions to 203.78.103.109:443 started at 02:25:18 and 02:29:49.*
*- **Pivot to the workstation:** RDP from the DC to the workstation started at 02:35:55, and Administrator logged on there for the first time at 02:36:24. Edge downloaded the same file at 02:39:58, and a C2 session started at 02:40:49.*
*- **Last contact from 194.61.24.102:** a second RDP session ran from 02:56:03 to 02:57:41. A DC C2 session started inside it at 02:56:38.*
*- **Later activity:** the FileShare root on the DC was modified at 03:34:18. ricksanchez logged on at 04:08:15, and a new C2 session started 30 s later at 04:08:45.*
*- **Last event seen:** at 05:38:42–05:38:51, three C2 sessions were still ESTABLISHED when the capture ended.*

Recorded in this step:
- Timeline events T-022 (last attacker RDP), T-023 (end of capture with live C2) and T-024 (first DC Meterpreter session closed, then re-

staged).
- Consolidated finding F-022.
- IOCs I-011 (landing URL http://194.61.24.102/) and I-012 (RDP cookies nmap/Administrator, plus the Nmap JA3).
- Earlier IOCs I-001..I-010 cover the IPs, the payload URL, the SHA256/SHA1/MD5 hashes, the file name, the Administrator account, the brute-force JA3/JA4 and the landing-page hash.

***Time consistency:** the raw EVE times are in +0200, and their conversion to UTC is consistent across event types (pcap_cnt order matches). In raw.suricata_flow the 'timestamp' field is always the first packet of the pcap, so only flow_start/flow_end are valid; the timeline uses them correctly. Kerberos times and packet times agree within seconds.*

F-022: Consolidated incident chronology (network-only): RDP brute force → DC compromise → Meterpreter → RDP pivot to workstation → logon-triggered C2 persistence, ongoing at end of capture

Severity	**Critical**
Confidence	probable
Independent sources	network
Category	context
MITRE ATT&CK	T1595, T1110.001, T1078.002, T1021.001, T1105, T1204.002, T1059, T1620, T1571, T1547.001
Period (UTC)	2020-09-19 02:19:26.792622 → 2020-09-19 04:08:45.483780

Observed facts

UTC chronology from case001.pcap: 21:59:39 workstation boot / 21:59:58 mortysmith logon (baseline). 02:19:26 Nmap RDP probe (cookie mstshash=nmap, ET SCAN 2036252) from 194.61.24.102 to DC 10.42.85.10:3389; 02:19-02:20 ~29k TLS RDP connections; 02:21:26-02:21:46 95 CredSSP failures (mstshash=Administrator); 02:21:47 first successful NLA; 02:22:36-02:52:47 interactive session. 02:23:09 IE11 first launch, 02:23:41 GET http://194.61.24.102/, 02:24:06 GET /coreupdater.exe (7168 B, SHA256 10f3b920…cfda6). 02:25:18 DC→203.78.103.109:443 Meterpreter stage (933 KB), closed 02:29:36; 02:29:49 second session (closed 02:52:46). 02:35:55 RDP DC→DESKTOP-SDN1RPT 10.42.85.115 (until 02:52:14); 02:36:24 Administrator Kerberos AS-REP / new profile. 02:39:26-02:39:58 workstation Edge downloads the same file; 02:40:45/02:40:49 SmartScreen checkappexec; 02:40:49.936 workstation C2 session (alive at end). 02:56:03-02:57:41 last attacker RDP session (last contact from 194.61.24.102); 02:56:38 DC C2 session (alive at end). 03:34:18 FileShare root modified on DC. 04:08:15 ricksanchez console logon on workstation; 04:08:45 new workstation C2 session (alive). 05:38:42-05:38:51 last packets of the 3 established C2 sessions; capture ends 05:38:57.

Analysis

Initial access = valid account (domain Administrator) obtained by RDP password guessing on an Internet-exposed DC (dwell from first probe to C2 ≈ 6 min). Execution via user-launched stager downloaded over HTTP; Meterpreter loaded in memory from a second IP (203.78.103.109). Lateral movement only by RDP from the DC. Re-staging 30-35 s after new logons without re-download indicates logon-triggered persistence on both hosts (mechanism not observable). No exfiltration or impact seen on the wire, but the C2 was still active when the capture ended, so post-05:38 activity is unknown. Times across EVE event types are consistent (+0200 normalised to UTC); certainty limited by the single network source (no host artefacts), hence probable.

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:19:26.792622	Detection Suricata	low ET SCAN RDP Connection Attempt from Nmap | TCP 194.61.24.102:38100 -> 10.42.85.10:3389 | Detection of a Network Scan | 1:BJ5fdE2KUXlcHa7o2AcT6LKStwM= | case001.pcap	suricata/case001.pcap/eve_alert.jsonl#25	case001.pcap
2020-09-19 02:22:36.664454	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40238 -> 10.42.85.10:3389 rdp state=closed bytes=4802218/12756305	suricata/case001.pcap/eve_flow.jsonl#30035	case001.pcap

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:24:06.940653	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.10 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/eve_fileinfo.jsonl#100	case001.pcap
2020-09-19 02:25:18.565676	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62414 -> 203.78.103.109:443 failed state=closed bytes=51044/932653	suricata/case001.pcap/eve_flow.jsonl#28844	case001.pcap
2020-09-19 02:35:55.347240	Network events (Suricata)	network event rdp TCP 10.42.85.115:3389 -> 10.42.85.10:62514	suricata/case001.pcap/eve_rdp.jsonl#297	case001.pcap
2020-09-19 02:36:24.913767	Network events (Suricata)	network event krb5 TCP 10.42.85.115:50695 -> 10.42.85.10:88	suricata/case001.pcap/eve_krb5.jsonl#60	case001.pcap
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/eve_flow.jsonl#27604	case001.pcap
2020-09-19 02:56:03.538780	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 194.61.24.102:40240 -> 10.42.85.10:3389 rdp state=closed bytes=266261/1008697	suricata/case001.pcap/eve_flow.jsonl#22204	case001.pcap
2020-09-19 02:56:38.402867	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62613 -> 203.78.103.109:443 failed state=established bytes=89314/960911	suricata/case001.pcap/eve_flow.jsonl#18312	case001.pcap
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed state=established bytes=39409/488552	suricata/case001.pcap/eve_flow.jsonl#19978	case001.pcap

5. Indicators of compromise

ID	Type	Value	Context	First seen (UTC)	Last seen (UTC)	References
I-001	ip	194.61.24.102	Attacker source: Nmap RDP scan, RDP brute force (cookie Administrator), interactive RDP sessions to DC, HTTP server hosting /coreupdater.exe	2020-09-19 02:19:26.792622	2020-09-19 02:24:06.940063	suricata/case001.pcap/eve_alert.jsonl#25 suricata/case001.pcap/eve_flow.jsonl#30035 suricata/case001.pcap/eve_http.jsonl#116
I-002	ip	203.78.103.109	Metasploit/Meterpreter handler, raw TCP port 443 (non-TLS), stage delivery + C2 to 10.42.85.10 and 10.42.85.115	2020-09-19 02:25:19.071794	2020-09-19 02:40:49.936293	suricata/case001.pcap/eve_alert.jsonl#439 suricata/case001.pcap/eve_flow.jsonl#27604
I-003	url	http://194.61.24.102/coreupdater.exe	Stager download URL (DC 02:24:06 UTC, workstation 02:39:58 UTC)	2020-09-19 02:24:06.940063	2020-09-19 02:39:58.411571	suricata/case001.pcap/eve_http.jsonl#116 suricata/case001.pcap/eve_http.jsonl#156
I-004	sha256	10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6	coreupdater.exe (7168 B) – presumed Metasploit stager	2020-09-19 02:24:06.940653	2020-09-19 02:39:58.412735	suricata/case001.pcap/eve_fileinfo.jsonl#100 suricata/case001.pcap/eve_fileinfo.jsonl#136
I-005	sha1	fd153c66386ca93ec9993d66a84d6f0d129a3a5c	coreupdater.exe SHA1	2020-09-19 02:24:06.940653	2020-09-19 02:24:06.940653	suricata/case001.pcap/eve_fileinfo.jsonl#100
I-006	md5	eed41b4500e473f97c50c7385ef5e374	coreupdater.exe MD5	2020-09-19 02:24:06.940653	2020-09-19 02:24:06.940653	suricata/case001.pcap/eve_fileinfo.jsonl#100
I-007	file	coreupdater.exe	Masquerading stager file name downloaded on CITADEL-DC01 and DESKTOP-SDN1RPT	2020-09-19 02:24:06.940653	2020-09-19 02:39:58.412735	suricata/case001.pcap/eve_fileinfo.jsonl#100 suricata/case001.pcap/eve_fileinfo.jsonl#136
I-008	user	C137\Administrator	Domain admin account compromised via RDP brute force; used interactively on DC and on DESKTOP-SDN1RPT (first logon 02:36:24 UTC) – context of coreupdater.exe execution.	2020-09-19 02:36:24.913767	2020-09-19 02:36:24.913767	suricata/case001.pcap/eve_krb5.jsonl#60
I-009	command	JA3 75fe51990656df4f7a249d5b86aa29ae / JA4 t13i781000_ab95583b6d39_d41ae481755e	TLS client fingerprint of the RDP brute-force tool used from 194.61.24.102 against CITADEL-DC01:3389 (29,204 handshakes, 02:19:26–02:20:26 UTC). This is a	2020-09-18 21:58:07.470323	2020-09-18 21:58:07.470323	suricata/case001.pcap/eve_tls.jsonl#932

ID	Type	Value	Context	First seen (UTC)	Last seen (UTC)	References
			hunting fingerprint, not unique on its own.			
I-010	sha256	6978f8fa2bf5afbe1a074401b923ead519ce89471d83376f5cfda50c12bc2d77	Attacker landing page http://194.61.24.102/ (228 B text/html, MD5 614645af648513231c283090faa5c469) served to DC 02:23:41 and workstation 02:39:26 UTC; referer of coreupdater.exe download.	2020-09-19 02:23:41.735412	2020-09-19 02:39:26.941704	suricata/case001.pcap/eve_fileinfo.jsonl#98 suricata/case001.pcap/eve_fileinfo.jsonl#135
I-011	url	http://194.61.24.102/	Attacker landing page (228 B, SHA256 6978f8fa…2d77) browsed manually with IE11 on the DC at 02:23:41 UTC and Edge on the workstation at 02:39:26 UTC; Referer of the coreupdater.exe downloads.	2020-09-19 02:24:06.940063	2020-09-19 02:39:58.411571	suricata/case001.pcap/eve_http.jsonl#116 suricata/case001.pcap/eve_http.jsonl#156
I-012	command	RDP X.224 cookie "mstshash=nmap" / "mstshash=Administrator" from 194.61.24.102 (JA3 3bdfb64d53404bacd8a47056c6a756be for Nmap SSL probe)	Nmap RDP probe (ET SCAN 2036252) at 02:19:26 UTC followed by RDP/NLA brute force against Administrator on CITADEL-DC01:3389.	2020-09-19 02:19:26.792622	2020-09-19 02:19:26.792649	suricata/case001.pcap/eve_alert.jsonl#25 suricata/case001.pcap/eve_tls.jsonl#920

6. MITRE ATT&CK mapping

Technique	Findings / events
T1005	F-017
T1021.001	F-003, F-006, F-008, F-013, F-016, F-021, F-022, T-002, T-006, T-011, T-022
T1021.002	T-017
T1036.005	F-019
T1037	F-009, F-012
T1039	F-017, T-018
T1041	F-010, F-021
T1046	F-003, F-011, T-001
T1053.005	F-009, F-012
T1059	F-005, F-007, F-022, T-024
T1071	F-005, F-010, F-021, T-004, T-023
T1071.001	F-018
T1078.002	F-003, F-008, F-013, F-014, F-015, F-016, F-022, T-002, T-015, T-022
T1095	F-005, F-010
T1105	F-004, F-005, F-007, F-010, F-018, F-022, T-003, T-007, T-019
T1110.001	F-003, F-011, F-013, F-022, T-001, T-014, T-015
T1133	F-003, F-011, F-016, T-015
T1135	F-017
T1204.002	F-004, F-007, F-018, F-022, T-020
T1547.001	F-009, F-012, F-022, T-012, T-013
T1558	F-015
T1562.001	F-020
T1570	F-016
T1571	F-005, F-007, F-010, F-019, F-022, T-004, T-023
T1595	F-022
T1608.001	F-018
T1620	F-019, F-022, T-024

7. Answers to the questions

(no question asked)

8. Recommendations

Immediate actions (containment)

- Isolate CITADEL-DC01 (10.42.85.10) and DESKTOP-SDN1RPT (10.42.85.115) from the network. Before any reboot, capture volatile memory (WinPMem/Magnet RAM Capture) and run a full KAPE triage on each host: the Meterpreter stage is memory-resident and three C2 sessions were live at the end of the capture.
- Block 194.61.24.102 and 203.78.103.109 (all ports) at the perimeter firewall and proxy, and search firewall, proxy and NetFlow logs for any other internal host talking to them.
- Remove Internet exposure of TCP 3389 on the domain controller immediately.
- Treat the AD domain C137.LOCAL as fully compromised. Reset the domain Administrator password (to a long random value), then reset all privileged and service accounts. Reset the krbtgt password twice, with an interval longer than the maximum ticket lifetime, after eviction is confirmed.
- Reset the passwords of mortysmith and ricksanchez, and invalidate the cached credentials and DPAPI material of the profiles used on DESKTOP-SDN1RPT.

Eradication and recovery

- Hunt for and remove coreupdater.exe (SHA256 10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6, SHA1 fd153c66386ca93ec9993d66a84d6f0d129a3a5c, MD5 eed41b4500e473f97c50c7385ef5e374) and any copies or renames on both hosts. Check the Administrator Downloads folders and the IE/Edge INetCache.
- Inspect and remove the logon persistence on both hosts: HKLM/HKCU\Software\Microsoft\Windows\CurrentVersion\Run and RunOnce, the All Users and per-user Startup folders, scheduled tasks with logon triggers (C:\Windows\System32\Tasks), services, WMI subscriptions, and Winlogon Userinit/Shell values.
- Rebuild DESKTOP-SDN1RPT from a known-good image. Given full Administrator control, plan to rebuild or forest-recover CITADEL-DC01 (promote a clean DC, then demote and rebuild the compromised one) rather than clean it in place.
- Audit AD for attacker changes made between 2020-09-19 02:21 and the end of the incident: new or modified accounts, group membership changes (Domain/Enterprise Admins, AdminSDHolder), GPO changes, delegation changes and SPNs. Review the contents of the FileShare share modified at 03:34:18 UTC.
- Only restore service after the C2 indicators show no beacons on full-packet or NetFlow monitoring for an agreed observation period.

Hardening

- Never expose RDP directly to the Internet. Put it behind a VPN or RD Gateway with MFA, and restrict RDP to domain controllers to a dedicated admin jump host (tiered administration model).
- Enforce an account lockout policy and strong passphrases for privileged accounts. Rename or disable the built-in Administrator for interactive use and use individual named admin accounts. Add privileged accounts to Protected Users.
- Rotate the Administrator password so AES Kerberos keys are generated, and disable RC4-HMAC for Kerberos where possible (the RC4 AS-REP was observed).
- Block web browsing and direct Internet egress from domain controllers. Enforce egress filtering so that only the proxy may reach external 80/443, which would also block raw TCP C2 on 443.
- Deploy AppLocker or WDAC to block execution from user-writable paths (Downloads, %TEMP%, %APPDATA%). Enable Defender tamper protection, cloud-delivered protection, ASR rules and SmartScreen in block mode.
- Use LAPS for local administrator accounts and restrict lateral RDP between hosts (DC → workstation) with host firewall rules.

Detection and monitoring

- Network IDS: keep the ET rules that fired (Nmap RDP cookie, RDP brute force, EXE download from a dotted-quad host, Metasploit Bind_API, PE over raw TCP). Add a rule for non-TLS payloads on TCP/443 and for HTTP GET of .exe from a bare IP address.
- SIEM: alert on more than 20 failed RDP/NLA logons (4625 LogonType 10/3 with failure 0xC000006A/0xC000006D) from one source in 1 minute, followed by a 4624 LogonType 10 from the same IP, especially against a DC.
- SIEM: alert on interactive logons (4624 type 2/10) of Administrator or Tier-0 accounts on DCs from non-jump-host sources, and on RDP from a DC to workstations.
- EDR/Sysmon: alert on iexplore.exe/msedge.exe writing a PE that is then executed and makes an outbound connection to a public IP on 443 without SNI/TLS (Sysmon 11 → 1 → 3 correlation).
- Alert on process creation from Run keys, the Startup folder or logon-triggered scheduled tasks pointing to user-writable paths (Sysmon 12/13, 4698, 7045).
- Add IOCs to watchlists: IPs 194.61.24.102 and 203.78.103.109; URLs http://194.61.24.102/ and /coreupdater.exe; the stager hashes; the landing-page SHA256 6978f8fa2bf5afbe1a074401b923ead519ce89471d83376f5cfda50c12bc2d77; JA3 75fe51990656df4f7a249d5b86aa29ae and 3bdfb64d53404bacd8a47056c6a756be; RDP cookies 'mstshash=nmap'/'mstshash=Administrator' from external sources.

Further investigation

- Acquire and analyse host evidence from both machines:
  - Security.evtx (4624/4625/4648/4672/4688/4720/4728/4732/4698/7045/1102)
  - TerminalServices-RemoteConnectionManager / LocalSessionManager
  - Microsoft-Windows-Windows Defender/Operational
  - Prefetch (COREUPDATER.EXE-*.pf), Amcache, ShimCache, BAM, UserAssist
  - $MFT/$J, LNK/JumpLists, IE/Edge history, Run keys, scheduled tasks

  This would confirm execution, persistence and post-exploitation that are only inferred today.
- Analyse memory images with Volatility (pslist/psscan/netscan/malfind/cmdline) to find the process hosting Meterpreter, migration targets, and evidence of credential dumping (lsass access, mimikatz/kiwi).
- Check the DC for NTDS.dit or SAM/SYSTEM hive access, shadow copies (vssadmin/ntdsutil), DCSync, and new accounts or privileged group changes between 02:21 and 05:38 UTC and later.
- Determine what was changed in \\CITADEL-DC01\FileShare at 2020-09-19 03:34:18 UTC (MFT/USN of the share folder).
- Reverse-engineer coreupdater.exe (7168 B) in a sandbox to confirm it is a msfvenom reverse_tcp stager and extract its configuration (LHOST 203.78.103.109, LPORT 443).
- Extend network review beyond the capture window: perimeter logs and NetFlow before 2020-09-18 21:58 and after 2020-09-19 05:38 UTC, plus the 22:10:48–00:34:48 gap. Look for continued C2, other internal hosts reaching the attacker IPs, and exfiltration.
- Enrich 194.61.24.102 and 203.78.103.109 with threat intelligence (ASN, hosting, related campaigns), and review whether the Administrator password is reused on other systems.

Appendix A: Summary of the analysis steps

Step 1: System context and scope

Scope: one pcap (case001.pcap, SHA256 09ABF49E…791CB) parsed by Suricata 8.0.7, window 2020-09-18 21:58:07 → 2020-09-19 05:38:57 UTC (raw EVE in +0200). Environment: AD domain C137.LOCAL; DC CITADEL-DC01 10.42.85.10 (Windows Server 2012 R2/NT 6.3, RDP exposed to Internet); workstation DESKTOP-SDN1RPT 10.42.85.115 (Windows 10 build 19041). Domain accounts observed in Kerberos: DESKTOP-SDN1RPT$, mortysmith, Administrator, ricksanchez. External 194.61.24.102 dominates traffic (≈29k RDP/TLS flows to DC:3389, cookies 'Administrator'/'nmap') and served

/coreupdater.exe to the DC – to be analysed later. Findings F-001 (system description) and F-002 (scope limitation: no host artefacts).

Not verified: No registry/EVTX/MFT/memory: OS install date, time zone, local accounts/SIDs, profiles, USB/volume history, Panther logs unavailable. OS versions inferred from User-Agents/SMB only. Corroboration limited to a single source class (network).

Step 2: Triage of automatic detections

Detections = Suricata only (461: 7 high, 4 medium, 450 low). No Hayabusa/Chainsaw/Zircolite/Defender data (no EVTX); hashset empty so hash_hits meaningless. True positives: (F-003) Nmap RDP scan + ~29k RDP brute-force connections with 'Administrator' cookie from 194.61.24.102 to DC 10.42.85.10:3389 at 02:19-02:21 UTC, followed by 30-min 17.5 MB interactive RDP session 02:22:36-02:52:47 and a second one 02:56:03-02:57:41 (successful Administrator logon inferred). (F-004) /coreupdater.exe (7168 B, SHA256 10f3b920…1cfda6, MD5 eed41b45…) downloaded from 194.61.24.102 by DC (IE11, 02:24:06) and workstation 10.42.85.115 (Edge, 02:39:58). (F-005) 5 Meterpreter stage deliveries/C2 sessions over raw TCP 203.78.103.109:443 (DC 02:25:18, 02:29:49, 02:56:38; workstation 02:40:49, 04:08:45), three still established at end of capture 05:38 UTC. (F-006, from pivot) RDP DC -> workstation 02:35:55-02:52:14. Timeline T-001..T-010, IOCs I-001..I-007. False positives/context: device-metadata/USB-metadata UA alerts, UA-CPU header (IE11 on DC), Discord traffic at 22:05 (pre-attack, low volume).

Not verified: All evidence comes from a single network capture: critical findings cannot reach 'confirmed' (no host logs, Prefetch, Amcache, memory). RDP is TLS-encrypted: authentication result and accounts in sessions are inferred from volume/timing. Execution of coreupdater.exe is inferred from subsequent Meterpreter staging, not observed. C2 content is not decoded.

Step 3: Program execution

Host execution artefacts (Prefetch, Amcache, ShimCache, BAM, UserAssist, PCA, SRUM, 4688/Sysmon, PowerShell) are not in the case. The only evidence is one pcap, so I inferred execution from network behaviour. F-007 (critical/probable): coreupdater.exe (7168 B, SHA256 10f3b920…cfda6) was most likely executed as a reverse-TCP Meterpreter stager. The DC downloaded it at 02:24:06 and opened its own outbound connection to 203.78.103.109:443 at 02:25:18. The workstation downloaded it at 02:39:58 and connected out at 02:40:49. F-008 (high/probable): domain Administrator logged on interactively to DESKTOP-SDN1RPT for the first time at 02:36:24, seen as Kerberos AS-REP, a NETLOGON 'Default User.V6' lookup and the DPAPI protected_storage pipe. So the workstation execution most likely ran as Administrator. F-009 (high/probable): user ricksanchez logged on at the console at 04:08:15 (no RDP at that time), and a new Meterpreter session started 30 s later at 04:08:45. This points to something that starts automatically at logon (Run key, Startup folder or scheduled task). I found no sign of remote execution between machines: no SVCCTL/ATSVC/WMI in DCERPC, and no other executable transferred over HTTP or SMB. Added timeline events T-011 and T-012 and IOC I-008 (C137\Administrator).

Not verified: I could not establish the process name, path, PID, parent, run count, SHA1 from Amcache, or the persistence mechanism, because no disk, EVTX, registry or memory data was provided. Encrypted traffic (TLS, RDP, the Meterpreter channels) hides any commands the attacker ran. hash_hits is empty because no hashset was loaded. Each conclusion rests on the pcap alone, so it cannot be raised to 'confirmed'.

Step 4: Memory (RAM)

The case has no memory image. I queried information_schema.tables for raw.vol_* tables (vol_info, vol_pslist, vol_psscan, vol_netscan, vol_malfind and so on) and got 0 rows. That matches the artifact_summary and the sources recorded in steps 1–2, which hold only one pcap (case001.pcap) parsed by Suricata. So there was no capture time to compare with the KAPE collection, no process list or parent/child check, and no checks for injection (malfind/hollowing), sockets (netscan), drivers/rootkits, services, console history or privileges. The Meterpreter sessions (F-005, F-007, F-009) and the coreupdater.exe execution can still only be inferred from network behaviour. No findings were added in this step.

Not verified: No memory image was provided, so this step could not show which process owned the Meterpreter C2 sockets to 203.78.103.109:443 on CITADEL-DC01 (10.42.85.10) or DESKTOP-SDN1RPT (10.42.85.115). It also could not reveal any injected or fileless code, a migrated Meterpreter process, a logon-triggered persistence mechanism (suspected in F-009), console command history or

credential-access activity (e.g. lsass access). Live captures of both hosts would close these gaps; given the three C2 sessions still established at 05:38 UTC, memory acquisition with WinPmem/DumpIt is recommended before any remediation reboot.

Step 5: Network traffic (PCAP)

Network analysis of case001.pcap. Capture window: 2020-09-18 21:58:07 → 2020-09-19 05:38:57 UTC. Anything before or after that window is invisible on the wire.

Alerts: All 461 Suricata alerts were re-qualified, and the step-2 qualification stands.
- True positives: Nmap RDP scan, RDP brute force with the 'Administrator' cookie, PE download from a dotted-quad host, Metasploit Bind_API, and PE over raw TCP.
- Noise: Discord, device-metadata/USB-metadata user agents, the UA-CPU header, and RDP responses to an external host.

Conversations:
- External peers are only the two attacker IPs plus Microsoft, CDN, OCSP and browsing sites.
- 194.61.24.102:
  - RDP brute force at 02:19–02:20 (29,208 TLS flows).
  - 100 RDP flows totalling 5.35 MB up / 14 MB down, until 02:57:41.
  - HTTP / and /coreupdater.exe at 02:23:41/02:24:06 (DC) and 02:39:26/02:39:58 (workstation).
- 203.78.103.109:443: five raw-TCP Meterpreter sessions (new finding F-010).
  - Each starts with a ~0.5–0.9 MB stage download.
  - Total upload is ≈0.5 MB, so no bulk exfiltration is visible.
  - Three sessions were still ESTABLISHED at the end of the capture.
- Neither attacker IP was resolved through DNS.

Lateral movement: Internal traffic shows only DC→workstation RDP (already F-006). SMB carried only sysvol GPO files, NETLOGON profile lookups and FileShare browsing after ricksanchez logged on. There was no admin$/C$, no executable transferred over SMB, and no SVCCTL/ATSVC/DCSync.

DNS: Nothing algorithmic or newly seen that is malicious. 10.90.90.90 is a DNSBL sinkhole.

TLS: New finding F-011 fingerprints the brute-force tool: JA3 75fe5199…/JA4 t13i781000_… (OpenSSL-based client offering TLS 1.3), plus three Nmap ssl-probe JA3s. Added as IOC I-009. RDP uses the DC's self-signed CN=CITADEL-DC01.C137.local certificate.

Files carved: The only suspicious one is coreupdater.exe (already IOCs I-004..I-006). hash_hits is empty because no hash lists were loaded.

Notes: Context was recorded for the unexplained Authenticode/Intel CRL check by the DC at 04:02 UTC during its live C2 session, the RC4 AS-REP for Administrator, and the OneDrive and telemetry traffic.

Not verified: The case has no disk, EVTX, registry, SRUM, browser or memory artefacts. So no connection can be tied to an owning process (vol_netscan, Sysmon 3, SRUM), and the creation and execution of coreupdater.exe cannot be checked on disk (Amcache SHA1, Zone.Identifier, Prefetch). For that reason high findings stay 'probable' with a single network source. The Meterpreter TLV content and the RDP session content are encrypted, so commands run and data moved through RDP clipboard or drive redirection cannot be seen. What the attacker did after 05:38:57 UTC is outside the capture.

Step 6: Persistence

Persistence analysis, network-only case. I confirmed that no host persistence artefact table exists (registry, EVTX, tasks, services, WMI, Startup folders, MFT). On the wire I checked:
- DCERPC interfaces: only EPM, DRSUAPI, LSARPC and NETLOGON, all normal for a domain workstation. No SVCCTL, ATSVC, WINREG or WMI, so no remote creation of a service, task or registry value.
- Default Domain Policy files over SMB: gpt.ini, Registry.pol and GptTmpl.inf have the same MD5 across the refreshes at

21:59, 02:16, 03:30 and 05:03, so the GPO was not tampered with.
- Reboots: machine-account Kerberos shows a single boot at 21:59:39, so no reboot happened during the capture.
- HTTP: no new download before the later C2 sessions.

New finding F-012 (high/probable): Meterpreter is re-staged 30-35 s after interactive logons, with no re-download:
- DC: RDP #2 at 02:56:03, C2 at 02:56:38. The previous C2 session died with the RDP logoff at 02:52:46/47 (timeline event T-013).
- Workstation: ricksanchez console logon at 04:08:15, C2 at 04:08:45.

This fits logon-triggered autostart persistence: a Run key (HKLM most likely), the All-Users Startup folder, or a logon scheduled task. The alternative is a manual relaunch, which is plausible on the DC but much less likely on the workstation.

Not verified: I could not identify the persistence mechanism itself (key, value, path, task or service name), when it was created, or whether it survives a reboot. That needs the SOFTWARE/NTUSER hives, C:\Windows\System32\Tasks, the Security, System and TaskScheduler EVTX, the Startup folders and the $MFT, and none of these were collected. Because the evidence comes from a single network source, F-012 stays probable.

Step 7: Accounts, authentication and privileges

Accounts and authentication were rebuilt from network data only. The case has no Security/TerminalServices EVTX and no SAM (raw.reg_useraccounts absent), so I worked from the RDP, Kerberos, SMB2 and LDAP records in case001.pcap.

RDP brute force: The 29,202 TLS-only flows at 02:19-02:20 never reached authentication (no X.224/CredSSP exchange). The real password guessing was 95 identical CredSSP failure flows with mstshash=Administrator at 02:21:26-02:21:46. A larger, different flow at 02:21:47 most likely marks the first successful login, i.e. the moment the password was found. Short sessions followed at 02:22:08, then the interactive session at 02:22:36 (F-013, critical/probable; T-014, T-015).

Per-account summary (F-014):
- DESKTOP-SDN1RPT$: boot at 21:59:39.
- mortysmith: legitimate console logon at 21:59:58 (T-016).
- Administrator: Kerberos logon on the workstation at 02:36:24 via RDP from the DC, which created a new profile and requested the DPAPI ProtectedStorage ticket (F-008).
- ricksanchez: console logon at 04:08:15.
- No Kerberos failures other than the normal PREAUTH_REQUIRED, and no lockout/revoked errors.
- All 24 SMB sessions and all 95 LDAP binds used Kerberos and succeeded. No NTLM and no SAMR were seen.

Encryption (F-015, low/possible): Administrator's AS-REP used RC4-HMAC while every other account used AES256. The likely cause is a legacy, never-rotated password with no AES keys.

Noise: I recorded a note on the TLS flood so it is not miscounted as failed logons.

Not verified: No host logs were available: 4624/4625/4634/4648/4672/4720-4738/4740/4768-4776, RDP 21/22/24/25/1149/1024, the SAM, local accounts and the DC's own interactive logons. Kerberos for those logons stays on the DC, so it never appears on the wire. That means no conclusion is possible on account creation or group changes made locally on the DC during the attacker's session, or on privilege escalation (4672). The account behind the successful NLA at 02:21:47 is inferred from the RDP cookie, not from an authenticated identity. The RC4 interpretation could not be checked against pwdLastSet or msDS-SupportedEncryptionTypes.

Step 8: Remote access and lateral movement

The DC was controlled remotely from the Internet, and the attacker then used it as a pivot to reach the workstation over RDP. RDP was the only remote-control channel seen on the wire, apart from the Meterpreter C2 already covered (F-005/F-010). New finding F-016 (high/probable), timeline event T-017, one note.

How this was checked (network data only, case001.pcap):
- Listed every internal flow pair and every flow on admin ports (445, 3389, 5985/5986, 135, 139, 22, 5900, 23).
- Read all eve_rdp negotiation records, the SMB tree connects, named pipes and file opens, the DCERPC interfaces, and the Kerberos service tickets.
- Searched DNS and TLS server names for RMM / remote-access tools.

Results:
1. **Internet → DC (inbound RDP):** the RDP sessions from 194.61.24.102 to the DC used the cookies 'nmap' (1) and 'Administrator' (99), and always negotiated 'hybrid' (CredSSP). That fits the non-Windows client already fingerprinted in F-011.
2. **DC → workstation (lateral movement):** the RDP hop from 10.42.85.10 to 10.42.85.115:3389 ran from 02:35:55 to 02:52:14 UTC.
   - It negotiated 'hybrid_ex', sent no cookie, and added an RDP-UDP channel (UDP flow #31428, 6.7 MB to the client). That is typical of the native Windows mstsc.exe running inside the attacker's session on the DC.
   - The workstation server also advertised restricted_admin, which does not show that mode was used.
   - The Kerberos ticket for Administrator on the workstation was issued at 02:36:24.
   - This refines F-006.
3. **No other remote-execution or admin traffic:**
   - No SVCCTL, task-scheduler (ATSVC), WMI/DCOM or registry (WINREG) RPC.
   - No ADMIN$/C$ share access and no PSEXESVC pipe.
   - No WinRM (5985/5986), SSH, VNC or Telnet.
   - The DC opened no SMB/RPC connection to the workstation.
   - No AnyDesk/TeamViewer/ScreenConnect/RustDesk/Splashtop/Atera/ngrok or similar tool in DNS or TLS.
4. **Shares used:** only IPC$, sysvol and NETLOGON (Administrator profile creation at 02:36:24), plus FileShare after ricksanchez logged on at 04:08:16. On FileShare there was only Explorer browsing (Desktop.ini/AutoRun.inf), so most likely the user's normal drive mapping.
5. **Ruled out (note):** 10.90.90.90 is a DNS-blocklist sinkhole (its certificate reads 'dnsbl@example.com') and 192.168.45.1 is the DC's DNS forwarder; neither is a lateral-movement peer. The capture shows no other internal hosts.

Not verified: Several sources asked for in this step do not exist in the case, which holds only the pcap:
- No host artefacts: TerminalServices LSM/RCM EVTX, Security 4624 type 10 / 4648 / 5140 / 5145 / 7045, WinRM/PowerShell logs, mstsc MRU / Terminal Server Client registry, Default.rdp, bitmap cache, raw.dissect_remoteaccess_* and raw.dissect_mru_networkdrive, ShellBags, $MFT (for RMM folders).
- No memory image (vol_netscan).
So session IDs, RDP accounts and source names on the hosts cannot be confirmed, and Restricted Admin use cannot be checked.

Some activity is invisible on the wire:
- anything carried inside the encrypted/raw Meterpreter channel (pivoting, port forwarding, token or credential use);
- the content of the RDP sessions;
- any activity outside 21:58–05:38 UTC.

The DC's own CredSSP Kerberos request for TERMSRV/DESKTOP-SDN1RPT is not on the wire because the KDC is local to the DC.

Step 9: User activity on files and folders

Host artefacts for user file and folder activity are not in the case. information_schema has only raw.pcap_files and the raw.suricata_* tables, so there is no LNK, Jump List, ShellBag, MRU, WordWheelQuery, Windows Timeline, SIDR, thumbcache, Notepad, MFT, UsnJrnl, Recycle Bin, or USB/EMDMgmt/setupapi data. Instead I rebuilt file activity from SMB2 and HTTP on the wire:
- All SMB CREATE/READ/WRITE records and all SMB fileinfo records were reviewed.
- Sysvol GPO reads are routine, with constant MD5s.

- The NETLOGON 'Default User.V6' lookup and the protected_storage pipe were already covered in F-008.
- The samr pipe writes at 22:10 are pre-attack and normal. This corrects step 7's statement that no SAMR was seen.

New finding F-017 (medium/possible), from the ricksanchez session on the workstation:
- The workstation connected to \\CITADEL-DC01\FileShare at 04:08:16 UTC and only enumerated the root (Desktop.ini/AutoRun.inf probes, 12.7 KB to the client in 90 min). No file was read or written on the share.
- The share root's modified/changed time was 03:34:18 UTC. No SMB client was active then, and the DC Meterpreter session was live, so the directory was changed locally on the DC. Whether the attacker did it cannot be determined from the pcap (T-018).

Pre-attack browsing by mortysmith around 22:00 UTC is benign context (note recorded).

Not verified: The case has no disk, registry or EVTX artefacts, so I could not check files opened, created, renamed or deleted on either host, Recycle Bin contents, removable media, or the coreupdater.exe on-disk path and Zone.Identifier. Suricata does not log SMB2 QUERY_DIRECTORY results, so the contents of FileShare and the entry changed at 03:34:18 cannot be identified. Anything done locally on the hosts or inside the Meterpreter channel is not visible.

Step 10: Web browsing, downloads and e-mail

The case has no browser, MFT or Zone.Identifier tables, so I rebuilt web activity from Suricata HTTP, DNS, TLS SNI and fileinfo records in case001.pcap.

New finding F-018 (high/probable): coreupdater.exe was delivered by hand-typed web browsing, with no search, phishing or e-mail step.

DC:
- 02:23:09: IE11 started for the first time in the attacker's RDP session.
- 02:23:41: GET http://194.61.24.102/, a 228 B landing page (SHA256 6978f8fa…2d77, IOC I-010).
- 02:24:06: /coreupdater.exe downloaded with that page as Referer.
- The DC made no SmartScreen lookups.

Workstation:
- 02:39:04: Edge started for the first time in the new Administrator profile.
- 02:39:26: same landing page loaded, with a SmartScreen nav check at 02:39:27.
- 02:39:58: download with the same Referer, followed by a download-reputation check at 02:40:01.
- 02:40:45 and 02:40:49.938: SmartScreen checkappexec lookups. The second is 2 ms after the Meterpreter connection to 203.78.103.109:443. This is strong network evidence that the file was launched from the shell, which supports F-007.

Also checked:
- No SMTP/IMAP/POP3 traffic and no webmail or cloud-storage use during the attack window.
- Browsing by mortysmith before the attack (Bing images, CNN, Office, Discord) is benign, as are checkappexec lookups at logon. Both are recorded in a note.

Added timeline events T-019 and T-020.

Not verified: Without Edge/IE history and download databases, MFT/$J, Zone.Identifier ADS or Prefetch, I can't establish where coreupdater.exe was saved, its MOTW HostUrl/ReferrerUrl, or a host-side execution time. HTTPS content (search terms, pages) can't be seen in the pcap. The content of the landing page wasn't extracted (Suricata gave only hashes and size). Every conclusion rests on one source, so it stays probable.

Step 11: Defense evasion and anti-forensics

Defense evasion was reviewed in a network-only case. The host checks this step asks for cannot be done: there is no EVTX (1102/104/4616/Defender 5001-5012), no MFT/$J/$LogFile (timestomping, mass deletion, wiping tools), no Defender exclusions table, no registry and no memory image. On the wire:

(1) New finding F-019 (medium/probable): masquerading. The stager is named coreupdater.exe, and Meterpreter C2 runs as raw non-TLS TCP on port 443 with no DNS resolution of the attacker IPs. The stage is loaded reflectively in memory (T1036.005/T1571/T1620).
(2) New finding F-020 (low/possible): the only Defender MAPS (wdcp.microsoft.com) lookup came from the workstation at 02:39:46 UTC, during the download. Defender was therefore active but did not stop the C2 at 02:40:49. Whether the stager went undetected or the attacker tampered with Defender (T1562.001) cannot be determined. The DC shows no AV cloud traffic. Telemetry on the workstation kept running until the end of the capture. Timeline event added for 02:39:46.
(3) No SMB access to log files, Defender files or admin shares. The GPO files were unchanged (step 6).
(4) Capture continuity: a 2 h 24 min pre-attack gap (22:10:48 to 00:34:48 UTC) is recorded as a note and is not attacker activity. The attack window is covered continuously. NTP shows nothing anomalous.

Not verified: Log clearing (1102/104), RecordID gaps, Defender events and exclusions, registry-based disabling, timestomping (SI<FN), USN deletions, Prefetch deletion, PowerShell logging changes, time changes (4616) and memory injection or hidden processes (malfind/psxview) could not be assessed. No EVTX, registry, MFT/$J or memory image is in the case. Anti-forensics done locally on the DC or workstation over the established Meterpreter sessions would not show up on the wire.

Step 12: Collection, exfiltration and communications

Collection/exfiltration review (network only). I ranked all internal→external flows by bytes_toserver, reviewed POST/PUT in HTTP, all fileinfo transfers, DNS (length, distinct names, cloud-storage domains) and the RDP volumes. I found no bulk exfiltration, no archive transfer, no cloud-storage upload and no DNS tunnelling. Meterpreter C2 upload totals ≈0.50 MB across 5 sessions; the largest is the workstation Administrator session, 280 KB. The attacker RDP session sent 12.76 MB DC→attacker, which fits screen graphics; it is encrypted, so clipboard or drive redirection cannot be excluded. The OneDrive traffic (oneclient.sfx.ms, 39 MB down) is client setup for the new Administrator profile. New finding F-021, plus a noise note. Network IOCs (194.61.24.102, 203.78.103.109, URL) were already recorded.

Not verified: There are no host artefacts: no MFT/UsnJrnl (archives, staging), 7-Zip history, SRUM network usage, BITS events, OneDrive ODL/SyncEngineDatabase, Sysmon 3/22, 5156 or firewall logs. The content of the Meterpreter and RDP channels is opaque (raw/encrypted). Three C2 sessions were still alive at the end of the capture (05:38 UTC), so later exfiltration cannot be assessed.

Step 13: Incident timeline and correlation

I put the incident timeline together from the network capture alone (case001.pcap, Suricata).

Chronology (UTC):
- **Start of the incident:** at 02:19:26, 194.61.24.102 sent an Nmap RDP probe to the domain controller (DC). RDP password guessing followed: 95 failed CredSSP logons between 02:21:26 and 02:21:46, then the first successful Administrator logon at 02:21:47.
- **DC compromise:** the attacker opened an interactive RDP session at 02:22:36. They then downloaded coreupdater.exe from http://194.61.24.102/ with IE11 at 02:24:06. Meterpreter sessions to 203.78.103.109:443 started at 02:25:18 and 02:29:49.
- **Pivot to the workstation:** RDP from the DC to the workstation started at 02:35:55, and Administrator logged on there for the first time at 02:36:24. Edge downloaded the same file at 02:39:58, and a C2 session started at 02:40:49.
- **Last contact from 194.61.24.102:** a second RDP session ran from 02:56:03 to 02:57:41. A DC C2 session started inside it at 02:56:38.
- **Later activity:** the FileShare root on the DC was modified at 03:34:18. ricksanchez logged on at 04:08:15, and a new C2 session started 30 s later at 04:08:45.
- **Last event seen:** at 05:38:42–05:38:51, three C2 sessions were still ESTABLISHED when the capture ended.

Recorded in this step:
- Timeline events T-022 (last attacker RDP), T-023 (end of capture with live C2) and T-024 (first DC Meterpreter session closed, then re-staged).
- Consolidated finding F-022.

- IOCs I-011 (landing URL http://194.61.24.102/) and I-012 (RDP cookies nmap/Administrator, plus the Nmap JA3).
- Earlier IOCs I-001..I-010 cover the IPs, the payload URL, the SHA256/SHA1/MD5 hashes, the file name, the Administrator account, the brute-force JA3/JA4 and the landing-page hash.

Time consistency: the raw EVE times are in +0200, and their conversion to UTC is consistent across event types (pcap_cnt order matches). In raw.suricata_flow the 'timestamp' field is always the first packet of the pcap, so only flow_start/flow_end are valid; the timeline uses them correctly. Kerberos times and packet times agree within seconds.

Not verified: There are no host artefacts (EVTX, MFT, registry, Prefetch, memory). Because of that I could not measure clock skew between hosts or check for timestomping, and the execution and persistence mechanisms are inferred from network behaviour, not observed. Anything after 05:38:57 UTC, including the outcome of the three live C2 sessions, is unknown. The RDP and Meterpreter content is encrypted or opaque, so the attacker's actions inside those sessions are not visible.

Step 14: Summary, answers and recommendations

Final report elements recorded:
- set_summary: verdict, executive summary, detailed technical assessment and limitations. The assessment covers recon, RDP brute force, the Administrator compromise, the coreupdater.exe Meterpreter stager, five raw-TCP C2 sessions to 203.78.103.109:443, the RDP pivot DC→workstation, logon-triggered re-staging, the FileShare change, the exfiltration assessment and three C2 sessions live at the end of the capture. It is cross-referenced to F-001..F-022 and T-001..T-024.
- set_recommendations: immediate containment, eradication and recovery, hardening, detection rules with IOC watchlists, and further investigation (host triage, memory, AD audit, sample reverse-engineering, extended network logs).

No questions were answered with answer_question because the analyst guidance file contains none.

Not verified: The case is network-only (one pcap). There are no EVTX, registry, filesystem or memory artefacts, so execution, persistence mechanism, privileges and post-exploitation remain inferred, and all critical/high findings stay 'probable'. RDP and C2 content is encrypted. Nothing is visible outside 21:58:07–05:38:57 UTC or during the 22:10:48–00:34:48 gap. The machine time zone is unknown.

Appendix B: Automatic detections

Engine	Level	Count
Suricata	low	450
Suricata	high	7
Suricata	medium	4

Critical and high detections (to compare with the findings; a detection is not a finding until it has been qualified):

UTC	Engine	Level	Rule	Host	Channel	EventID	RecordID	Reference
2020-09-19 02:24:06.940653	Suricata	high	ET INFO PE EXE or DLL Windows file download HTTP	10.42.85.10	http	2018959	178050707123 6130	suricata/case001.pcap/eve_alert.jsonl#370
2020-09-19 02:25:19.071794	Suricata	high	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	10.42.85.10	failed	2025644	186661126815 0811	suricata/case001.pcap/eve_alert.jsonl#439
2020-09-19 02:29:49.580408	Suricata	high	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	10.42.85.10	failed	2025644	152833364753 4182	suricata/case001.pcap/eve_alert.jsonl#403
2020-09-19 02:39:58.412735	Suricata	high	ET INFO PE EXE or DLL Windows file download HTTP	10.42.85.115	http	2018959	176186070649 5548	suricata/case001.pcap/eve_alert.jsonl#443
2020-09-19 02:40:50.282950	Suricata	high	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	10.42.85.115	failed	2025644	362175423385 991	suricata/case001.pcap/eve_alert.jsonl#446
2020-09-19 02:56:38.758942	Suricata	high	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	10.42.85.10	failed	2025644	173030239598 0396	suricata/case001.pcap/eve_alert.jsonl#452
2020-09-19 04:08:45.783080	Suricata	high	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	10.42.85.115	failed	2025644	151486936731 1013	suricata/case001.pcap/eve_alert.jsonl#458

Appendix C: Working notes (leads ruled out, false positives)

- [step 2] Step 2 alert triage – benign / context alerts: 'ET USER_AGENTS Microsoft Device Metadata Retrieval Client User-Agent' (24) and 'ET INFO Windows OS Submitting USB Metadata to Microsoft' (4) from 10.42.85.115 to Microsoft IPs = normal Windows device-metadata traffic (FP). 'ET INFO Observed UA-CPU Header' (1, DC -> 204.79.197.200:80 at 02:23:09 UTC) = IE11 on the DC contacting Bing/Microsoft at first browser launch during the attacker RDP session (context, not malicious by itself). Discord alerts (14, 22:05 UTC, 10.42.85.115 -> 162.159.135.233/136.232, ~52 kB up/676 kB down, 4h before the intrusion) = probable user activity, no link with the attack found; low priority lead. No Hayabusa/Chainsaw/Zircolite/Defender detections and empty hashset: only Suricata (ET Open) alerts exist in this case.
- [step 3] Step 3 (execution): DCERPC interfaces seen workstation→DC are only DRSUAPI (e3514235, opnums 0/1/12/13), EPM, NETLOGON (12345678…cffb), LSARPC (12345778…) and SAMR pipe at 22:10 UTC – normal domain-member traffic. No SVCCTL (367abb81), ATSVC/ITaskSchedulerService, WMI/DCOM (remote execution, PsExec-style) observed in cleartext. SMB activity limited to SYSVOL GPO reads, NETLOGON profile lookup, protected_storage and FileShare browsing (Desktop.ini/AutoRun.inf). DC CryptoAPI requests to intel.com/symcd/verisign at 04:02 UTC (and workstation at 05:06 UTC) are certificate validation, not attributable to a specific program without host data.
- [step 5] Network context / ruled-out leads (step 5): (1) 10.90.90.90 is a DNS-blocklist sinkhole (pfBlockerNG-style DNSBL certificate 'CN=CN_DNSBL, Email=dnsbl@example.com'). It answers ad/telemetry domains (settings-win.data.microsoft.com, reddit/wikia trackers) for the workstation (eve_tls#1). Resolver 192.168.45.1 is the gateway. This is benign environment context. (2) At 04:02:07–04:02:10 UTC the DC (10.42.85.10) queried ocsp.digicert.com, ocsp.verisign.com, sf.symcd.com, crl.geotrust.com, www.intel.com and certificates.intel.com (eve_dns#2183, #2185, #2187, #2246, #2547, #2549) and fetched the 'Intel External Basic Issuing CA 3A' CRL. This is an Authenticode chain check of an Intel-signed binary, made while the DC's Meterpreter session (eve_flow#18312) was active. It is unexplained, but legitimate driver/update activity cannot be excluded without host artefacts, so it is not recorded as a finding. (3) DCERPC from the workstation to the DC (DRSUAPI opnums 0/12/1, LSARPC 76/77, NETLOGON 29, EPM 3) and LDAP binds/searches are routine domain-member traffic. There is no DsGetNCChanges (DCSync) and no SVCCTL/ATSVC. (4) The Kerberos AS-REP for Administrator at 02:36:24 used an RC4-HMAC enc-part (eve_krb5#60, weak_encryption anomaly). This suggests the account lacks AES keys or the client requested RC4. It is hygiene context only. (5) The workstation's large TLS transfers (oneclient.sfx.ms 39 MB at 02:37 UTC = OneDrive client update for the new Administrator profile; *.events.data.microsoft.com telemetry) are benign.
- [step 6] Step 6 (persistence): there are no host persistence sources in the case. information_schema shows no raw.reg_*, dissect_*, evtx, taskcache, startup, mft or registry tables. So none of these could be checked: services (7045/4697, reg_services), scheduled tasks (4698/106/200, TaskCache), Run/RunOnce/Winlogon/IFEO, AppInit_DLLs, COM hijack (HKCU CLSID), Active Setup, print monitors, BootExecute, WMI consumers (5861), Startup folders, hosts file, Office startup, PendingFileRenameOperations. On the wire, the following were ruled out: remote service/task/registry creation (no SVCCTL/ATSVC/WINREG/WMI DCERPC) and modification of Default Domain Policy (stable MD5s across 4 GPO refreshes). Only one boot was observed (DESKTOP-SDN1RPT$ AS-REQ at 21:59:39 UTC), so reboot survival cannot be assessed.
- [step 7] Step 7 – Authentication: the 29,202 TLS-only flows from 194.61.24.102 to DC:3389 at 02:19:26-02:20:26 (723/134 B, no X.224/CredSSP) are handshakes the server rejected (tool misconfiguration / flood). They are not credential attempts, so a raw count of 29k must not be reported as '29k failed logons'. The real guessing was 95 CredSSP failures at 02:21:26-02:21:46 (F-013). Normal Kerberos KDC_ERR_PREAUTH_REQUIRED errors are not failures.
- [step 8] Ruled out as a lateral-movement or C2 peer: 10.90.90.90. The workstation sent 354 TLS flows to 443 and 1 HTTP flow to 80 there, from 21:59 to 05:28 UTC. Every TLS session presents a 'CN=CN_DNSBL, Email=dnsbl@example.com' certificate for many ad/telemetry SNIs (settings-win.data.microsoft.com, alb.reddit.com, ib.adnxs.com…) (eve_tls #1, #170). This is a DNS-blocklist/sinkhole service in the environment. 192.168.45.1 is the DC's upstream DNS forwarder (559 UDP/53 flows). Neither host received SMB, RDP or WinRM traffic. Also note that Suricata eve_rdp reverses src/dst labels on the X.224 request/response records. Use eve_flow for the real direction (the client is the DC for the internal hop, and 194.61.24.102 for the Internet hop).

- [step 9] Step 9 (user file activity): none of the host artefact tables (LNK, Jump Lists, ShellBags, RecentDocs/OpenSave/LastVisited, WordWheelQuery, Windows Timeline, Search/SIDR, thumbcache, Notepad tabs, MFT, UsnJrnl, LogFile, RecycleBin, USB/USBSTOR/EMDMgmt/setupapi) exists; information_schema shows only raw.pcap_files plus raw.suricata_*. Pre-attack context: at ~22:00-22:05 UTC mortysmith on 10.42.85.115 browsed Rick and Morty fan images (turner/deviantart/wikia), which is benign (eve_http #14-#29). Correction to step 7: 16 SMB2 WRITEs to the 'samr' pipe from the workstation at 22:10:08 UTC (eve_smb #55-#71), pre-attack, routine domain-member behaviour.
- [step 10] Step 10 (web/e-mail): no browser artefact tables (Chromium/Firefox/IE WebCache/hindsight/dissect_browser, MFT, Zone.Identifier) in case. Pre-attack browsing by mortysmith on workstation (~22:00-22:08 UTC: Bing image search Rick and Morty images, cnn.com, office.com, Discord) is benign context. No e-mail protocol flows (SMTP/IMAP/POP3) and no webmail/cloud-storage SNI during the attack window. Logon-time checkappexec lookups (22:00:15, 02:36:31, 04:08:34) are baseline Explorer behaviour, not attributed. Post-02:41 HTTP is Microsoft telemetry, device metadata, weather tiles and Delivery Optimization (05:23) – noise.
- [step 11] Step 11 – Capture continuity: the flow start times show no new flow between 2020-09-18 22:10:48.96 and 2020-09-19 00:34:48.92 UTC (2 h 24 min), and no flow spans that interval. This is a pre-attack gap in capture or traffic (idle hosts or a capture interruption), not attacker anti-forensics. Other gaps are at most ~6 min (periodic idle). The attack window 02:19–05:38 is covered continuously. The only NTP traffic is workstation→DC UDP/123 at the start of the capture; nothing on the wire suggests a system time change, but 4616 / Kernel-General 1 cannot be checked.
- [step 12] Step 12 noise ruled out:
  - oneclient.sfx.ms (104.119.185.124), 02:37 UTC, 39.4 MB down: OneDrive client setup for the new Administrator profile on DESKTOP-SDN1RPT.
  - v10.events.data.microsoft.com (52.114.132.91), 02:44 UTC, 180 KB up: Windows telemetry.
  - HTTP POSTs to dmd.metaservices.microsoft.com/metadata.svc and go.microsoft.com/fwlink?LinkID=252669: device-metadata retrieval.
  - POST to o.ss2.us at 22:01: pre-attack.
  - Files that crossed the wire: Appx/PNG/JPG downloads from browsing and Store.

  None of these is exfiltration.
- [step 13] Time consistency (step 13): raw Suricata EVE timestamps are in +0200 and were normalised to UTC (-2 h) consistently across eve_alert/eve_tls/eve_http/eve_flow (e.g. alert #25 04:19:26.792622+0200 = timeline 02:19:26.792622 UTC; same pcap_cnt sequence 84372/84373 as tls #920). In raw.suricata_flow the 'timestamp' field is constant (2020-09-18T23:58:07.470323+0200 = first packet of the pcap) and must not be used: flow_start/flow_end carry the real times, and the timeline correctly uses them. No clock skew between hosts can be measured (no host logs); Kerberos ticket times and packet times line up within seconds (e.g. Administrator AS-REP 02:36:24 vs RDP hop 02:35:55). Timestomping cannot be assessed (no MFT). DC RDP self-signed certificate notBefore 2020-09-17 22:28:10 (tls #920) = context only (fresh certificate / recent host build), not an attacker artefact.

Appendix E: Artifact coverage

Artifact	Tool	Mapping	Events	First (UTC)	Last (UTC)
Network flows (Suricata)	Suricata	validated	65044	2020-09-18 21:58:07.470329	2020-09-19 05:38:57.828520
TLS (Suricata)	Suricata	validated	30453	2020-09-18 21:58:07.470323	2020-09-19 05:38:39.310228
DNS (Suricata)	Suricata	validated	2650	2020-09-18 21:58:49.238799	2020-09-19 05:38:39.007813
Network events (Suricata)	Suricata	validated	1233	2020-09-18 21:58:07.470323	2020-09-19 05:37:22.585366
Files transferred (Suricata)	Suricata	validated	241	2020-09-18 21:58:07.470323	2020-09-19 05:25:21.431188
HTTP (Suricata)	Suricata	validated	229	2020-09-18 22:00:11.462431	2020-09-19 05:23:57.308493
Network anomalies (Suricata)	Suricata	validated	18	2020-09-18 22:01:00.976791	2020-09-19 05:23:49.804096

Appendix F: Traceability

DuckDB database	C:\Sleuthline\cases\test03\db\case.duckdb SHA-256 6c3a35a8cf85d7b507d06fa2711b0ecd3487c5851fa52105064f29fa68d82dbf
Pipeline version	commit 5d46b65c0f4f; content sha256 507779379B984D32; deployed 2026-10-20 18:46
Collection integrity	not checked (run the inventory again with a recent version of the pipeline)
Normalization mappings	v2.2 sha256 8a32b59741d3439e43d8b015c92d36d65fbfff14a8f28ae100ad78305263d1ac
Model	anthropic / claude-opus-5-5
Agent runs	2026-10-20T16:53:58Z → 2026-10-20T17:56:10Z: anthropic/claude-opus-5-5, playbook sha256 0833d7d614f54875…
Usage	5,314,256 input tokens (of which 4,702,707 read from cache), 117,046 output tokens, 159 calls
Full trace	C:\Sleuthline\cases\test03\analysis\transcript.jsonl: every SQL query, result and model answer
Analysis state	C:\Sleuthline\cases\test03\analysis\findings.json
Exports	iocs.csv, key_timeline.csv, incident_timeline.csv
Reference format	<output relative to parsed>#<record number>: CSV data line (header excluded) or JSONL line
Report generated on	2026-10-20 17:56 UTC